```
=== 1. OVERVIEW ===
Shape: (79330, 31)

First rows:
    IsCanceled  LeadTime  ArrivalDateYear ArrivalDateMonth  ArrivalDateWeekNumber  ArrivalDateDayOfMonth  StaysInWeekendNights  StaysInWeekNights  Adults  Children  Babies  \
0           0         6             2015             July                     27                      1                     0                  2       1       0.0       0
1           1        88             2015             July                     27                      1                     0                  4       2       0.0       0
2           1        65             2015             July                     27                      1                     0                  4       1       0.0       0
3           1        92             2015             July                     27                      1                     2                  4       2       0.0       0
4           1       100             2015             July                     27                      2                     0                  2       2       0.0       0

        Meal Country  MarketSegment DistributionChannel  IsRepeatedGuest  PreviousCancellations  PreviousBookingsNotCanceled  ReservedRoomType  AssignedRoomType  BookingChanges  \
0  HB            PRT  Offline TA/TO               TA/TO                0                      0                            0  A                 A                               0
1  BB            PRT      Online TA               TA/TO                0                      0                            0  A                 A                               0
2  BB            PRT      Online TA               TA/TO                0                      0                            0  A                 A                               0
3  BB            PRT      Online TA               TA/TO                0                      0                            0  A                 A                               0
4  BB            PRT      Online TA               TA/TO                0                      0                            0  A                 A                               0

       DepositType        Agent      Company  DaysInWaitingList CustomerType   ADR  RequiredCarParkingSpaces  TotalOfSpecialRequests ReservationStatus ReservationStatusDate
0  No Deposit                6         NULL                  0    Transient   0.0                         0                       0         Check-Out           2015-07-03
1  No Deposit                9         NULL                  0    Transient  76.5                         0                       1          Canceled           2015-07-01
2  No Deposit                9         NULL                  0    Transient  68.0                         0                       1          Canceled           2015-04-30
3  No Deposit                9         NULL                  0    Transient  76.5                         0                       2          Canceled           2015-06-23
4  No Deposit                9         NULL                  0    Transient  76.5                         0                       1          Canceled           2015-04-02

Types:
 IsCanceled                      int64
LeadTime                        int64
ArrivalDateYear                 int64
ArrivalDateMonth                  str
ArrivalDateWeekNumber           int64
ArrivalDateDayOfMonth           int64
StaysInWeekendNights            int64
StaysInWeekNights               int64
Adults                          int64
Children                      float64
Babies                          int64
Meal                              str
Country                           str
MarketSegment                     str
DistributionChannel               str
IsRepeatedGuest                 int64
PreviousCancellations           int64
PreviousBookingsNotCanceled     int64
ReservedRoomType                  str
AssignedRoomType                  str
BookingChanges                  int64
DepositType                       str
Agent                             str
Company                           str
DaysInWaitingList               int64
CustomerType                      str
ADR                           float64
RequiredCarParkingSpaces        int64
TotalOfSpecialRequests          int64
ReservationStatus                 str
ReservationStatusDate             str
dtype: object

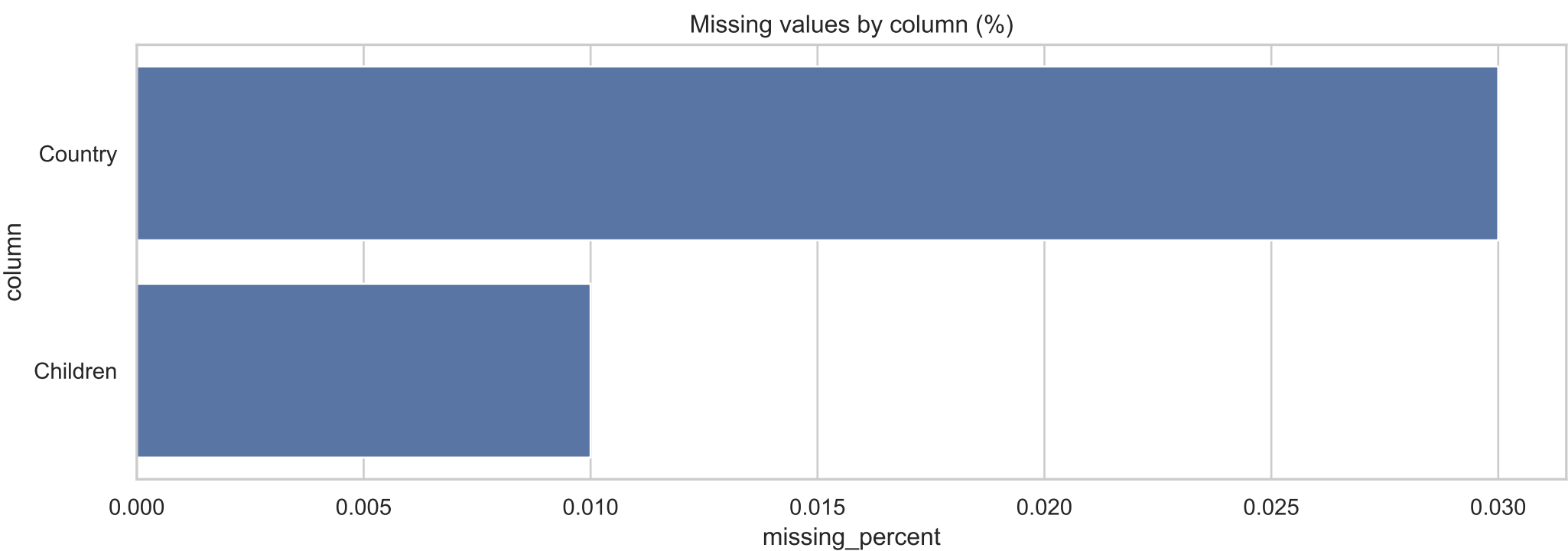
Summary:
                               count unique              top   freq         mean         std     min     25%     50%     75%     max
IsCanceled                   79330.0    NaN              NaN    NaN      0.41727    0.493111     0.0     0.0     0.0     1.0     1.0
LeadTime                     79330.0    NaN              NaN    NaN   109.735724  110.948526     0.0    23.0    74.0   163.0   629.0
ArrivalDateYear              79330.0    NaN              NaN    NaN  2016.174285    0.699181  2015.0  2016.0  2016.0  2017.0  2017.0
ArrivalDateMonth               79330     12           August   8983          NaN         NaN     NaN     NaN     NaN     NaN     NaN
ArrivalDateWeekNumber        79330.0    NaN              NaN    NaN    27.177449   13.398523     1.0    17.0    27.0    38.0    53.0
ArrivalDateDayOfMonth        79330.0    NaN              NaN    NaN    15.786625    8.728451     1.0     8.0    16.0    23.0    31.0
StaysInWeekendNights         79330.0    NaN              NaN    NaN     0.795185    0.885026     0.0     0.0     1.0     2.0    16.0
StaysInWeekNights            79330.0    NaN              NaN    NaN     2.182957    1.456416     0.0     1.0     2.0     3.0    41.0
```

Adults	79330.0	NaN	NaN	NaN	1.850977	0.509292	0.0	2.0	2.0	2.0	4.0
Children	79326.0	NaN	NaN	NaN	0.09137	0.372177	0.0	0.0	0.0	0.0	3.0
Babies	79330.0	NaN	NaN	NaN	0.004941	0.084323	0.0	0.0	0.0	0.0	10.0
Meal	79330	4	BB	62305	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Country	79306	166	PRT	30960	NaN	NaN	NaN	NaN	NaN	NaN	NaN
MarketSegment	79330	8	Online TA	38748	NaN	NaN	NaN	NaN	NaN	NaN	NaN
DistributionChannel	79330	5	TA/TO	68945	NaN	NaN	NaN	NaN	NaN	NaN	NaN
IsRepeatedGuest	79330.0	NaN	NaN	NaN	0.025615	0.157983	0.0	0.0	0.0	0.0	1.0
PreviousCancellations	79330.0	NaN	NaN	NaN	0.079743	0.415472	0.0	0.0	0.0	0.0	21.0
PreviousBookingsNotCanceled	79330.0	NaN	NaN	NaN	0.132371	1.693411	0.0	0.0	0.0	0.0	72.0
ReservedRoomType	79330	8	A	62595	NaN	NaN	NaN	NaN	NaN	NaN	NaN
AssignedRoomType	79330	9	A	57007	NaN	NaN	NaN	NaN	NaN	NaN	NaN
BookingChanges	79330.0	NaN	NaN	NaN	0.187369	0.60862	0.0	0.0	0.0	0.0	21.0
DepositType	79330	3	No Deposit	66442	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Agent	79330	224	9	31955	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Company	79330	208	NULL	75641	NaN	NaN	NaN	NaN	NaN	NaN	NaN
DaysInWaitingList	79330.0	NaN	NaN	NaN	3.226774	20.87089	0.0	0.0	0.0	0.0	391.0
CustomerType	79330	4	Transient	59404	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ADR	79330.0	NaN	NaN	NaN	105.304465	43.602954	0.0	79.2	99.9	126.0	5400.0
RequiredCarParkingSpaces	79330.0	NaN	NaN	NaN	0.024367	0.154919	0.0	0.0	0.0	0.0	3.0
TotalOfSpecialRequests	79330.0	NaN	NaN	NaN	0.546918	0.780776	0.0	0.0	0.0	1.0	5.0
ReservationStatus	79330	3	Check-Out	46228	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ReservationStatusDate	79330	864	2015-10-21	1416	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Exact duplicate rows: 25902

=== 2. MISSING VALUES ===

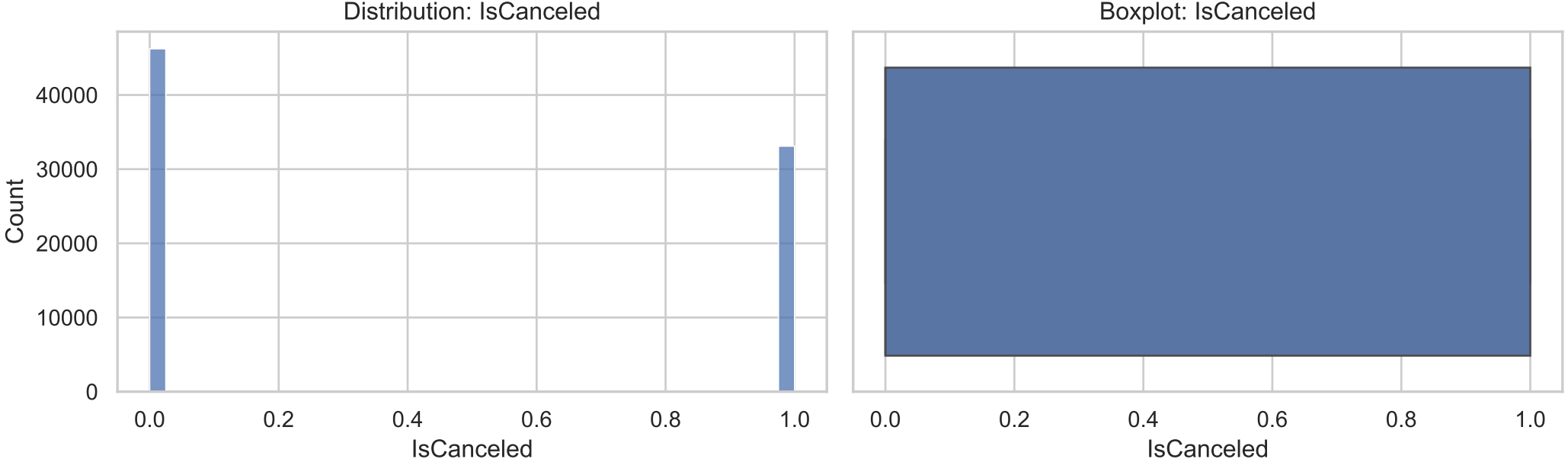
	missing_count	missing_percent	dtype
Country	24	0.03	str
Children	4	0.01	float64
IsCanceled	0	0.00	int64
ArrivalDateMonth	0	0.00	str
ArrivalDateWeekNumber	0	0.00	int64
LeadTime	0	0.00	int64
ArrivalDateYear	0	0.00	int64
StaysInWeekendNights	0	0.00	int64
ArrivalDateDayOfMonth	0	0.00	int64
Adults	0	0.00	int64
StaysInWeekNights	0	0.00	int64
Babies	0	0.00	int64
Meal	0	0.00	str
MarketSegment	0	0.00	str
DistributionChannel	0	0.00	str
IsRepeatedGuest	0	0.00	int64
PreviousCancellations	0	0.00	int64
PreviousBookingsNotCanceled	0	0.00	int64
ReservedRoomType	0	0.00	str
AssignedRoomType	0	0.00	str
BookingChanges	0	0.00	int64
DepositType	0	0.00	str
Agent	0	0.00	str
Company	0	0.00	str
DaysInWaitingList	0	0.00	int64
CustomerType	0	0.00	str
ADR	0	0.00	float64
RequiredCarParkingSpaces	0	0.00	int64
TotalOfSpecialRequests	0	0.00	int64
ReservationStatus	0	0.00	str
ReservationStatusDate	0	0.00	str

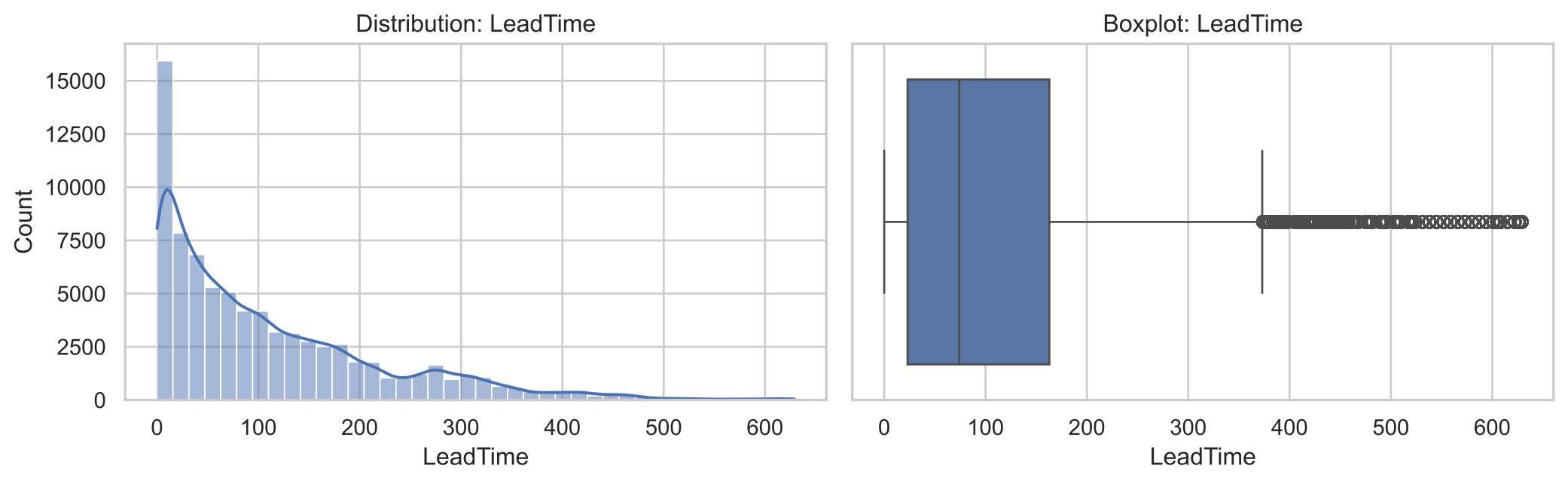

Missing values by column (%)
Country
Children
column
0.000
0.005
0.010
0.015
0.020
0.025
0.030
missing_percent

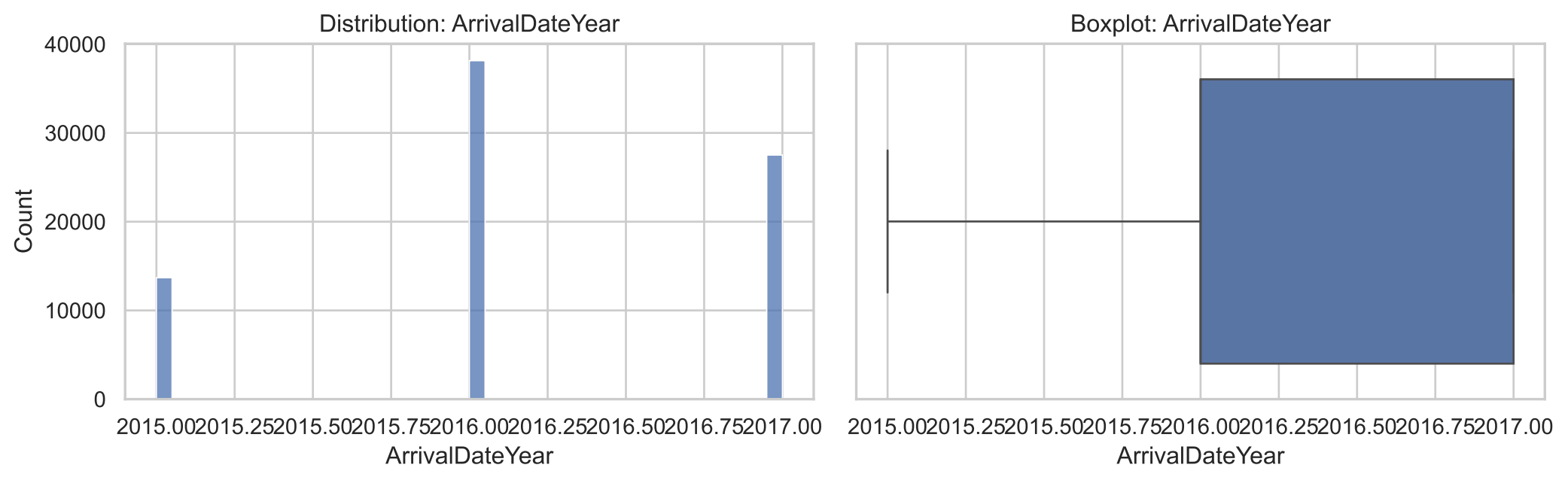
```
=== 3. VARIANCE / SPREAD ===
                                variance     std_dev  unique_values     min     max
LeadTime                       12309.575369  110.948526             453     0.0   629.0
ADR                             1901.217583   43.602954           5405     0.0  5400.0
DaysInWaitingList                435.594049   20.870890            115     0.0   391.0
ArrivalDateWeekNumber            179.520416   13.398523             53     1.0    53.0
ArrivalDateDayOfMonth             76.185849    8.728451             31     1.0    31.0
PreviousBookingsNotCanceled        2.867639    1.693411             73     0.0    72.0
StaysInWeekNights                  2.121148    1.456416             29     0.0    41.0
StaysInWeekendNights               0.783272    0.885026             14     0.0    16.0
TotalOfSpecialRequests             0.609612    0.780776              6     0.0     5.0
ArrivalDateYear                    0.488855    0.699181              3  2015.0  2017.0
BookingChanges                     0.370419    0.608620             21     0.0    21.0
Adults                             0.259379    0.509292              5     0.0     4.0
IsCanceled                         0.243159    0.493111              2     0.0     1.0
PreviousCancellations              0.172617    0.415472             10     0.0    21.0
Children                           0.138516    0.372177              4     0.0     3.0
IsRepeatedGuest                    0.024959    0.157983              2     0.0     1.0
RequiredCarParkingSpaces           0.024000    0.154919              4     0.0     3.0
Babies                             0.007110    0.084323              5     0.0    10.0

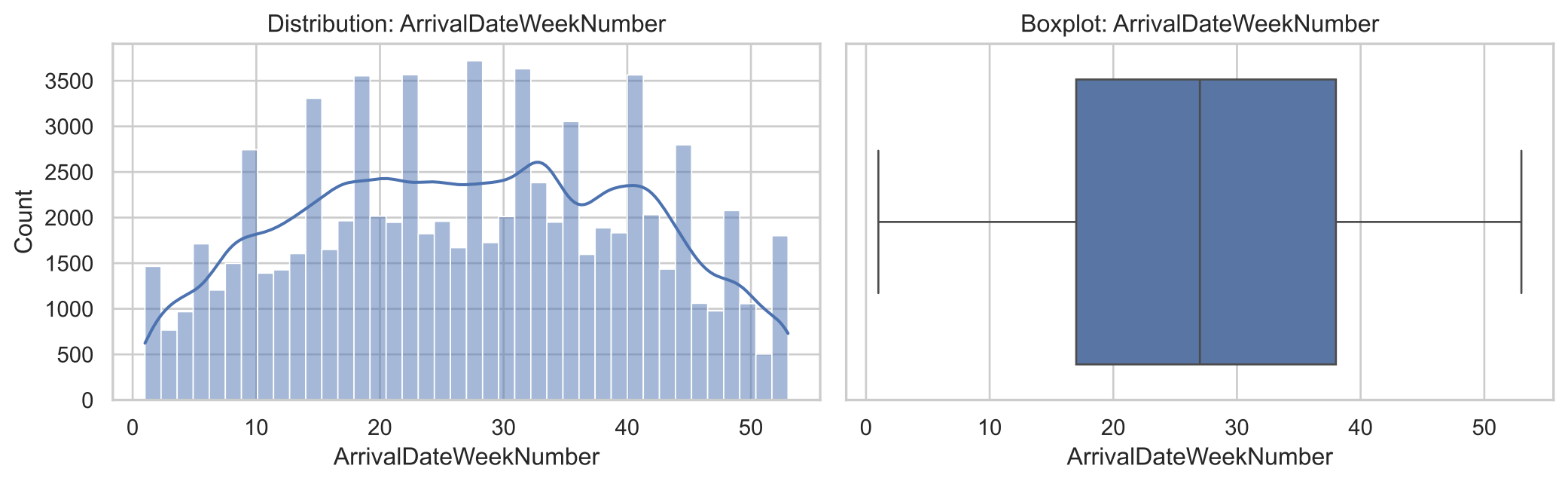
Constant columns: []
Low-cardinality columns (inspect, not automatically remove): ['IsCanceled', 'IsRepeatedGuest']

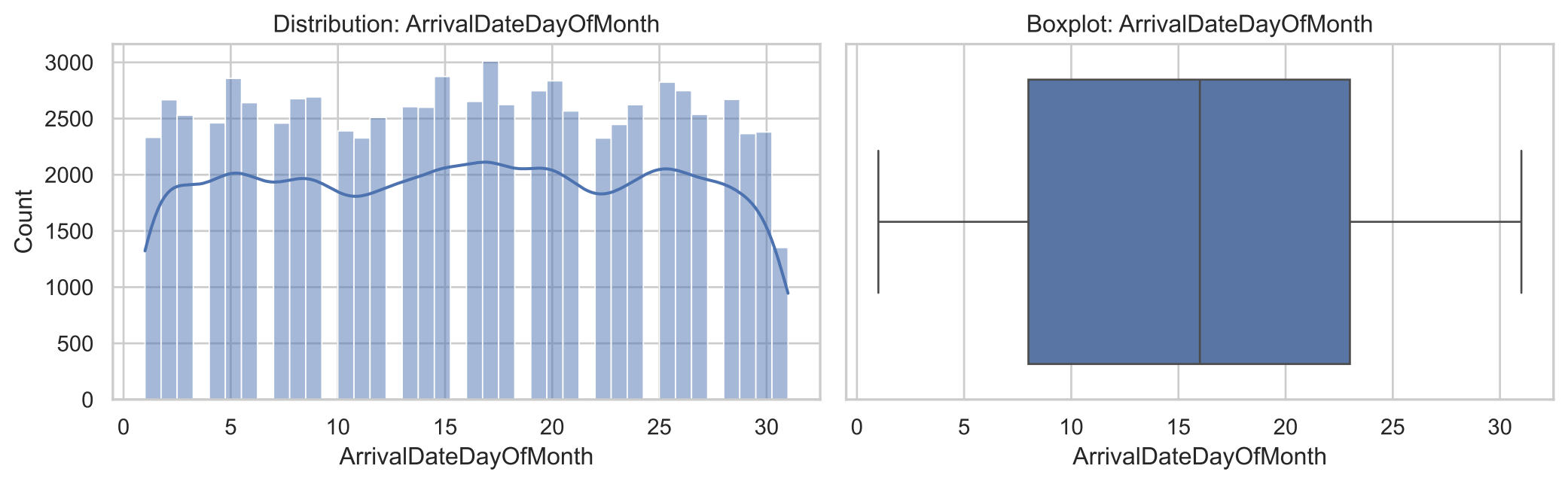
=== 4. NUMERIC DISTRIBUTIONS AND OUTLIERS ===
```

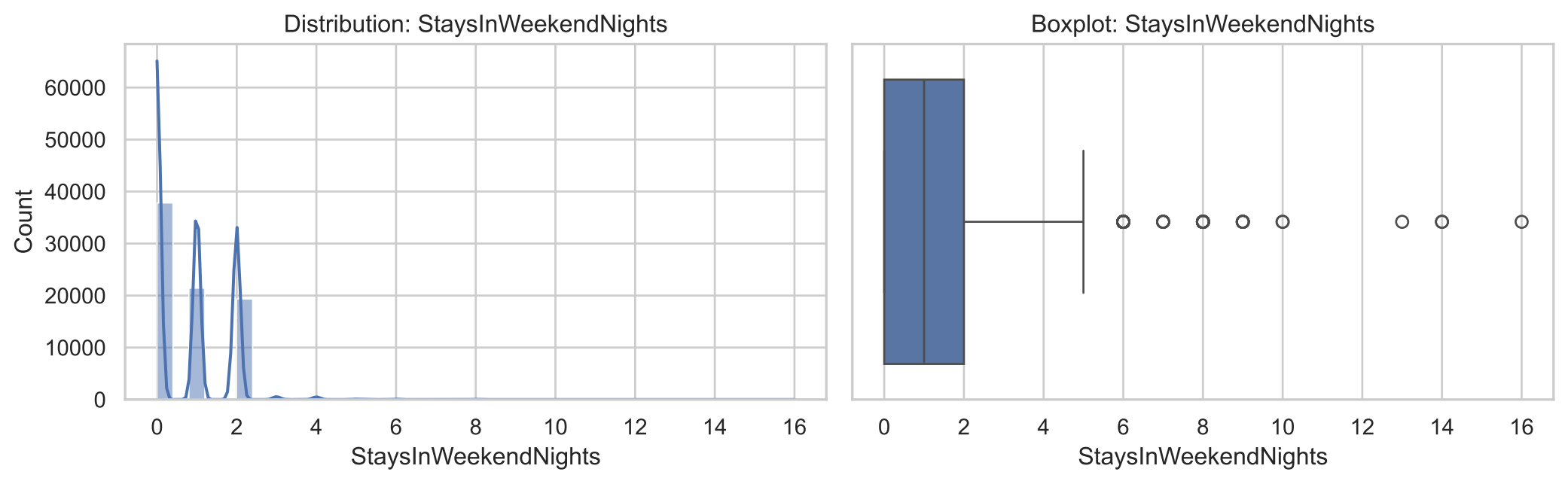

Distribution: IsCanceled
Count
40000
30000
20000
10000
0
0.0
0.2
0.4
0.6
0.8
1.0
IsCanceled
Boxplot: IsCanceled
0.0
0.2
0.4
0.6
0.8
1.0
IsCanceled

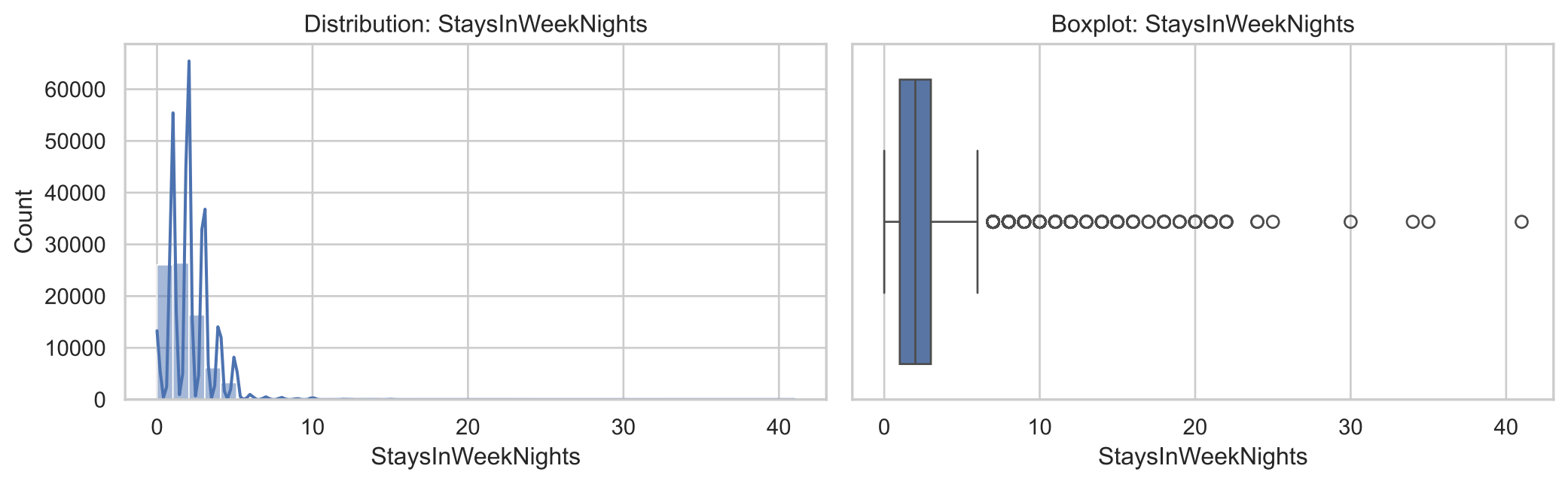

Distribution: LeadTime
Count
15000
12500
10000
7500
5000
2500
0
0
100
200
300
400
500
600
LeadTime
Boxplot: LeadTime
0
100
200
300
400
500
600
LeadTime

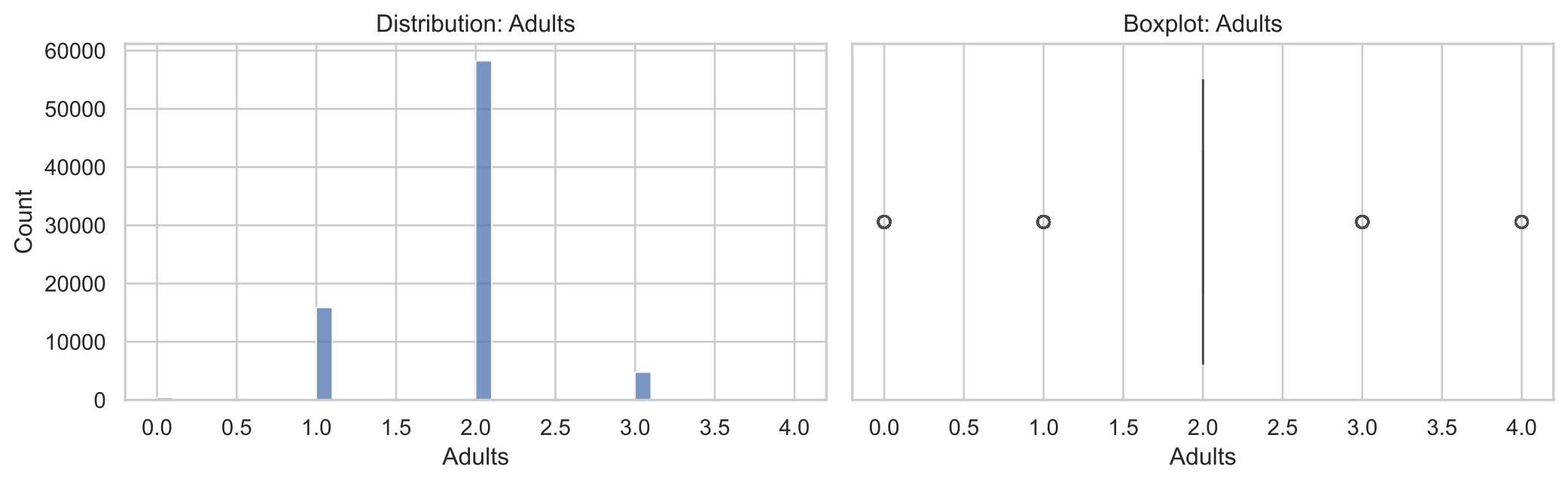

Distribution: ArrivalDateYear
Count
40000
30000
20000
10000
0
2015.00
2015.25
2015.50
2015.75
2016.00
2016.25
2016.50
2016.75
2017.00
ArrivalDateYear
Boxplot: ArrivalDateYear
2015.00
2015.25
2015.50
2015.75
2016.00
2016.25
2016.50
2016.75
2017.00
ArrivalDateYear

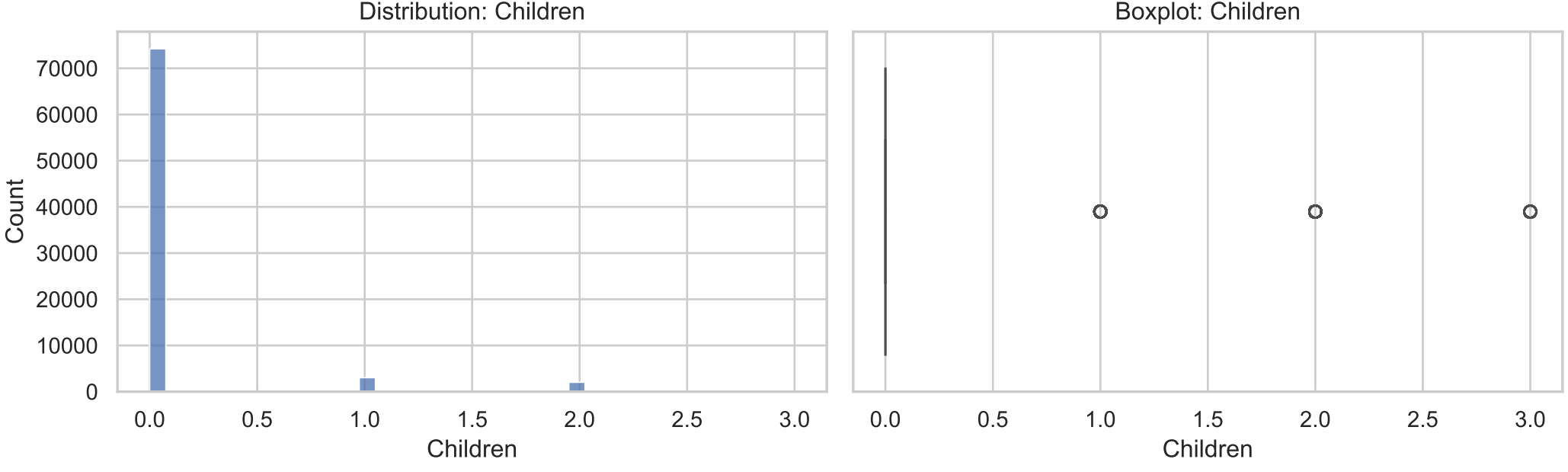

Distribution: ArrivalDateWeekNumber
Count
0
500
1000
1500
2000
2500
3000
3500
0
10
20
30
40
50
ArrivalDateWeekNumber
Boxplot: ArrivalDateWeekNumber
0
10
20
30
40
50
ArrivalDateWeekNumber

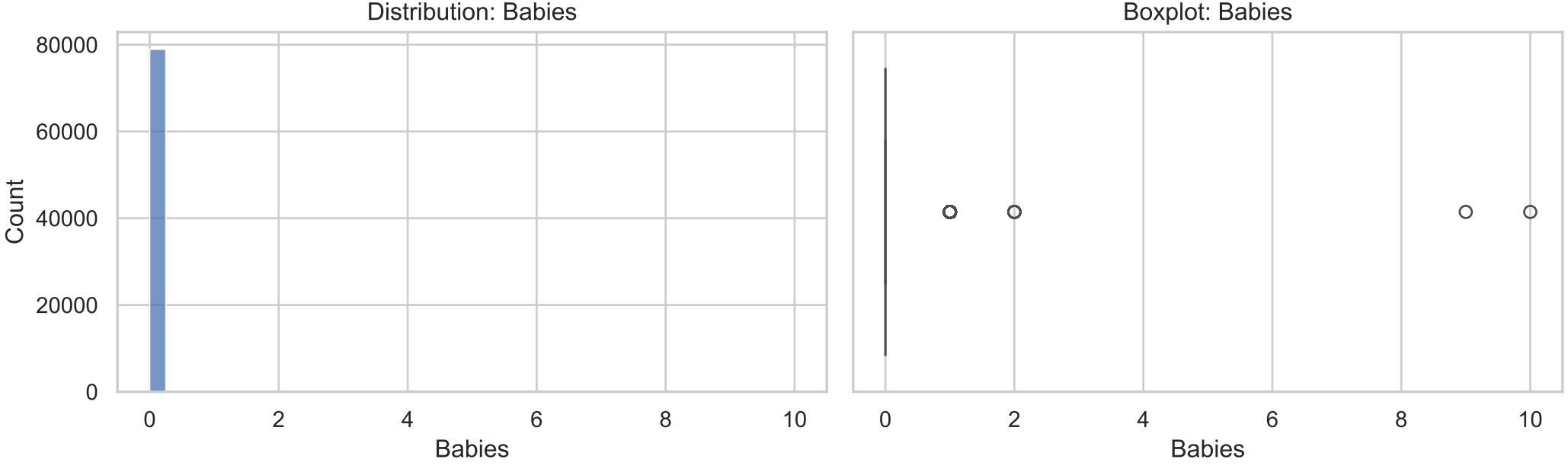

Distribution: ArrivalDateDayOfMonth
Count
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
25
30
ArrivalDateDayOfMonth
Boxplot: ArrivalDateDayOfMonth
0
5
10
15
20
25
30
ArrivalDateDayOfMonth

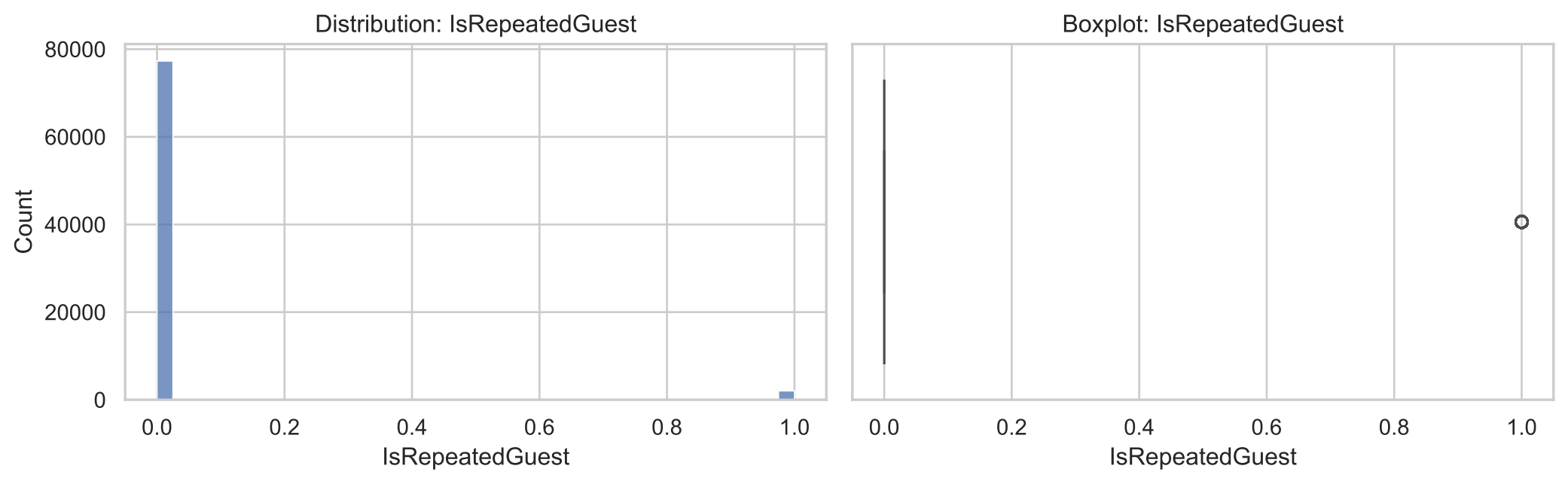

Distribution: StaysInWeekendNights
Count
60000
50000
40000
30000
20000
10000
0
0
2
4
6
8
10
12
14
16
StaysInWeekendNights
Boxplot: StaysInWeekendNights
0
2
4
6
8
10
12
14
16
StaysInWeekendNights

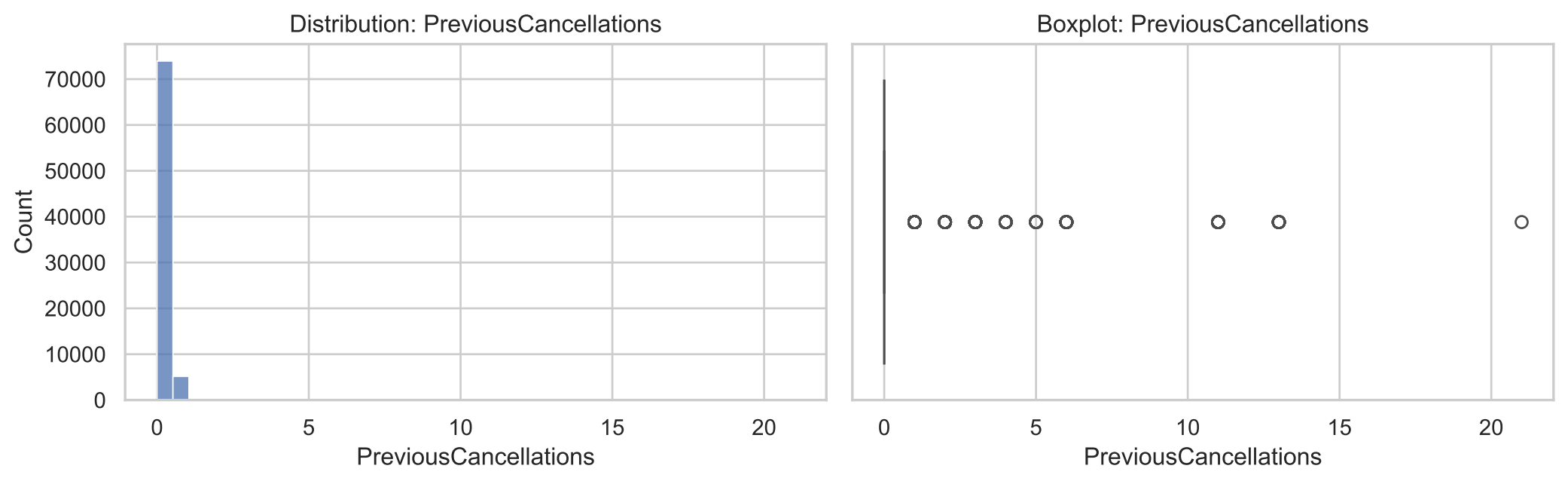

Distribution: StaysInWeekNights
Count
60000
50000
40000
30000
20000
10000
0
0
10
20
30
40
StaysInWeekNights
Boxplot: StaysInWeekNights
0
10
20
30
40
StaysInWeekNights

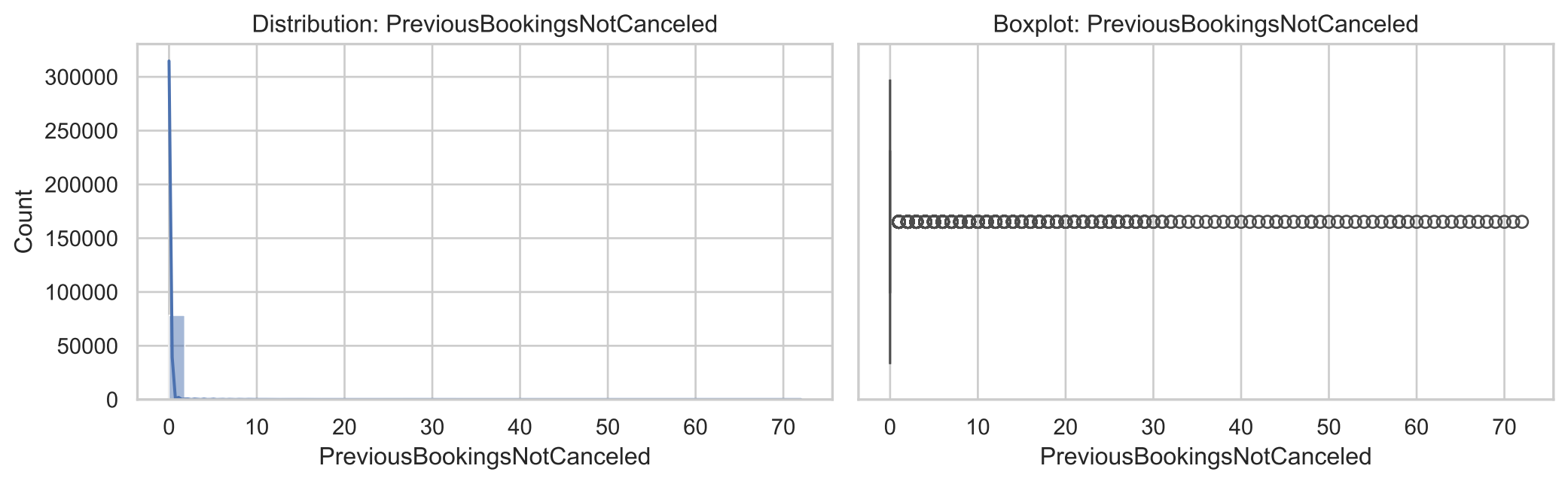

Distribution: Adults
Count
60000
50000
40000
30000
20000
10000
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Adults
Boxplot: Adults
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Adults

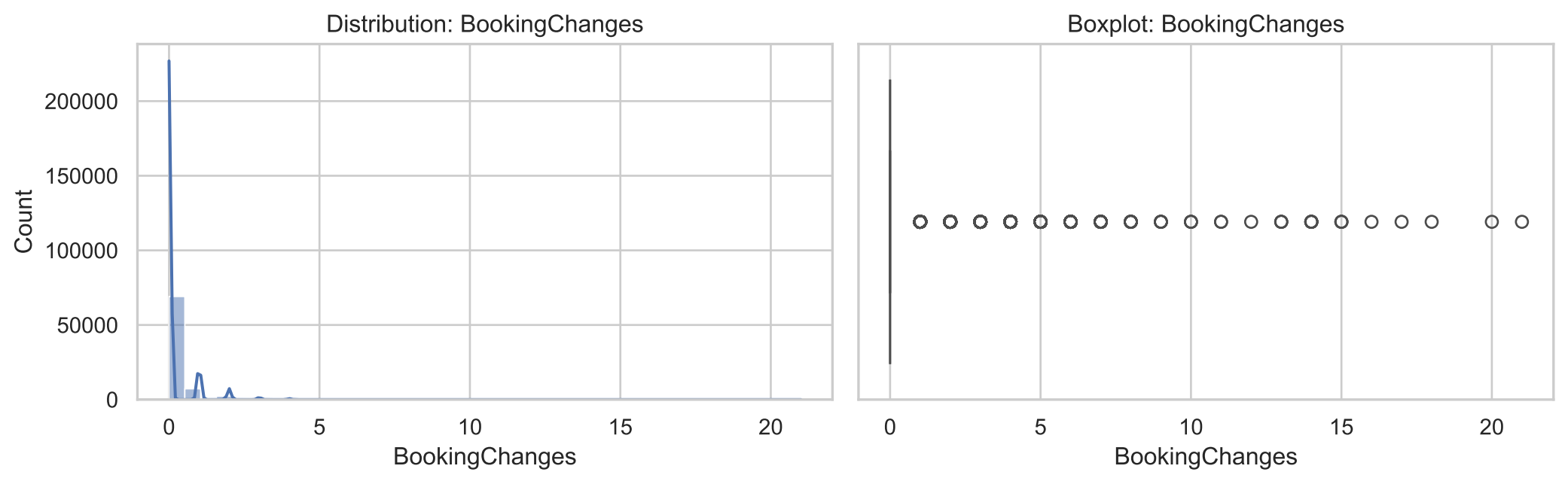

Distribution: Children
Count
70000
60000
50000
40000
30000
20000
10000
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Children
Boxplot: Children
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Children

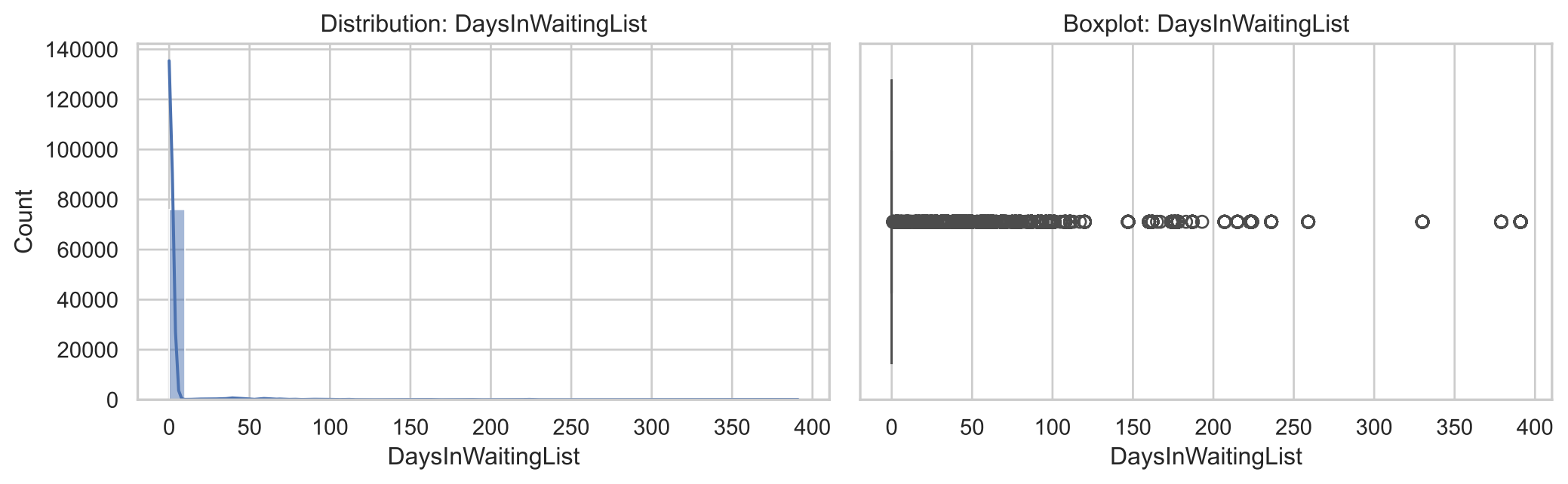

Distribution: Babies
Count
80000
60000
40000
20000
0
0
2
4
6
8
10
Babies
Boxplot: Babies
0
2
4
6
8
10
Babies

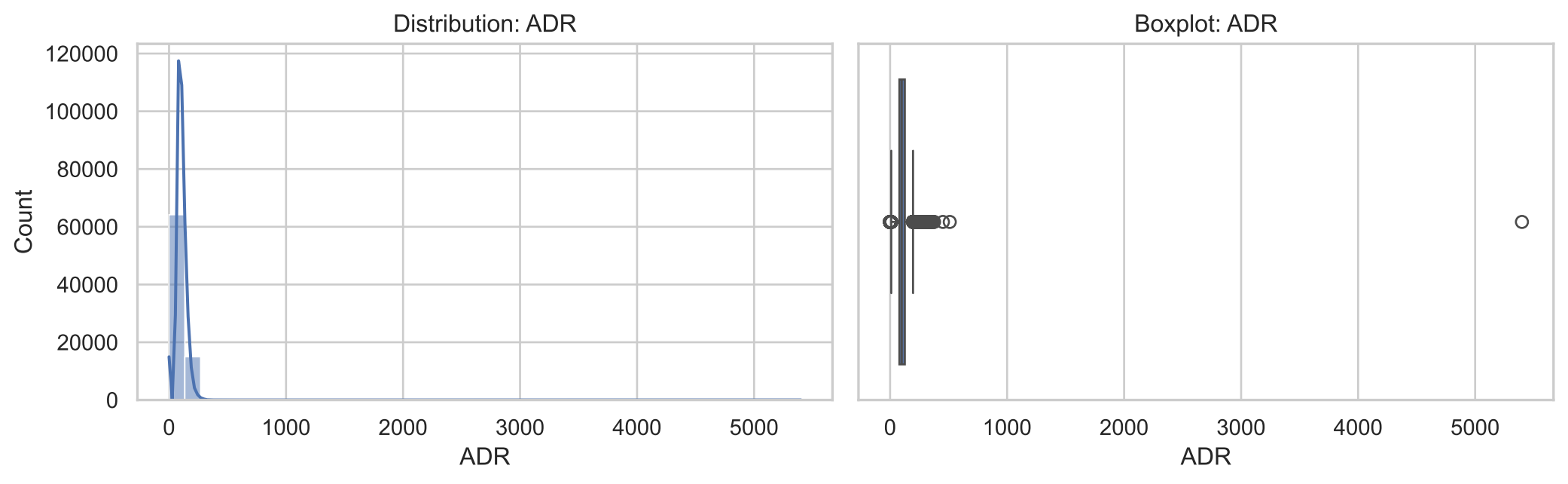

Distribution: IsRepeatedGuest
Count
80000
60000
40000
20000
0
0.0
0.2
0.4
0.6
0.8
1.0
IsRepeatedGuest
Boxplot: IsRepeatedGuest
0.0
0.2
0.4
0.6
0.8
1.0
IsRepeatedGuest

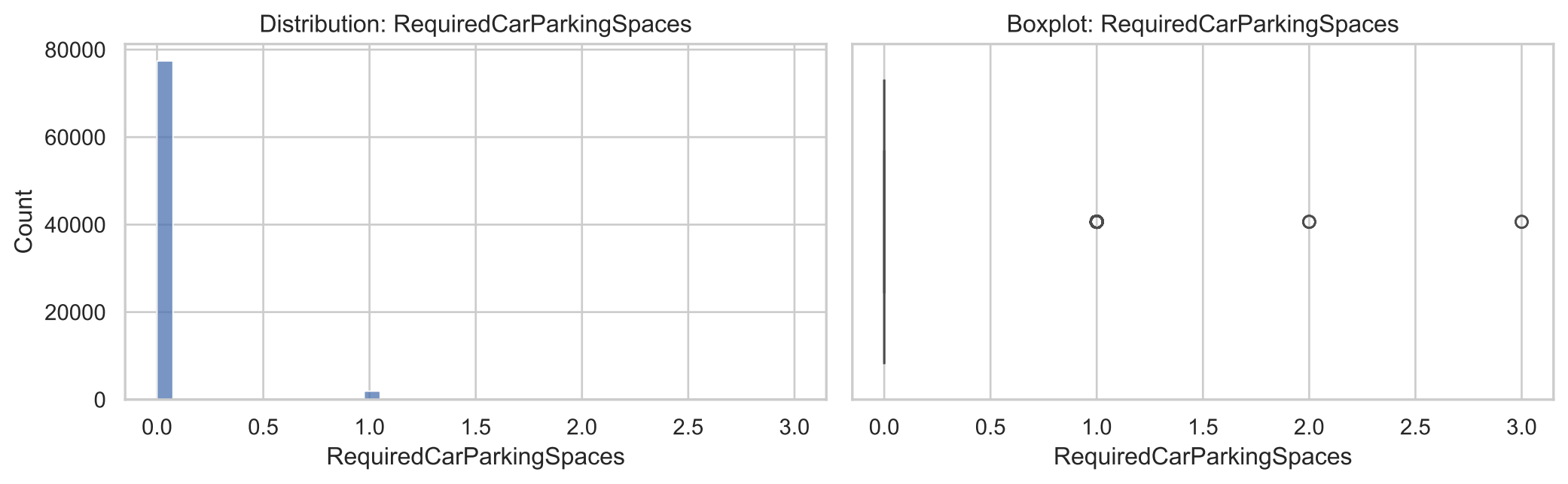

Distribution: PreviousCancellations
Count
70000
60000
50000
40000
30000
20000
10000
0
0
5
10
15
20
PreviousCancellations
Boxplot: PreviousCancellations
0
5
10
15
20
PreviousCancellations

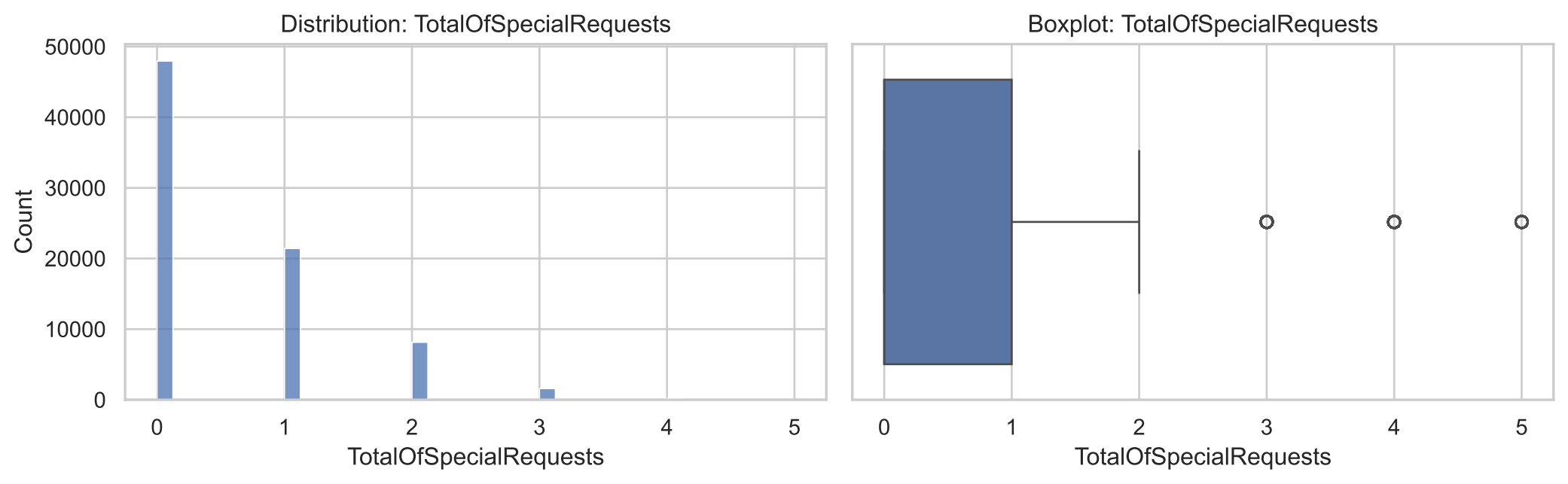

Distribution: PreviousBookingsNotCanceled
Count
300000
250000
200000
150000
100000
50000
0
0
10
20
30
40
50
60
70
PreviousBookingsNotCanceled
Boxplot: PreviousBookingsNotCanceled
0
10
20
30
40
50
60
70
PreviousBookingsNotCanceled


Distribution: BookingChanges
Count
200000
150000
100000
50000
0
0
5
10
15
20
BookingChanges
Boxplot: BookingChanges
0
5
10
15
20
BookingChanges


Distribution: DaysInWaitingList
Count
140000
120000
100000
80000
60000
40000
20000
0
0
50
100
150
200
250
300
350
400
DaysInWaitingList
Boxplot: DaysInWaitingList
0
50
100
150
200
250
300
350
400
DaysInWaitingList


Distribution: ADR
Count
120000
100000
80000
60000
40000
20000
0
0
1000
2000
3000
4000
5000
ADR
Boxplot: ADR
0
1000
2000
3000
4000
5000
ADR


Distribution: RequiredCarParkingSpaces
Count
80000
60000
40000
20000
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
RequiredCarParkingSpaces
Boxplot: RequiredCarParkingSpaces
0.0
0.5
1.0
1.5
2.0
2.5
3.0
RequiredCarParkingSpaces


Distribution: TotalOfSpecialRequests
Count
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
TotalOfSpecialRequests
Boxplot: TotalOfSpecialRequests
0
1
2
3
4
5
TotalOfSpecialRequests

IQR outlier summary (outliers need investigation, not automatic deletion):

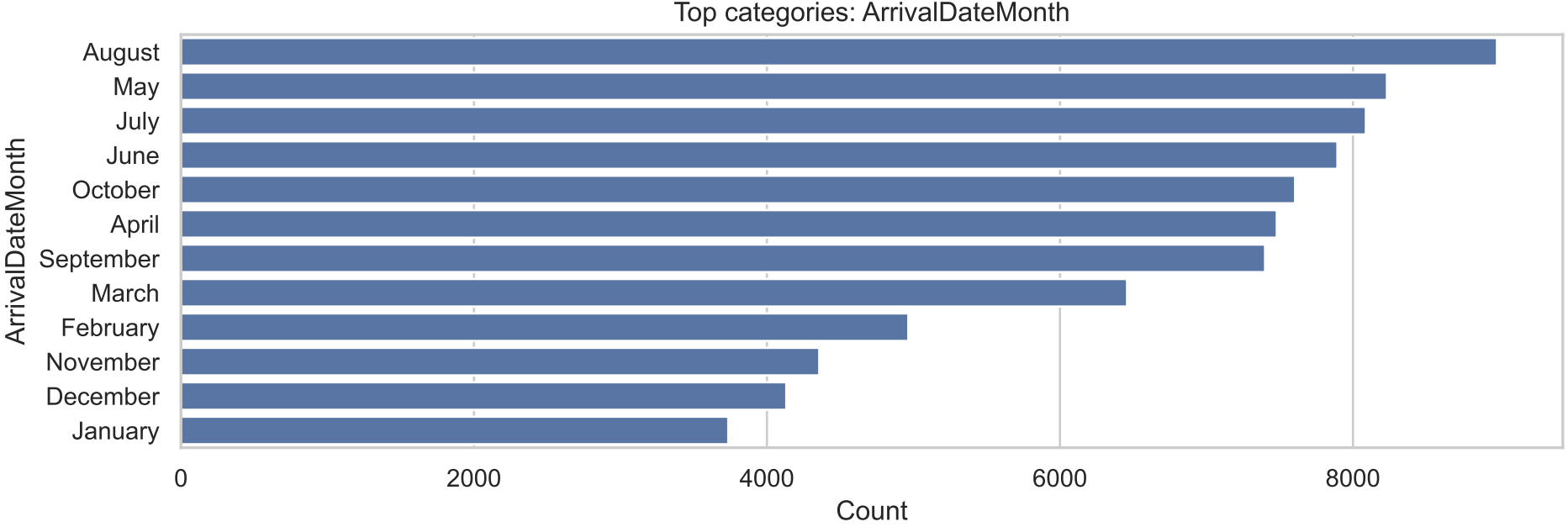
	feature	lower_bound	upper_bound	count	percent
6	Adults	2.0	2.0	21075	26.57
11	BookingChanges	0.0	0.0	10268	12.94
9	PreviousCancellations	0.0	0.0	5389	6.79
7	Children	0.0	0.0	5106	6.44
12	DaysInWaitingList	0.0	0.0	3443	4.34
13	ADR	9.0	196.2	3387	4.27
0	LeadTime	-187.0	373.0	2576	3.25
14	RequiredCarParkingSpaces	0.0	0.0	1926	2.43
15	TotalOfSpecialRequests	-1.5	2.5	1811	2.28
10	PreviousBookingsNotCanceled	0.0	0.0	1588	2.00
5	StaysInWeekNights	-2.0	6.0	719	0.91
8	Babies	0.0	0.0	369	0.47
4	StaysInWeekendNights	-3.0	5.0	82	0.10
2	ArrivalDateWeekNumber	-14.5	69.5	0	0.00
3	ArrivalDateDayOfMonth	-14.5	45.5	0	0.00
1	ArrivalDateYear	2014.5	2018.5	0	0.00

=== 5. CATEGORICAL DISTRIBUTIONS ===

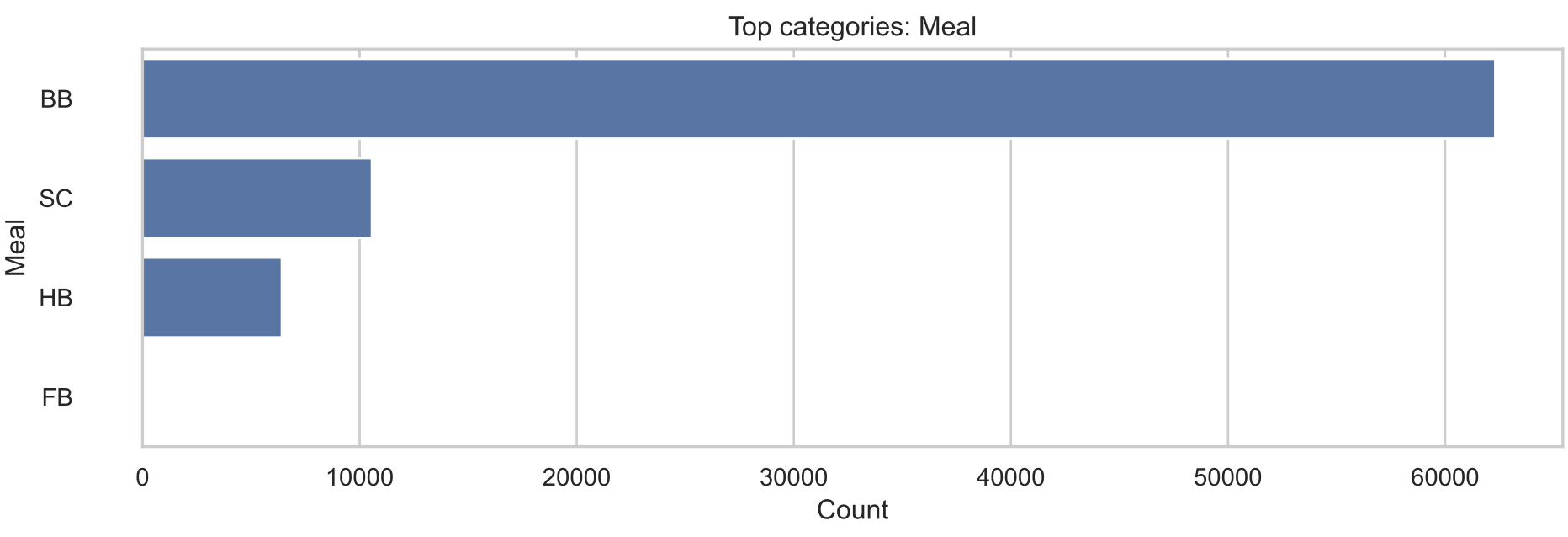
ArrivalDateMonth:

ArrivalDateMonth	
August	8983
May	8232
July	8088
June	7894
October	7605
April	7480
September	7400
March	6458
February	4965
November	4357
December	4132
January	3736

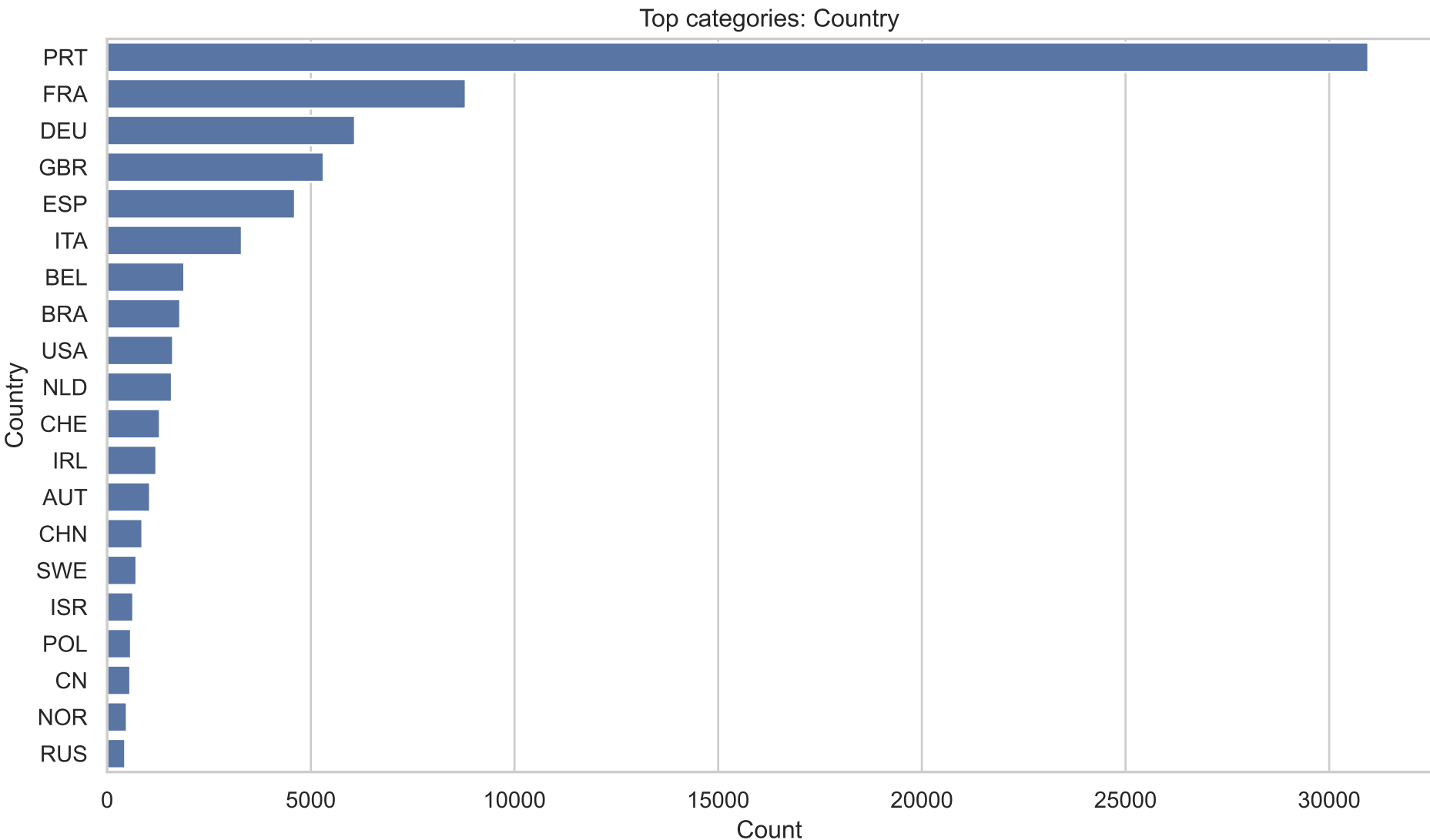
Name: count, dtype: Int64


Top categories: ArrivalDateMonth
August
May
July
June
October
April
September
March
February
November
December
January
ArrivalDateMonth
0
2000
4000
6000
8000
Count

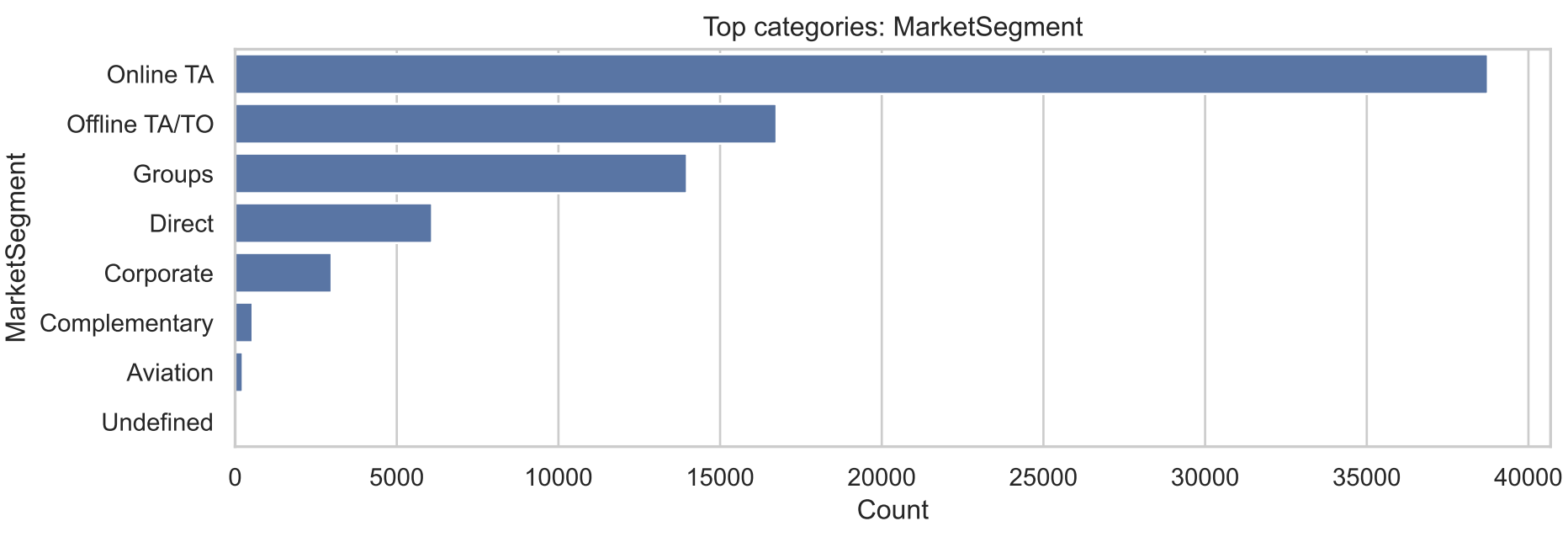
```
Meal:
Meal
BB           62305
SC           10564
HB            6417
FB              44
Name: count, dtype: Int64
```


Top categories: Meal
Meal
BB
SC
HB
FB
0
10000
20000
30000
40000
50000
60000
Count

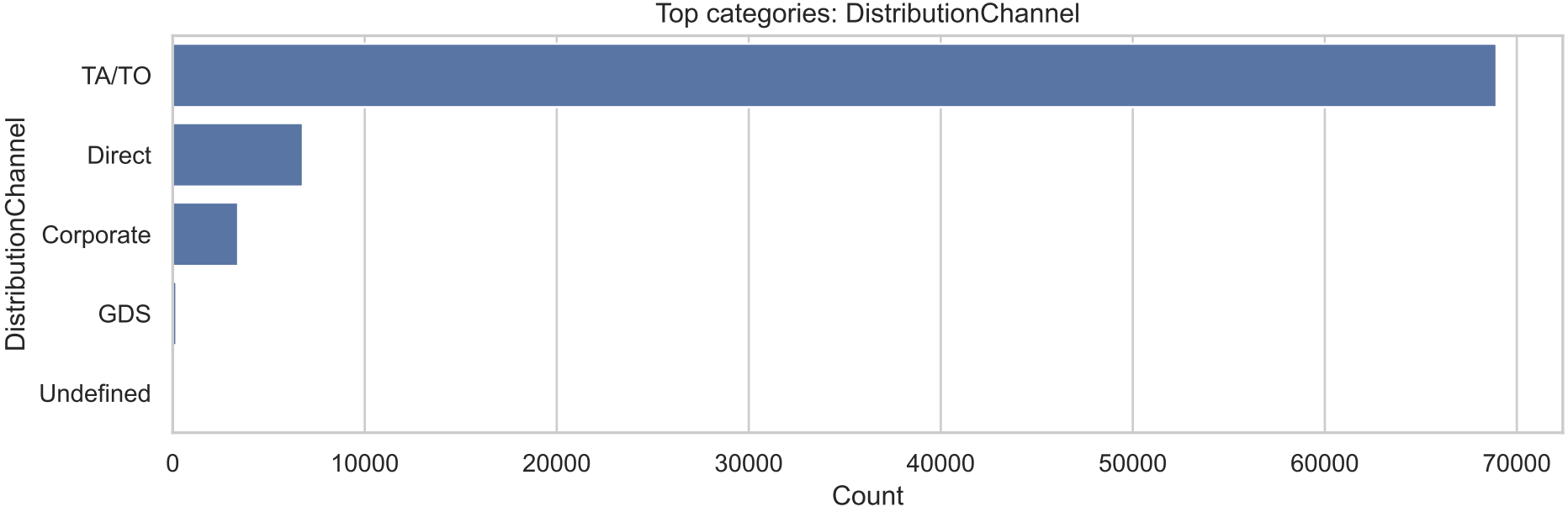
```
Country:
Country
PRT    30960
FRA     8804
DEU     6084
GBR     5315
ESP     4611
ITA     3307
BEL     1894
BRA     1794
USA     1618
NLD     1590
CHE     1295
IRL     1209
AUT     1053
CHN      865
SWE      720
ISR      641
POL      586
CN       569
NOR      484
RUS      443
Name: count, dtype: Int64
```


Top categories: Country
PRT
FRA
DEU
GBR
ESP
ITA
BEL
BRA
USA
NLD
CHE
IRL
AUT
CHN
SWE
ISR
POL
CN
NOR
RUS
Country
0
5000
10000
15000
20000
25000
30000
Count

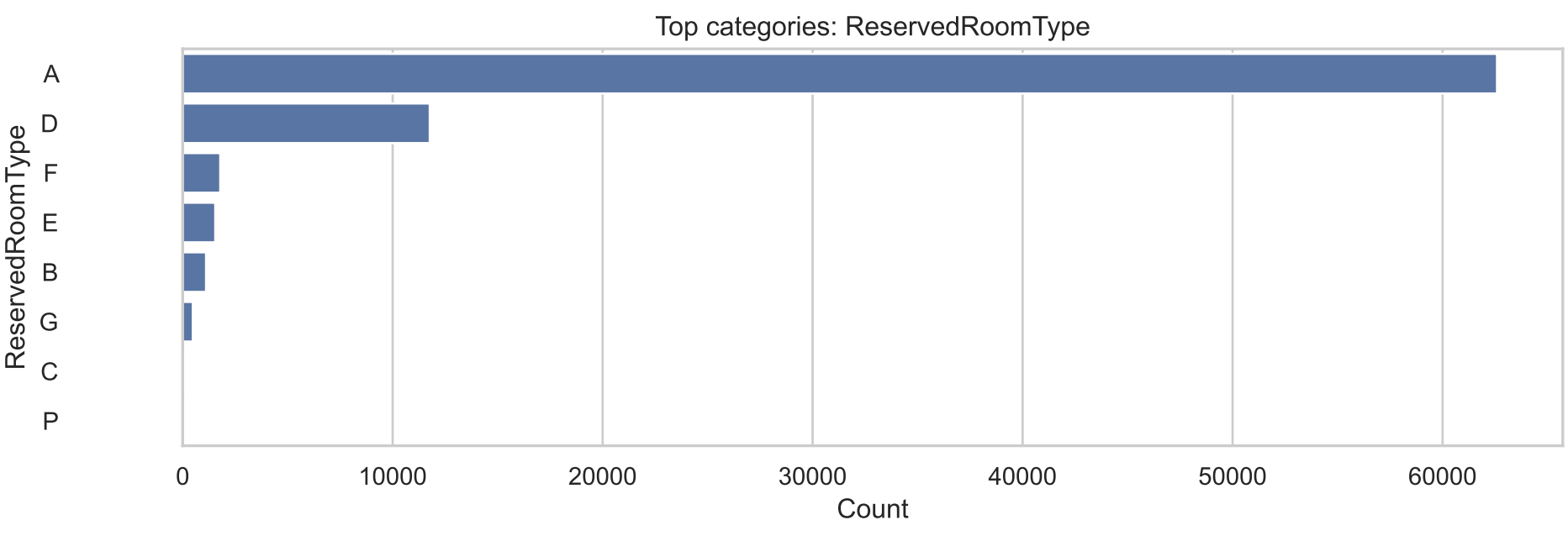
```
MarketSegment:
MarketSegment
Online TA        38748
Offline TA/TO    16747
Groups           13975
Direct            6093
Corporate         2986
Complementary      542
Aviation           237
Undefined            2
Name: count, dtype: Int64
```


Top categories: MarketSegment
MarketSegment
Online TA
Offline TA/TO
Groups
Direct
Corporate
Complementary
Aviation
Undefined
0
5000
10000
15000
20000
25000
30000
35000
40000
Count

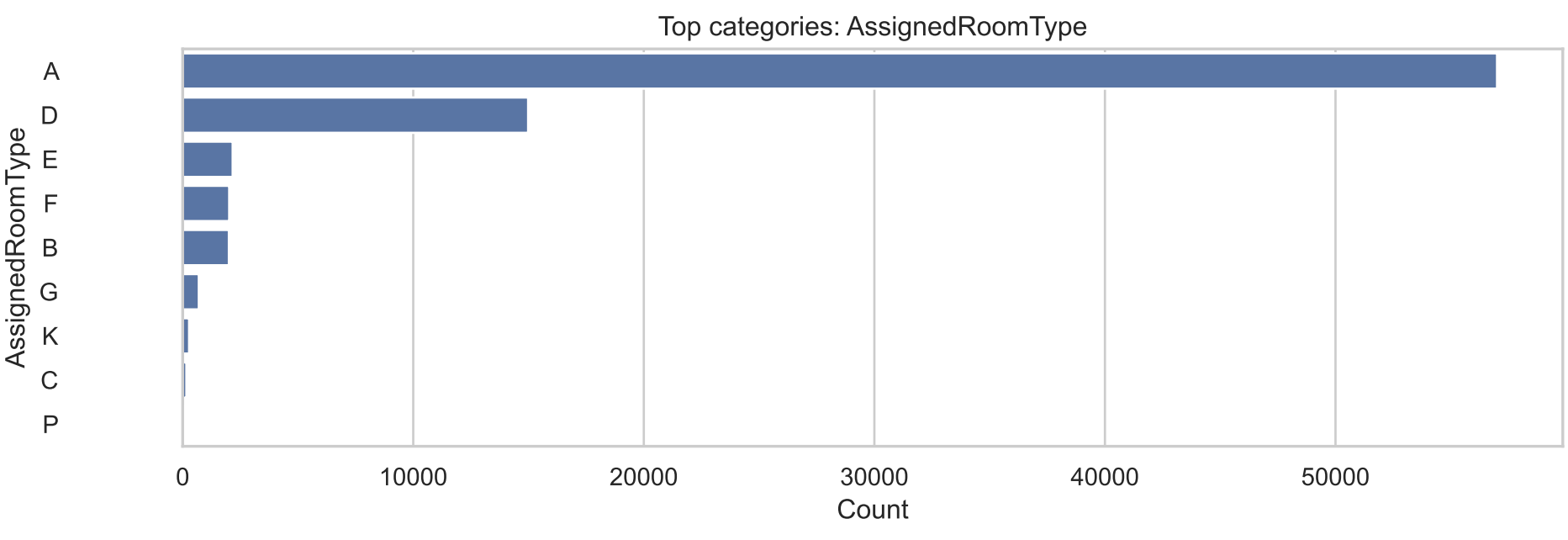
```
DistributionChannel:
DistributionChannel
TA/TO        68945
Direct        6780
Corporate     3408
GDS            193
Undefined        4
Name: count, dtype: Int64
```


Top categories: DistributionChannel
DistributionChannel
TA/TO
Direct
Corporate
GDS
Undefined
0
10000
20000
30000
40000
50000
60000
70000
Count

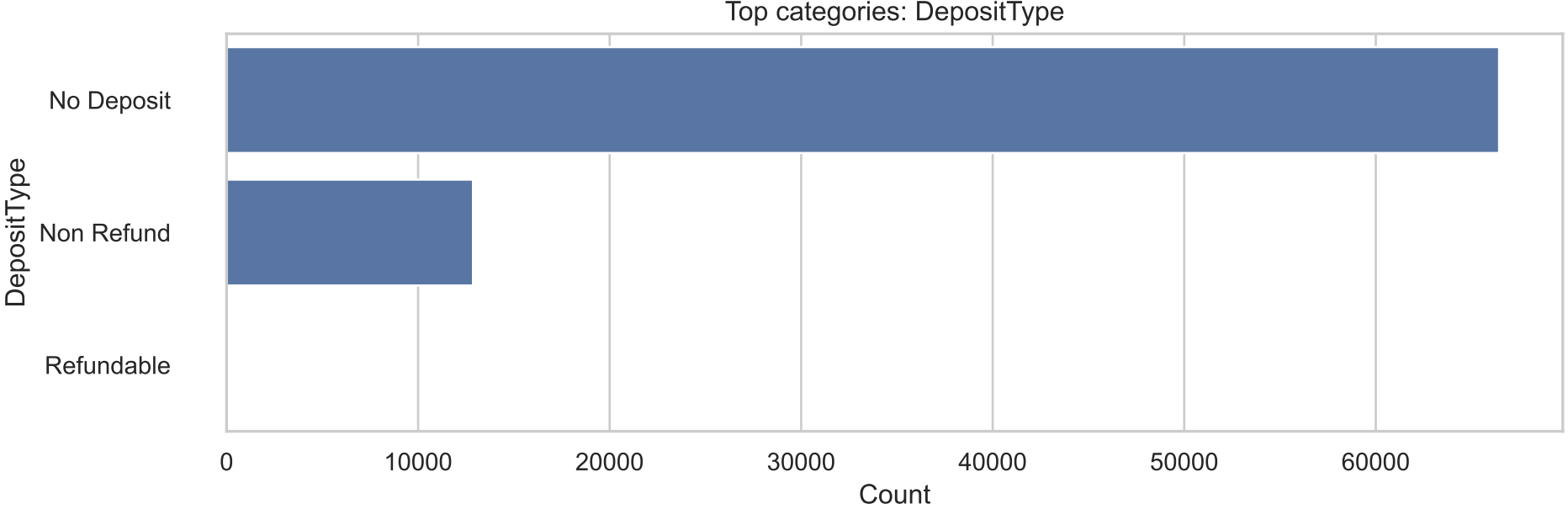
```
ReservedRoomType:
ReservedRoomType
A                   62595
D                   11768
F                    1791
E                    1553
B                    1115
G                     484
C                      14
P                      10
Name: count, dtype: Int64
```


Top categories: ReservedRoomType
ReservedRoomType
A
D
F
E
B
G
C
P
0
10000
20000
30000
40000
50000
60000
Count

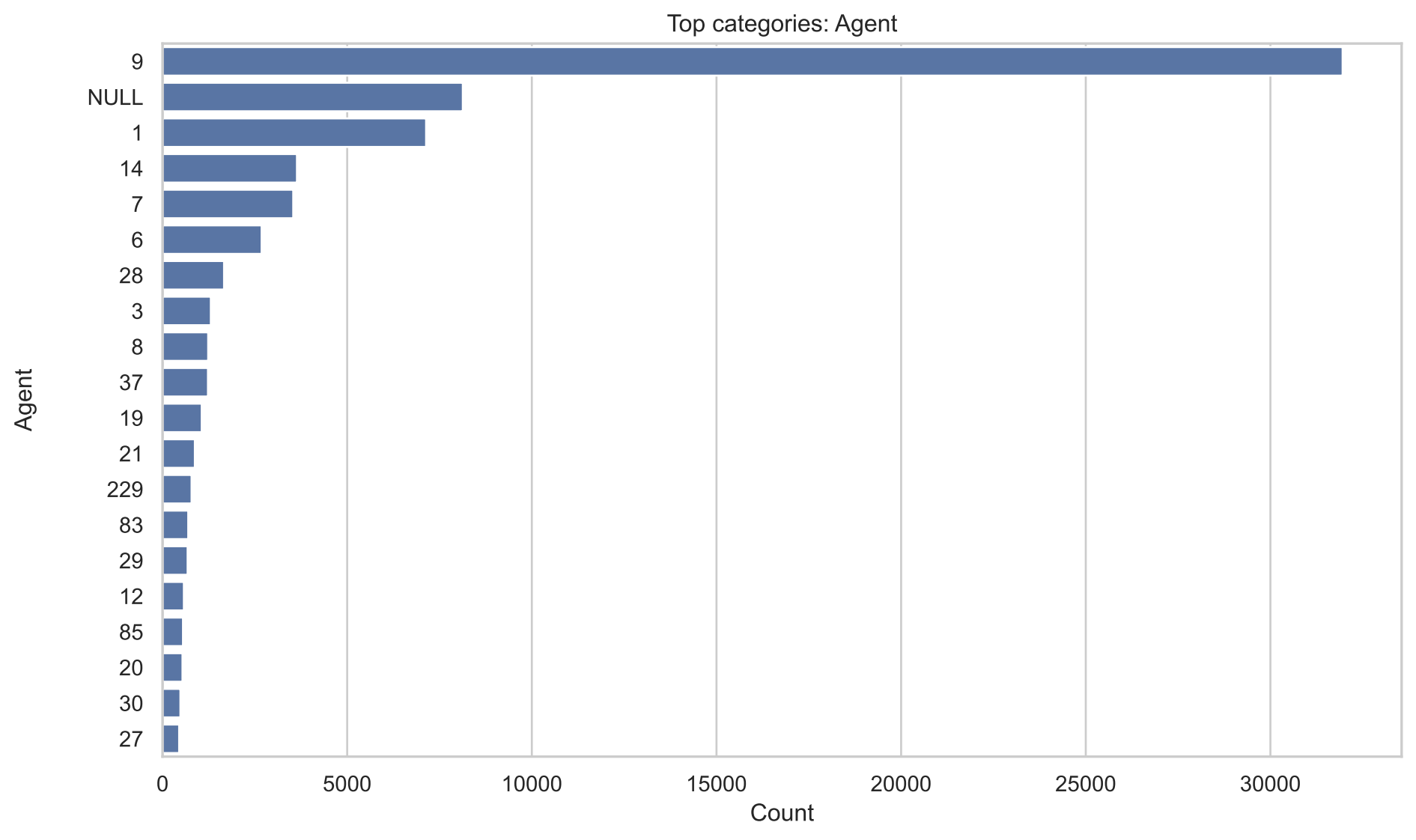
```
AssignedRoomType:
AssignedRoomType
A                   57007
D                   14983
E                    2168
F                    2018
B                    2004
G                     700
K                     279
C                     161
P                      10
Name: count, dtype: Int64
```


Top categories: AssignedRoomType
AssignedRoomType
A
D
E
F
B
G
K
C
P
0
10000
20000
30000
40000
50000
Count

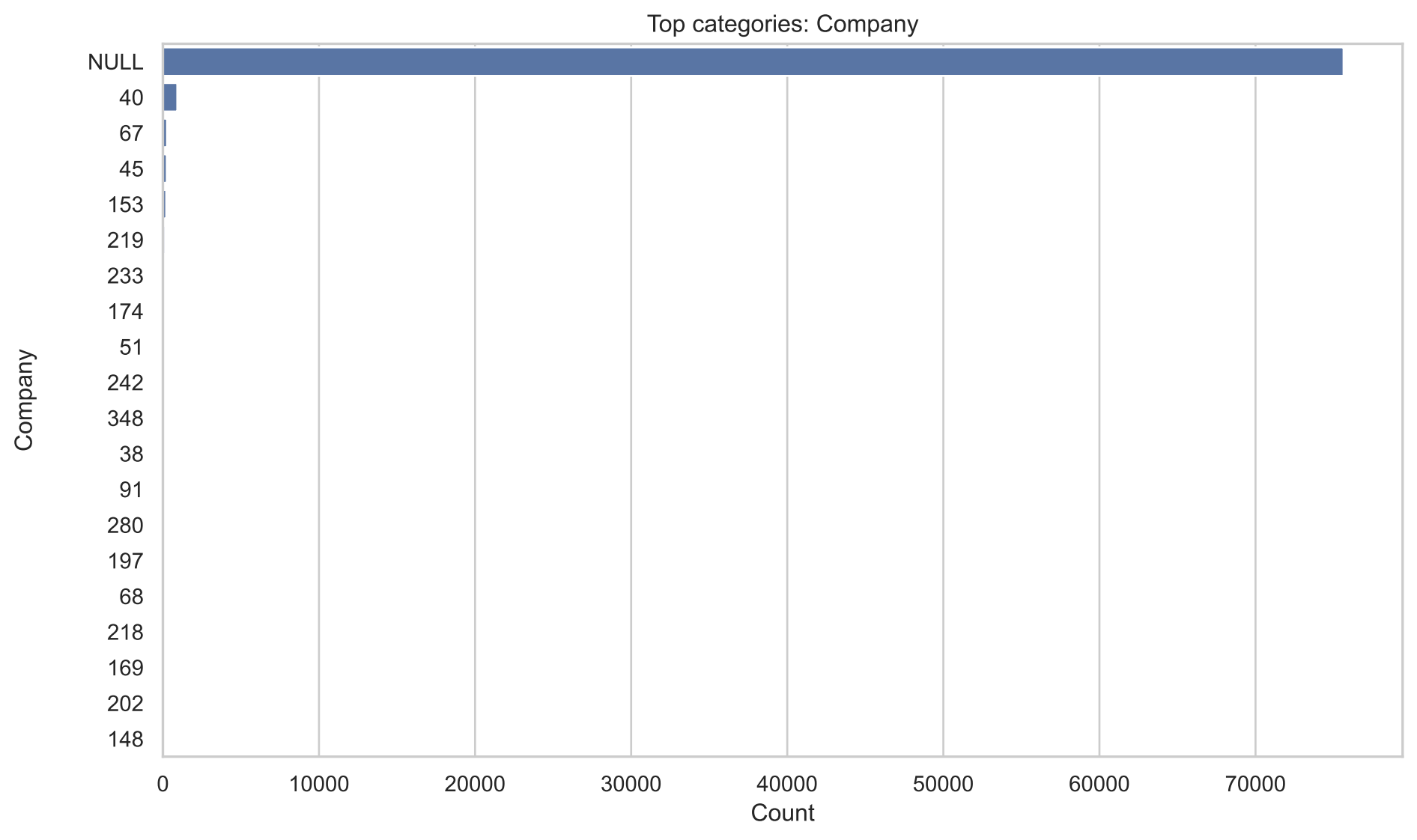
```
DepositType:
DepositType
No Deposit         66442
Non Refund         12868
Refundable            20
Name: count, dtype: Int64
```


Top categories: DepositType
No Deposit
Non Refund
Refundable
DepositType
0
10000
20000
30000
40000
50000
60000
Count

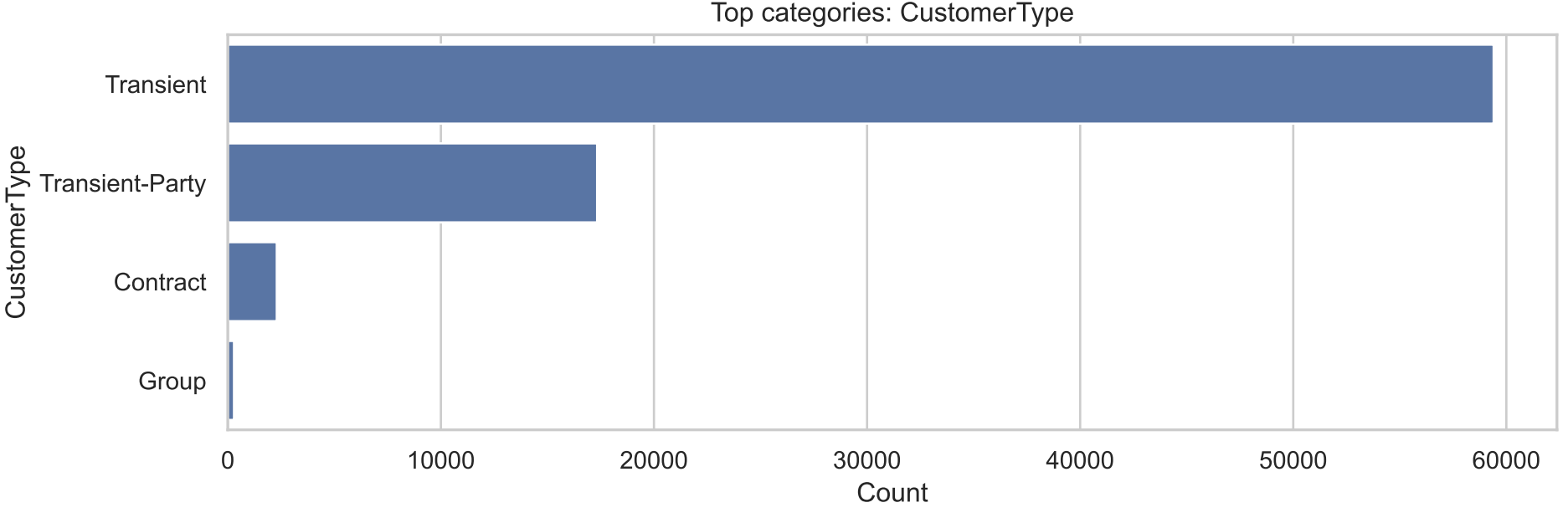
```
Agent:
Agent
   9    31955
NULL     8131
   1     7137
  14     3640
   7     3539
   6     2683
  28     1666
   3     1308
   8     1236
  37     1230
  19     1061
  21      875
 229      786
  83      696
  29      679
  12      578
  85      554
  20      540
  30      484
  27      450
Name: count, dtype: Int64
```


Top categories: Agent
Agent
9
NULL
1
14
7
6
28
3
8
37
19
21
229
83
29
12
85
20
30
27
0
5000
10000
15000
20000
25000
30000
Count

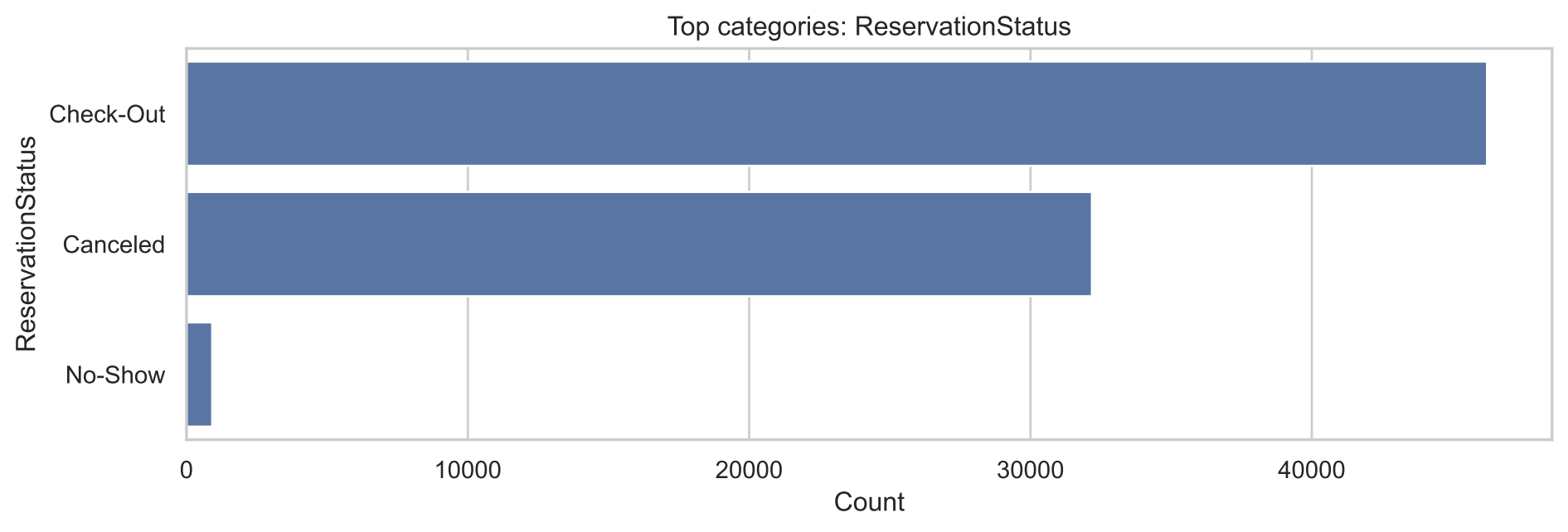
```
Company:
Company
NULL    75641
  40      924
  67      267
  45      250
 153      215
 219      141
 233      114
 174      113
  51       86
 242       61
 348       59
  38       51
  91       48
 280       48
 197       47
  68       46
 218       42
 169       39
 202       38
 148       33
Name: count, dtype: Int64
```


Top categories: Company
NULL
40
67
45
153
219
233
174
51
242
348
38
91
280
197
68
218
169
202
148
Company
0
10000
20000
30000
40000
50000
60000
70000
Count

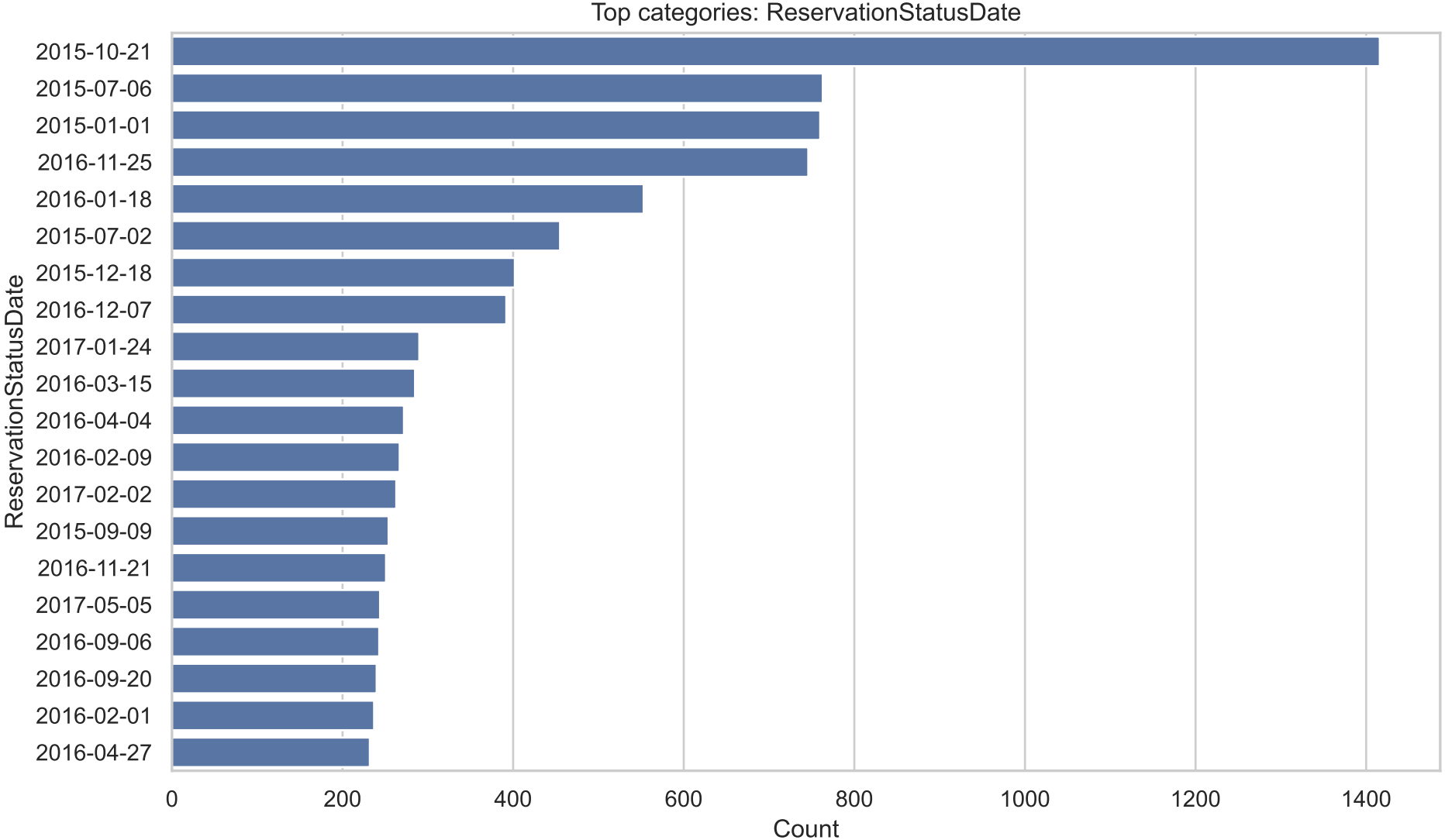
```
CustomerType:
CustomerType
Transient          59404
Transient-Party    17333
Contract            2300
Group                293
Name: count, dtype: Int64
```


Top categories: CustomerType
CustomerType
Transient
Transient-Party
Contract
Group
0
10000
20000
30000
40000
50000
60000
Count

```
ReservationStatus:
ReservationStatus
Check-Out    46228
Canceled     32186
No-Show        916
Name: count, dtype: Int64
```


Top categories: ReservationStatus
ReservationStatus
Check-Out
Canceled
No-Show
0
10000
20000
30000
40000
Count

```
ReservationStatusDate:
ReservationStatusDate
2015-10-21    1416
2015-07-06     763
2015-01-01     760
2016-11-25     746
2016-01-18     553
2015-07-02     455
2015-12-18     402
2016-12-07     392
2017-01-24     290
2016-03-15     285
2016-04-04     272
2016-02-09     267
2017-02-02     263
2015-09-09     254
2016-11-21     251
2017-05-05     244
2016-09-06     243
2016-09-20     240
2016-02-01     237
2016-04-27     232
Name: count, dtype: Int64
```


Top categories: ReservationStatusDate
ReservationStatusDate
2015-10-21
2015-07-06
2015-01-01
2016-11-25
2016-01-18
2015-07-02
2015-12-18
2016-12-07
2017-01-24
2016-03-15
2016-04-04
2016-02-09
2017-02-02
2015-09-09
2016-11-21
2017-05-05
2016-09-06
2016-09-20
2016-02-01
2016-04-27
0
200
400
600
800
1000
1200
1400
Count

=== 6. NUMERIC CORRELATION ===

Pearson correlation:

	IsCanceled	LeadTime	ArrivalDateYear	ArrivalDateWeekNumber	ArrivalDateDayOfMonth	StaysInWeekendNights	StaysInWeekNights	Adults	Children	\
IsCanceled	1.000	0.309	-0.003	0.002	-0.004	-0.007	0.049	0.053	-0.027	
LeadTime	0.309	1.000	0.028	0.130	0.015	-0.024	0.058	0.116	-0.054	
ArrivalDateYear	-0.003	0.028	1.000	-0.543	-0.003	0.035	0.066	0.068	0.056	
ArrivalDateWeekNumber	0.002	0.130	-0.543	1.000	0.062	-0.001	-0.009	0.007	-0.001	
ArrivalDateDayOfMonth	-0.004	0.015	-0.003	0.062	1.000	0.000	-0.012	-0.001	0.013	
StaysInWeekendNights	-0.007	-0.024	0.035	-0.001	0.000	1.000	0.220	0.082	0.038	
StaysInWeekNights	0.049	0.058	0.066	-0.009	-0.012	0.220	1.000	0.088	0.037	
Adults	0.053	0.116	0.068	0.007	-0.001	0.082	0.088	1.000	-0.006	
Children	-0.027	-0.054	0.056	-0.001	0.013	0.038	0.037	-0.006	1.000	
Babies	-0.030	-0.030	-0.008	0.005	0.000	0.006	0.005	0.012	0.024	
IsRepeatedGuest	-0.066	-0.108	-0.029	0.003	-0.012	-0.080	-0.094	-0.167	-0.029	
PreviousCancellations	0.167	0.123	-0.195	0.037	-0.011	-0.039	-0.051	-0.038	-0.044	
PreviousBookingsNotCanceled	-0.053	-0.065	0.021	-0.012	-0.002	-0.027	-0.041	-0.109	-0.017	
BookingChanges	-0.149	-0.031	0.032	0.002	0.014	0.046	0.080	-0.088	0.047	
DaysInWaitingList	0.061	0.187	-0.069	0.018	0.023	-0.056	0.016	-0.008	-0.036	
ADR	-0.012	-0.137	0.249	0.017	0.024	0.037	0.057	0.263	0.312	
RequiredCarParkingSpaces	-0.133	-0.078	-0.001	-0.004	0.004	-0.024	-0.038	0.011	0.055	
TotalOfSpecialRequests	-0.294	-0.132	0.135	0.016	-0.002	0.069	0.064	0.156	0.113	

	Babies	IsRepeatedGuest	PreviousCancellations	PreviousBookingsNotCanceled	BookingChanges	DaysInWaitingList	ADR	RequiredCarParkingSpaces	\
IsCanceled	-0.030	-0.066	0.167	-0.053	-0.149	0.061	-0.012	-0.133	
LeadTime	-0.030	-0.108	0.123	-0.065	-0.031	0.187	-0.137	-0.078	
ArrivalDateYear	-0.008	-0.029	-0.195	0.021	0.032	-0.069	0.249	-0.001	
ArrivalDateWeekNumber	0.005	0.003	0.037	-0.012	0.002	0.018	0.017	-0.004	
ArrivalDateDayOfMonth	0.000	-0.012	-0.011	-0.002	0.014	0.023	0.024	0.004	
StaysInWeekendNights	0.006	-0.080	-0.039	-0.027	0.046	-0.056	0.037	-0.024	
StaysInWeekNights	0.005	-0.094	-0.051	-0.041	0.080	0.016	0.057	-0.038	
Adults	0.012	-0.167	-0.038	-0.109	-0.088	-0.008	0.263	0.011	
Children	0.024	-0.029	-0.044	-0.017	0.047	-0.036	0.312	0.055	
Babies	1.000	-0.002	-0.009	-0.003	0.066	-0.009	0.014	0.020	
IsRepeatedGuest	-0.002	1.000	0.268	0.452	0.014	-0.024	-0.153	0.093	
PreviousCancellations	-0.009	0.268	1.000	0.392	-0.040	0.021	-0.129	-0.003	
PreviousBookingsNotCanceled	-0.003	0.452	0.392	1.000	0.017	-0.009	-0.075	0.068	
BookingChanges	0.066	0.014	-0.040	0.017	1.000	-0.012	0.024	0.042	
DaysInWaitingList	-0.009	-0.024	0.021	-0.009	-0.012	1.000	-0.067	-0.024	
ADR	0.014	-0.153	-0.129	-0.075	0.024	-0.067	1.000	0.060	
RequiredCarParkingSpaces	0.020	0.093	-0.003	0.068	0.042	-0.024	0.060	1.000	
TotalOfSpecialRequests	0.072	0.018	-0.077	0.046	0.063	-0.093	0.175	0.088	

	TotalOfSpecialRequests
IsCanceled	-0.294
LeadTime	-0.132
ArrivalDateYear	0.135
ArrivalDateWeekNumber	0.016
ArrivalDateDayOfMonth	-0.002
StaysInWeekendNights	0.069
StaysInWeekNights	0.064
Adults	0.156
Children	0.113
Babies	0.072
IsRepeatedGuest	0.018
PreviousCancellations	-0.077
PreviousBookingsNotCanceled	0.046
BookingChanges	0.063
DaysInWaitingList	-0.093
ADR	0.175
RequiredCarParkingSpaces	0.088
TotalOfSpecialRequests	1.000

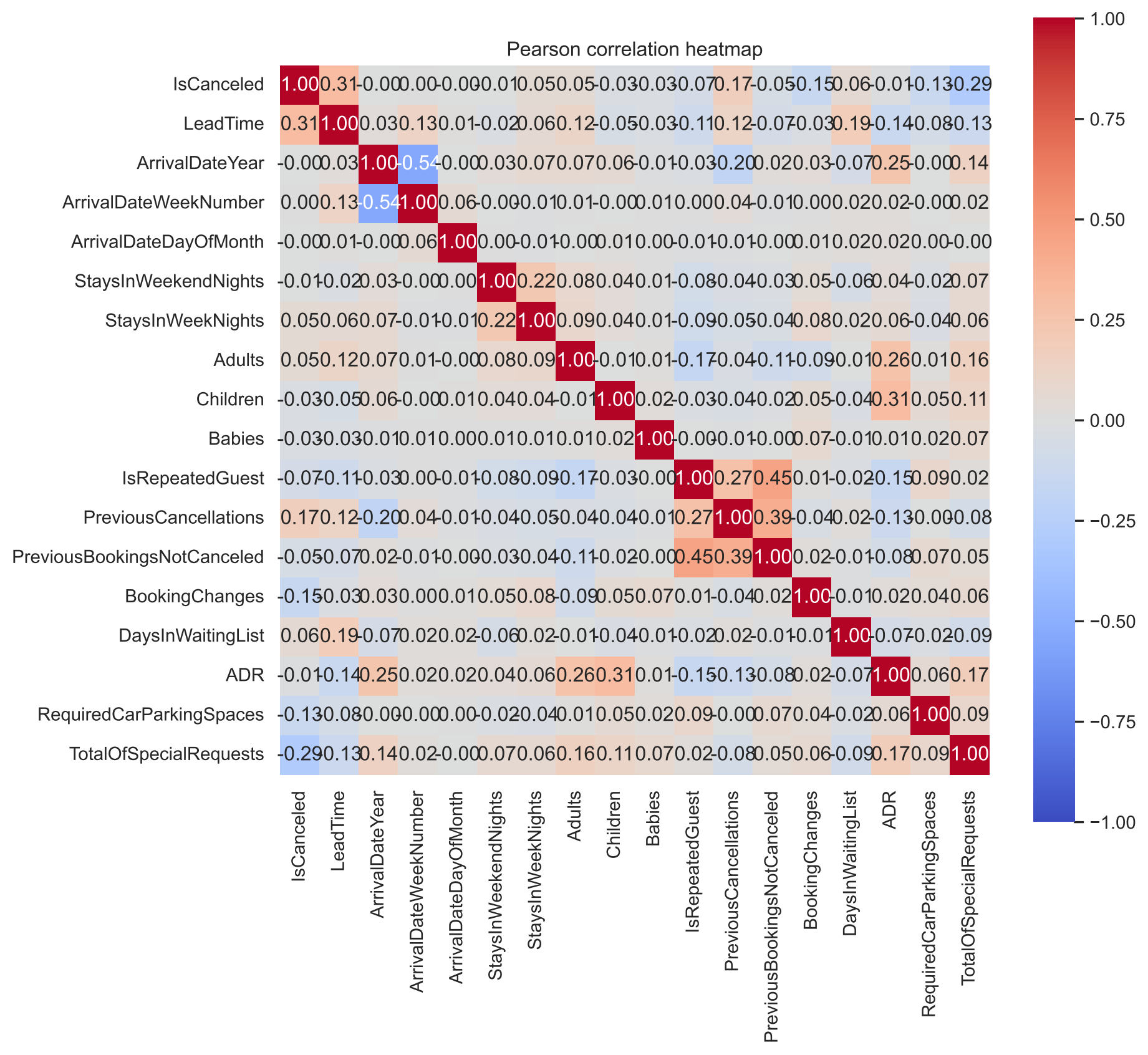
Spearman correlation:

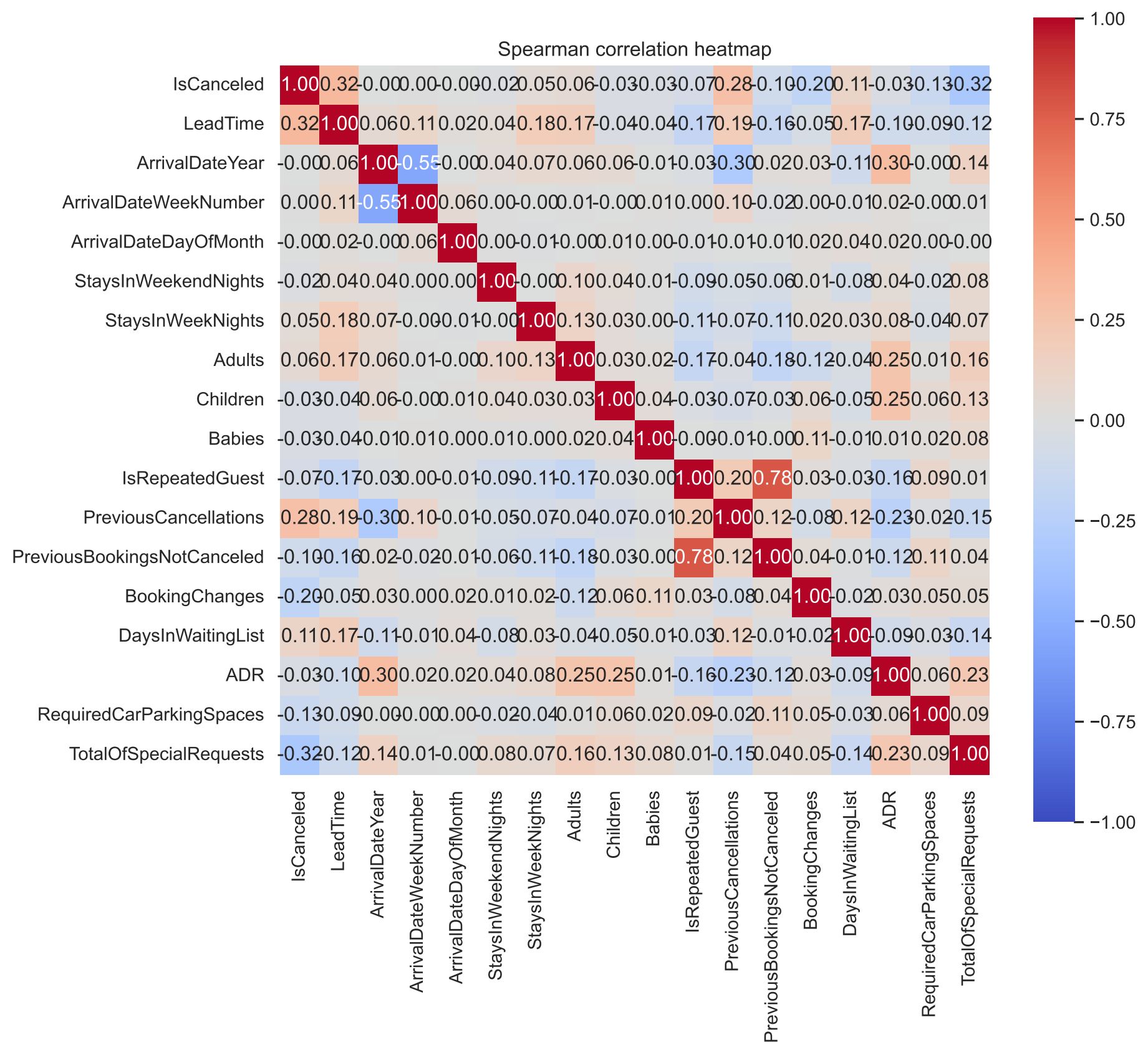
	IsCanceled	LeadTime	ArrivalDateYear	ArrivalDateWeekNumber	ArrivalDateDayOfMonth	StaysInWeekendNights	StaysInWeekNights	Adults	Children	\
IsCanceled	1.000	0.325	-0.001	0.001	-0.004	-0.018	0.046	0.056	-0.033	
LeadTime	0.325	1.000	0.061	0.106	0.019	0.044	0.183	0.175	-0.036	
ArrivalDateYear	-0.001	0.061	1.000	-0.548	-0.004	0.037	0.071	0.065	0.059	
ArrivalDateWeekNumber	0.001	0.106	-0.548	1.000	0.055	0.005	-0.005	0.006	-0.002	

ArrivalDateDayOfMonth	-0.004	0.019	-0.004	0.055	1.000	0.002	-0.011	-0.002	0.014
StaysInWeekendNights	-0.018	0.044	0.037	0.005	0.002	1.000	-0.003	0.101	0.040
StaysInWeekNights	0.046	0.183	0.071	-0.005	-0.011	-0.003	1.000	0.129	0.032
Adults	0.056	0.175	0.065	0.006	-0.002	0.101	0.129	1.000	0.033
Children	-0.033	-0.036	0.059	-0.002	0.014	0.040	0.032	0.033	1.000
Babies	-0.034	-0.038	-0.008	0.009	0.001	0.006	0.004	0.019	0.036
IsRepeatedGuest	-0.066	-0.174	-0.027	0.003	-0.012	-0.086	-0.113	-0.170	-0.031
PreviousCancellations	0.280	0.186	-0.302	0.097	-0.006	-0.053	-0.067	-0.038	-0.065
PreviousBookingsNotCanceled	-0.099	-0.164	0.021	-0.020	-0.007	-0.059	-0.111	-0.181	-0.027
BookingChanges	-0.196	-0.047	0.026	0.004	0.021	0.013	0.017	-0.119	0.057
DaysInWaitingList	0.112	0.174	-0.107	-0.015	0.036	-0.077	0.031	-0.043	-0.054
ADR	-0.031	-0.100	0.298	0.020	0.021	0.043	0.077	0.249	0.250
RequiredCarParkingSpaces	-0.133	-0.091	-0.005	-0.004	0.004	-0.024	-0.044	0.010	0.056
TotalOfSpecialRequests	-0.324	-0.120	0.141	0.007	-0.002	0.077	0.069	0.165	0.126

	Babies	IsRepeatedGuest	PreviousCancellations	PreviousBookingsNotCanceled	BookingChanges	DaysInWaitingList	ADR	RequiredCarParkingSpaces	\
IsCanceled	-0.034	-0.066	0.280	-0.099	-0.196	0.112	-0.031	-0.133	
LeadTime	-0.038	-0.174	0.186	-0.164	-0.047	0.174	-0.100	-0.091	
ArrivalDateYear	-0.008	-0.027	-0.302	0.021	0.026	-0.107	0.298	-0.005	
ArrivalDateWeekNumber	0.009	0.003	0.097	-0.020	0.004	-0.015	0.020	-0.004	
ArrivalDateDayOfMonth	0.001	-0.012	-0.006	-0.007	0.021	0.036	0.021	0.004	
StaysInWeekendNights	0.006	-0.086	-0.053	-0.059	0.013	-0.077	0.043	-0.024	
StaysInWeekNights	0.004	-0.113	-0.067	-0.111	0.017	0.031	0.077	-0.044	
Adults	0.019	-0.170	-0.038	-0.181	-0.119	-0.043	0.249	0.010	
Children	0.036	-0.031	-0.065	-0.027	0.057	-0.054	0.250	0.056	
Babies	1.000	-0.002	-0.013	-0.001	0.105	-0.014	0.014	0.023	
IsRepeatedGuest	-0.002	1.000	0.195	0.784	0.025	-0.030	-0.157	0.094	
PreviousCancellations	-0.013	0.195	1.000	0.123	-0.076	0.125	-0.231	-0.020	
PreviousBookingsNotCanceled	-0.001	0.784	0.123	1.000	0.039	-0.014	-0.122	0.112	
BookingChanges	0.105	0.025	-0.076	0.039	1.000	-0.019	0.025	0.055	
DaysInWaitingList	-0.014	-0.030	0.125	-0.014	-0.019	1.000	-0.090	-0.033	
ADR	0.014	-0.157	-0.231	-0.122	0.025	-0.090	1.000	0.062	
RequiredCarParkingSpaces	0.023	0.094	-0.020	0.112	0.055	-0.033	0.062	1.000	
TotalOfSpecialRequests	0.077	0.015	-0.151	0.044	0.054	-0.142	0.231	0.092	

	TotalOfSpecialRequests
IsCanceled	-0.324
LeadTime	-0.120
ArrivalDateYear	0.141
ArrivalDateWeekNumber	0.007
ArrivalDateDayOfMonth	-0.002
StaysInWeekendNights	0.077
StaysInWeekNights	0.069
Adults	0.165
Children	0.126
Babies	0.077
IsRepeatedGuest	0.015
PreviousCancellations	-0.151
PreviousBookingsNotCanceled	0.044
BookingChanges	0.054
DaysInWaitingList	-0.142
ADR	0.231
RequiredCarParkingSpaces	0.092
TotalOfSpecialRequests	1.000


Pearson correlation heatmap
IsCanceled 1.00 0.31 -0.00 0.00 -0.00 -0.01 0.05 0.05 -0.03 -0.03 -0.07 0.17 -0.05 -0.15 0.06 -0.01 -0.13 0.29
LeadTime 0.31 1.00 0.03 0.13 0.01 -0.02 0.06 0.12 -0.05 -0.03 -0.11 0.12 -0.07 -0.03 0.19 -0.14 0.08 0.13
ArrivalDateYear -0.00 0.03 1.00 -0.54 0.00 0.03 0.07 0.07 0.06 -0.01 -0.03 -0.20 0.02 0.03 -0.07 0.25 -0.00 0.14
ArrivalDateWeekNumber 0.00 0.13 -0.54 1.00 0.06 -0.00 -0.01 0.01 -0.00 0.01 0.00 0.04 -0.01 0.00 0.02 0.02 -0.00 0.02
ArrivalDateDayOfMonth -0.00 0.01 -0.00 0.06 1.00 0.00 -0.01 -0.00 0.01 0.00 -0.01 -0.01 -0.00 0.01 0.02 0.02 0.00 -0.00
StaysInWeekendNights -0.01 -0.02 0.03 -0.00 0.00 1.00 0.22 0.08 0.04 0.01 -0.08 -0.04 0.03 0.05 -0.06 0.04 -0.02 0.07
StaysInWeekNights 0.05 0.06 0.07 -0.01 -0.01 0.22 1.00 0.09 0.04 0.01 -0.09 -0.05 -0.04 0.08 0.02 0.06 -0.04 0.06
Adults 0.05 0.12 0.07 0.01 -0.00 0.08 0.09 1.00 -0.01 0.01 -0.17 -0.04 -0.11 -0.09 -0.01 0.26 0.01 0.16
Children -0.03 -0.05 0.06 -0.00 0.01 0.04 0.04 -0.01 1.00 0.02 -0.03 -0.04 0.02 0.05 -0.04 0.31 0.05 0.11
Babies -0.03 -0.03 0.01 0.01 0.00 0.01 0.01 0.01 0.02 1.00 -0.00 -0.01 -0.00 0.07 -0.01 0.01 0.02 0.07
IsRepeatedGuest -0.07 -0.11 -0.03 0.00 -0.01 -0.08 -0.09 -0.17 -0.03 -0.00 1.00 0.27 0.45 0.01 -0.02 -0.15 0.09 0.02
PreviousCancellations 0.17 0.12 -0.20 0.04 -0.01 -0.04 0.05 -0.04 0.04 -0.01 0.27 1.00 0.39 -0.04 0.02 -0.13 0.00 -0.08
PreviousBookingsNotCanceled -0.05 -0.07 0.02 -0.01 -0.00 -0.03 -0.04 -0.11 -0.02 -0.00 0.45 0.39 1.00 0.02 -0.01 -0.08 0.07 0.05
BookingChanges -0.15 -0.03 0.03 0.00 0.01 0.05 0.08 -0.09 0.05 0.07 0.01 -0.04 0.02 1.00 -0.01 0.02 0.04 0.06
DaysInWaitingList 0.06 0.19 -0.07 0.02 0.02 -0.06 0.02 -0.01 -0.04 -0.01 -0.02 0.02 -0.01 -0.01 1.00 -0.07 -0.02 -0.09
ADR -0.01 -0.14 0.25 0.02 0.02 0.04 0.06 0.26 0.31 0.01 -0.15 -0.13 0.08 0.02 -0.07 1.00 0.06 0.17
RequiredCarParkingSpaces -0.13 -0.08 0.00 -0.00 0.00 -0.02 0.04 0.01 0.05 0.02 0.09 -0.00 0.07 0.04 -0.02 0.06 1.00 0.09
TotalOfSpecialRequests -0.29 0.13 0.14 0.02 -0.00 0.07 0.06 0.16 0.11 0.07 0.02 -0.08 0.05 0.06 -0.09 0.17 0.09 1.00
IsCanceled
LeadTime
ArrivalDateYear
ArrivalDateWeekNumber
ArrivalDateDayOfMonth
StaysInWeekendNights
StaysInWeekNights
Adults
Children
Babies
IsRepeatedGuest
PreviousCancellations
PreviousBookingsNotCanceled
BookingChanges
DaysInWaitingList
ADR
RequiredCarParkingSpaces
TotalOfSpecialRequests
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

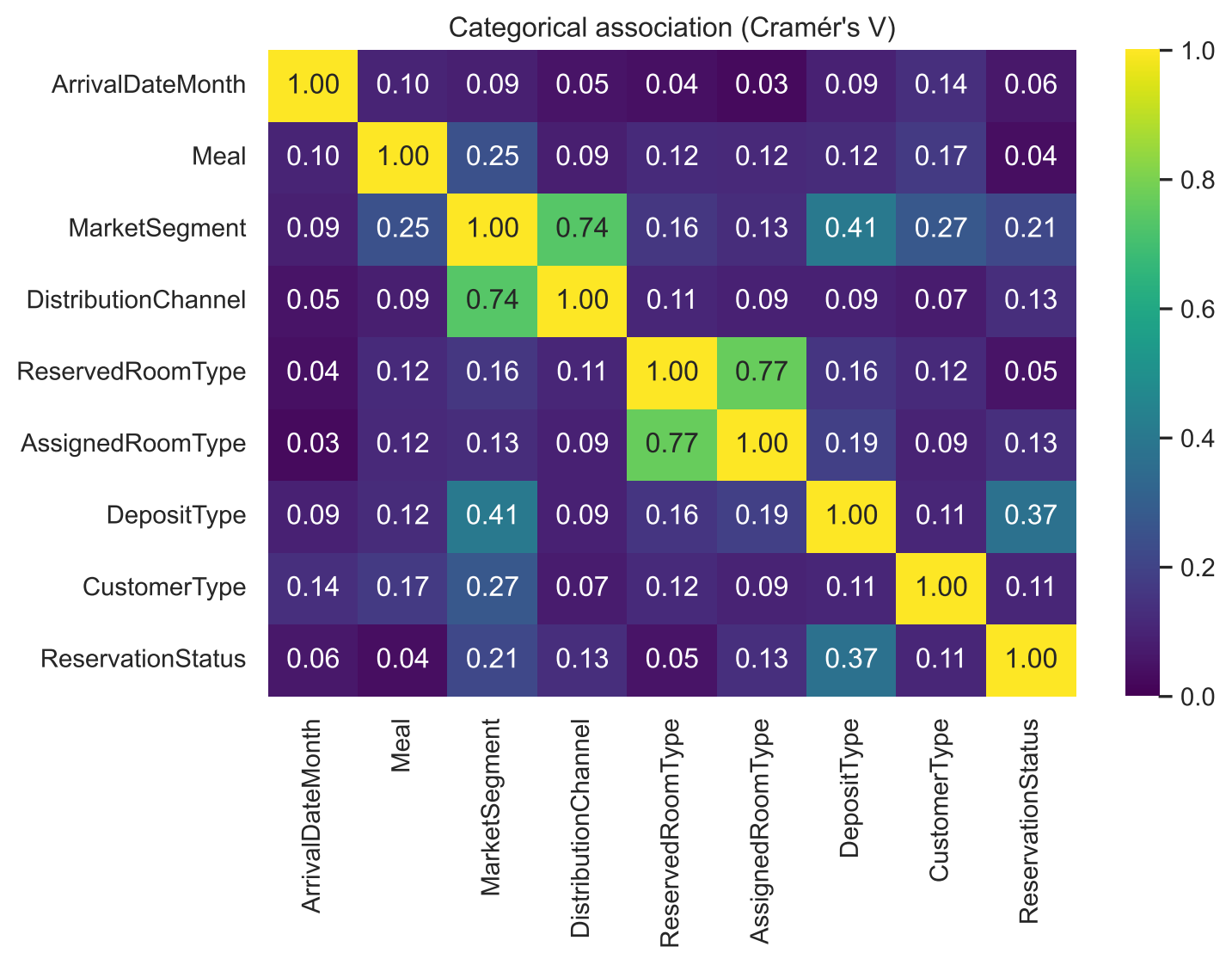

Spearman correlation heatmap
IsCanceled 1.00 0.32 -0.00 0.00 -0.00 -0.02 0.05 0.06 -0.03 -0.03 0.07 0.28 -0.10 0.20 0.11 -0.03 0.13 0.32
LeadTime 0.32 1.00 0.06 0.11 0.02 0.04 0.18 0.17 -0.04 0.04 0.17 0.19 -0.16 0.05 0.17 -0.10 0.09 0.12
ArrivalDateYear -0.00 0.06 1.00 -0.55 0.00 0.04 0.07 0.06 0.06 -0.01 0.03 0.30 0.02 0.03 -0.11 0.30 0.00 0.14
ArrivalDateWeekNumber 0.00 0.11 -0.55 1.00 0.06 0.00 -0.00 0.01 -0.00 0.01 0.00 0.10 -0.02 0.00 -0.01 0.02 0.00 0.01
ArrivalDateDayOfMonth -0.00 0.02 -0.00 0.06 1.00 0.00 -0.01 -0.00 0.01 0.00 -0.01 -0.01 -0.01 0.02 0.04 0.02 0.00 -0.00
StaysInWeekendNights -0.02 0.04 0.04 0.00 0.00 1.00 -0.00 0.10 0.04 0.01 -0.09 0.05 -0.06 0.01 -0.08 0.04 -0.02 0.08
StaysInWeekNights 0.05 0.18 0.07 -0.00 -0.01 -0.00 1.00 0.13 0.03 0.00 -0.11 -0.07 -0.11 0.02 0.03 0.08 -0.04 0.07
Adults 0.06 0.17 0.06 0.01 -0.00 0.10 0.13 1.00 0.03 0.02 -0.17 0.04 -0.18 -0.12 0.04 0.25 0.01 0.16
Children -0.03 0.04 0.06 -0.00 0.01 0.04 0.03 0.03 1.00 0.04 -0.03 0.07 -0.03 0.06 -0.05 0.25 0.06 0.13
Babies -0.03 0.04 0.01 0.01 0.00 0.01 0.00 0.02 0.04 1.00 -0.00 0.01 -0.00 0.11 -0.01 0.01 0.02 0.08
IsRepeatedGuest -0.07 -0.17 -0.03 0.00 -0.01 -0.09 -0.11 -0.17 -0.03 0.00 1.00 0.20 0.78 0.03 -0.03 0.16 0.09 0.01
PreviousCancellations 0.28 0.19 -0.30 0.10 -0.01 -0.05 -0.07 -0.04 0.07 -0.01 0.20 1.00 0.12 -0.08 0.12 -0.23 0.02 0.15
PreviousBookingsNotCanceled -0.10 -0.16 0.02 -0.02 0.01 -0.06 -0.11 -0.18 0.03 0.00 0.78 0.12 1.00 0.04 -0.01 0.12 0.11 0.04
BookingChanges -0.20 0.05 0.03 0.00 0.02 0.01 0.02 -0.12 0.06 0.11 0.03 -0.08 0.04 1.00 -0.02 0.03 0.05 0.05
DaysInWaitingList 0.11 0.17 -0.11 0.01 0.04 -0.08 0.03 -0.04 0.05 -0.01 -0.03 0.12 -0.01 -0.02 1.00 -0.09 0.03 0.14
ADR -0.03 0.10 0.30 0.02 0.02 0.04 0.08 0.25 0.25 0.01 -0.16 0.23 -0.12 0.03 -0.09 1.00 0.06 0.23
RequiredCarParkingSpaces -0.13 0.09 0.00 0.00 0.00 -0.02 0.04 0.01 0.06 0.02 0.09 -0.02 0.11 0.05 -0.03 0.06 1.00 0.09
TotalOfSpecialRequests -0.32 0.12 0.14 0.01 -0.00 0.08 0.07 0.16 0.13 0.08 0.01 -0.15 0.04 0.05 0.14 0.23 0.09 1.00
IsCanceled
LeadTime
ArrivalDateYear
ArrivalDateWeekNumber
ArrivalDateDayOfMonth
StaysInWeekendNights
StaysInWeekNights
Adults
Children
Babies
IsRepeatedGuest
PreviousCancellations
PreviousBookingsNotCanceled
BookingChanges
DaysInWaitingList
ADR
RequiredCarParkingSpaces
TotalOfSpecialRequests
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Strongest numeric correlation pairs:

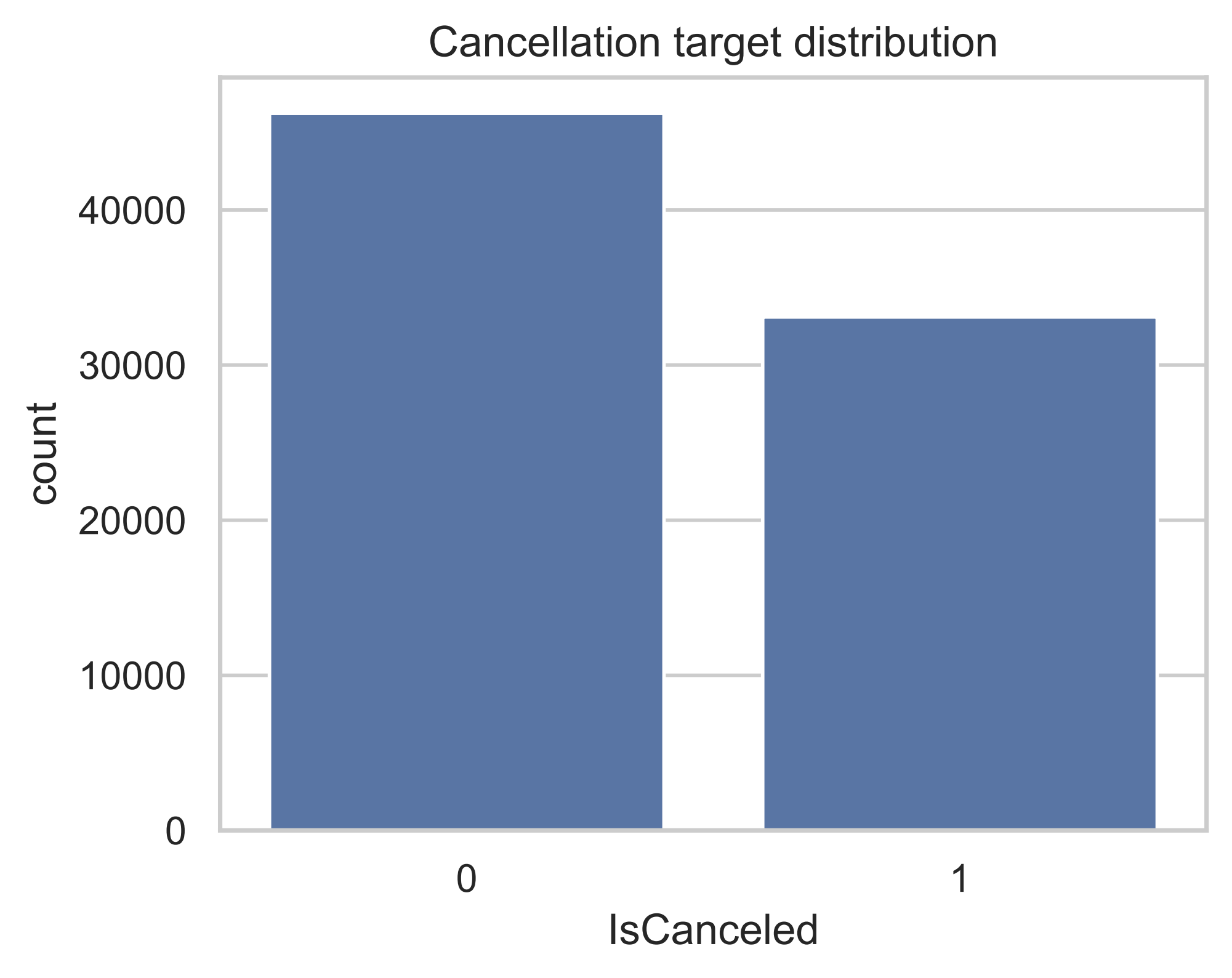
feature_1	feature_2	correlation	absolute_correlation
ArrivalDateYear	ArrivalDateWeekNumber	-0.543027	0.543027
IsRepeatedGuest	PreviousBookingsNotCanceled	0.451965	0.451965
PreviousCancellations	PreviousBookingsNotCanceled	0.392141	0.392141
Children	ADR	0.311961	0.311961
IsCanceled	LeadTime	0.309242	0.309242
IsCanceled	TotalOfSpecialRequests	-0.293889	0.293889
IsRepeatedGuest	PreviousCancellations	0.268287	0.268287
Adults	ADR	0.262895	0.262895
ArrivalDateYear	ADR	0.248757	0.248757
StaysInWeekendNights	StaysInWeekNights	0.219601	0.219601
ArrivalDateYear	PreviousCancellations	-0.195255	0.195255
LeadTime	DaysInWaitingList	0.187110	0.187110
ADR	TotalOfSpecialRequests	0.174968	0.174968
Adults	IsRepeatedGuest	-0.167354	0.167354
IsCanceled	PreviousCancellations	0.166643	0.166643
Adults	TotalOfSpecialRequests	0.155642	0.155642
IsRepeatedGuest	ADR	-0.152716	0.152716
IsCanceled	BookingChanges	-0.149500	0.149500
LeadTime	ADR	-0.137240	0.137240
ArrivalDateYear	TotalOfSpecialRequests	0.135439	0.135439
IsCanceled	RequiredCarParkingSpaces	-0.133096	0.133096
LeadTime	TotalOfSpecialRequests	-0.132450	0.132450
LeadTime	ArrivalDateWeekNumber	0.130392	0.130392
PreviousCancellations	ADR	-0.129247	0.129247
LeadTime	PreviousCancellations	0.122511	0.122511
LeadTime	Adults	0.115792	0.115792
Children	TotalOfSpecialRequests	0.112716	0.112716
Adults	PreviousBookingsNotCanceled	-0.109113	0.109113
LeadTime	IsRepeatedGuest	-0.107900	0.107900
StaysInWeekNights	IsRepeatedGuest	-0.094165	0.094165

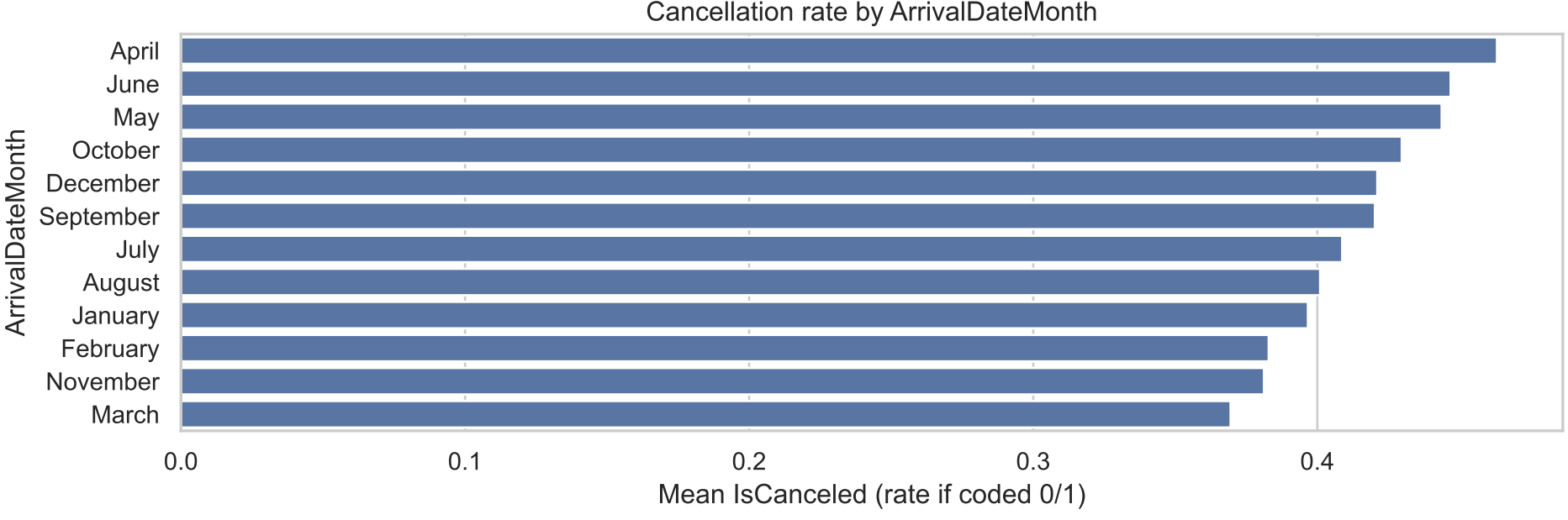
=== 7. CATEGORICAL ASSOCIATIONS (CRAMER'S V) ===

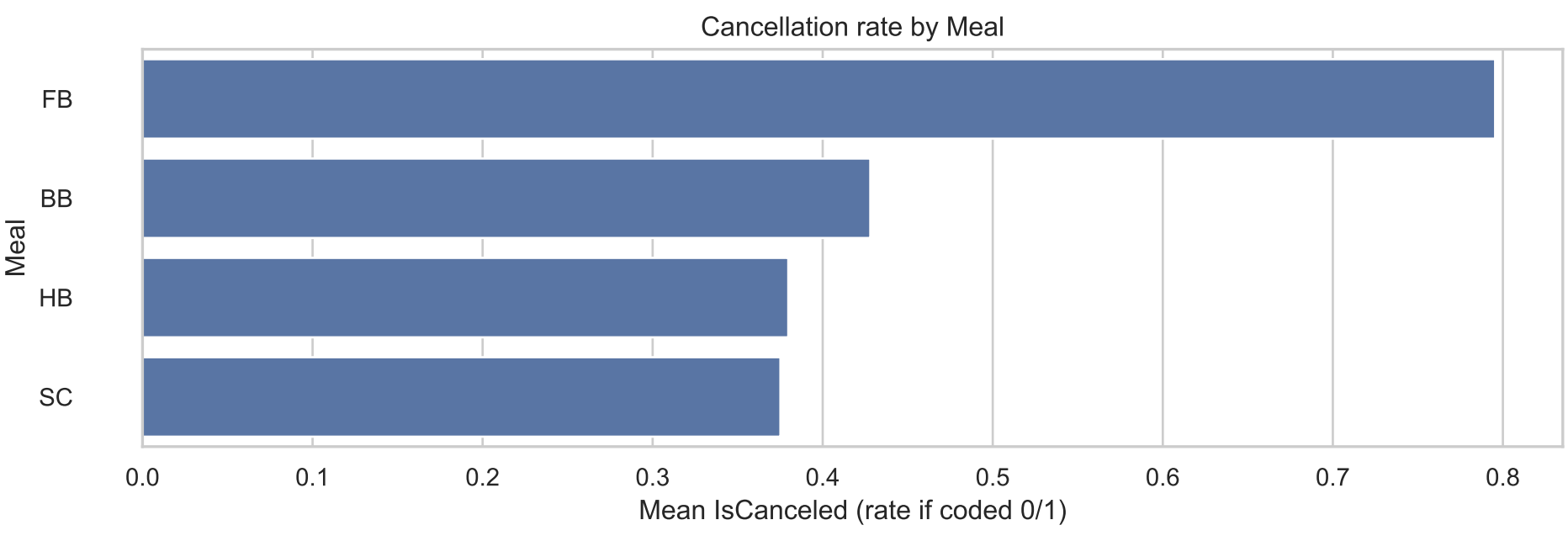
	ArrivalDateMonth	Meal	MarketSegment	DistributionChannel	ReservedRoomType	AssignedRoomType	DepositType	CustomerType	ReservationStatus
ArrivalDateMonth	1.000	0.099	0.086	0.054	0.043	0.025	0.091	0.135	0.064
Meal	0.099	1.000	0.247	0.085	0.118	0.116	0.122	0.171	0.037
MarketSegment	0.086	0.247	1.000	0.737	0.157	0.127	0.406	0.271	0.215
DistributionChannel	0.054	0.085	0.737	1.000	0.107	0.091	0.093	0.075	0.129
ReservedRoomType	0.043	0.118	0.157	0.107	1.000	0.768	0.155	0.121	0.055
AssignedRoomType	0.025	0.116	0.127	0.091	0.768	1.000	0.188	0.094	0.133
DepositType	0.091	0.122	0.406	0.093	0.155	0.188	1.000	0.107	0.375
CustomerType	0.135	0.171	0.271	0.075	0.121	0.094	0.107	1.000	0.109
ReservationStatus	0.064	0.037	0.215	0.129	0.055	0.133	0.375	0.109	1.000

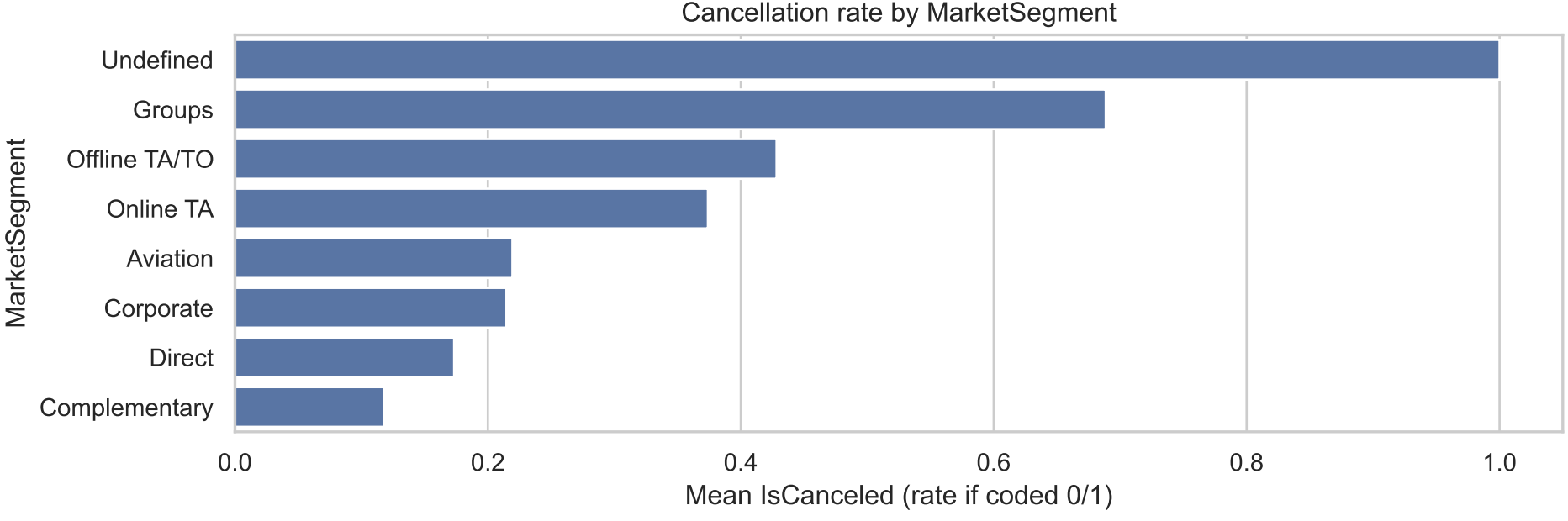

Categorical association (Cramér's V)
ArrivalDateMonth 1.00 0.10 0.09 0.05 0.04 0.03 0.09 0.14 0.06
Meal 0.10 1.00 0.25 0.09 0.12 0.12 0.12 0.17 0.04
MarketSegment 0.09 0.25 1.00 0.74 0.16 0.13 0.41 0.27 0.21
DistributionChannel 0.05 0.09 0.74 1.00 0.11 0.09 0.09 0.07 0.13
ReservedRoomType 0.04 0.12 0.16 0.11 1.00 0.77 0.16 0.12 0.05
AssignedRoomType 0.03 0.12 0.13 0.09 0.77 1.00 0.19 0.09 0.13
DepositType 0.09 0.12 0.41 0.09 0.16 0.19 1.00 0.11 0.37
CustomerType 0.14 0.17 0.27 0.07 0.12 0.09 0.11 1.00 0.11
ReservationStatus 0.06 0.04 0.21 0.13 0.05 0.13 0.37 0.11 1.00
ArrivalDateMonth
Meal
MarketSegment
DistributionChannel
ReservedRoomType
AssignedRoomType
DepositType
CustomerType
ReservationStatus
1.0
0.8
0.6
0.4
0.2
0.0

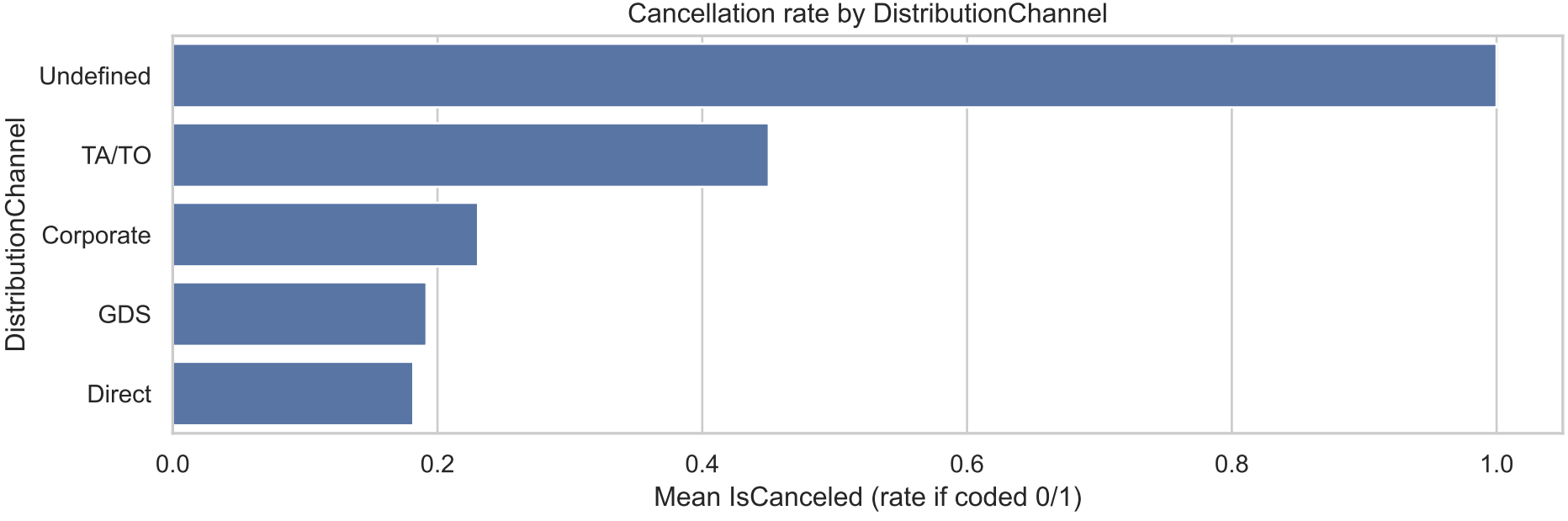
```
=== 8. TARGET ANALYSIS ===
IsCanceled counts:
 IsCanceled
0    46228
1    33102
Name: count, dtype: int64
IsCanceled percentages:
 IsCanceled
0    58.27
1    41.73
Name: proportion, dtype: float64
```

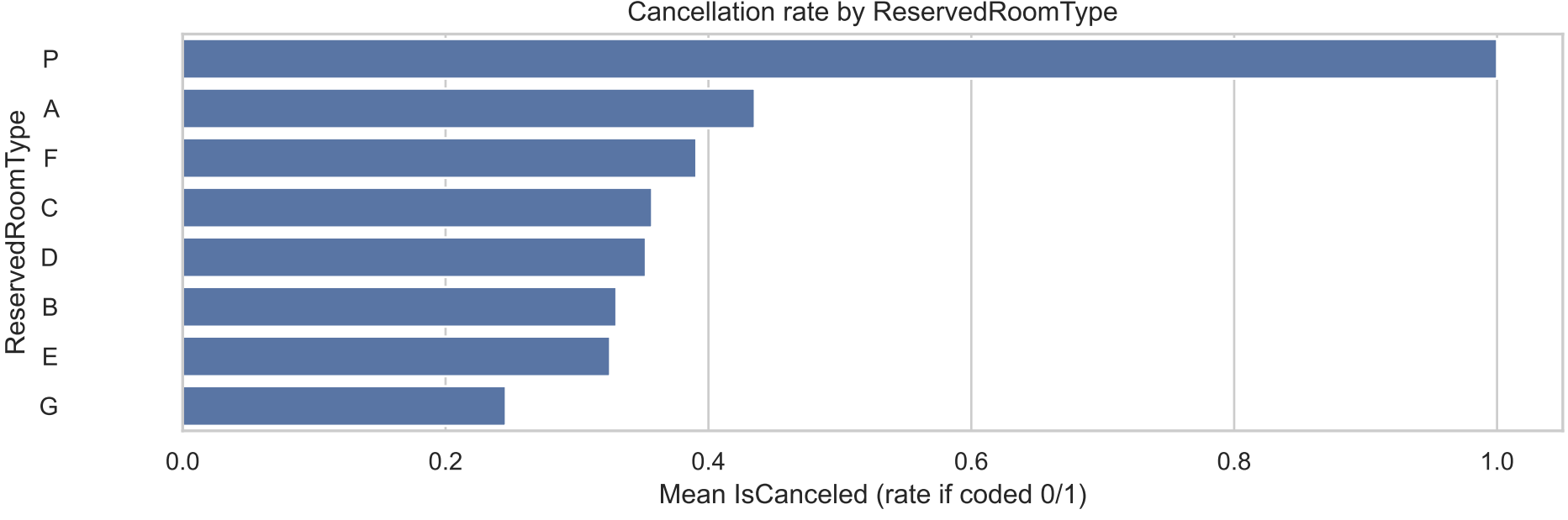

Cancellation target distribution
40000
30000
20000
10000
0
count
0
1
IsCanceled

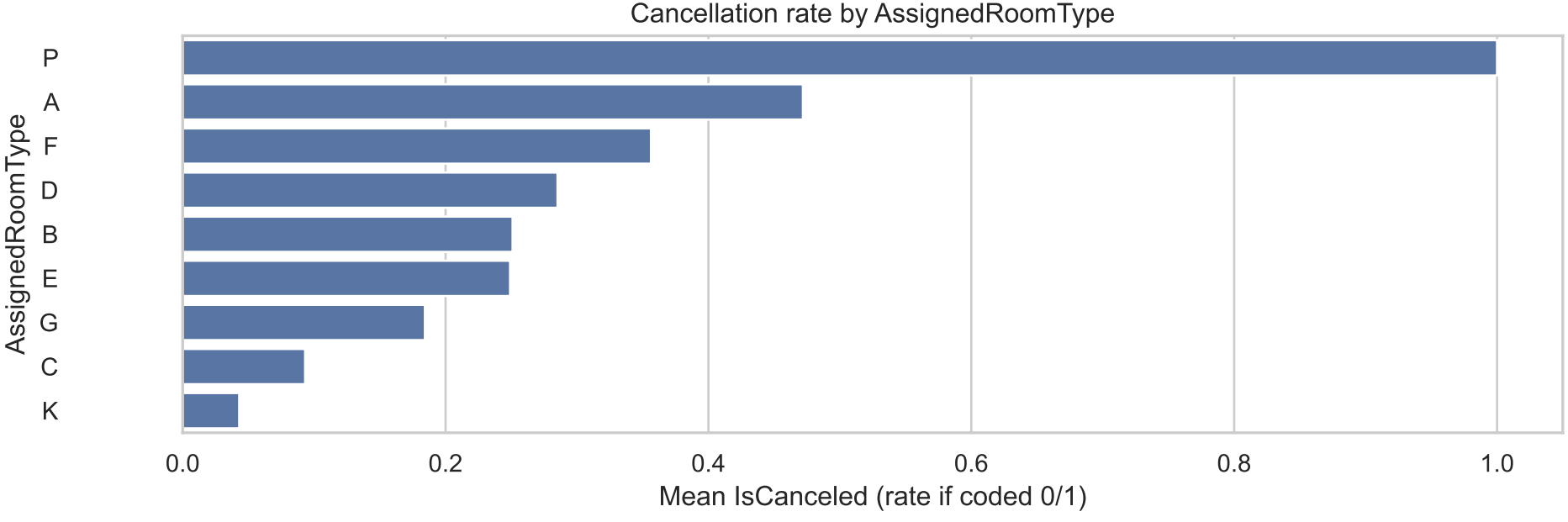

Cancellation rate by ArrivalDateMonth
ArrivalDateMonth
April
June
May
October
December
September
July
August
January
February
November
March
0.0
0.1
0.2
0.3
0.4
Mean IsCanceled (rate if coded 0/1)

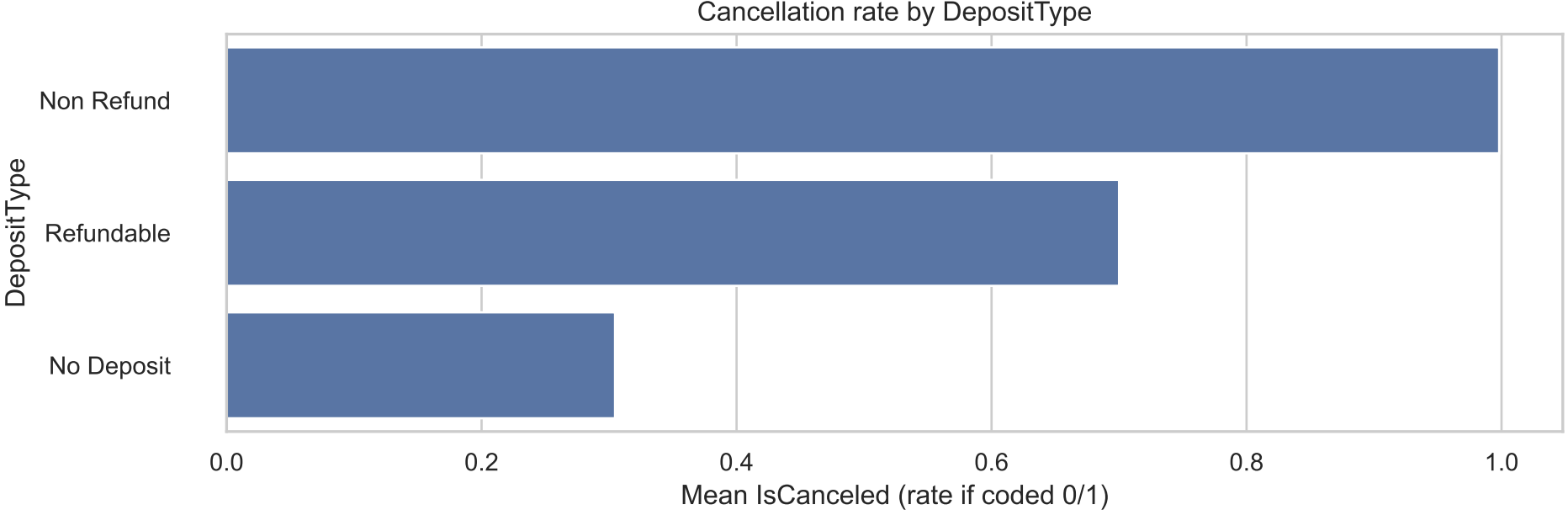

Cancellation rate by Meal
Meal
FB
BB
HB
SC
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Mean IsCanceled (rate if coded 0/1)

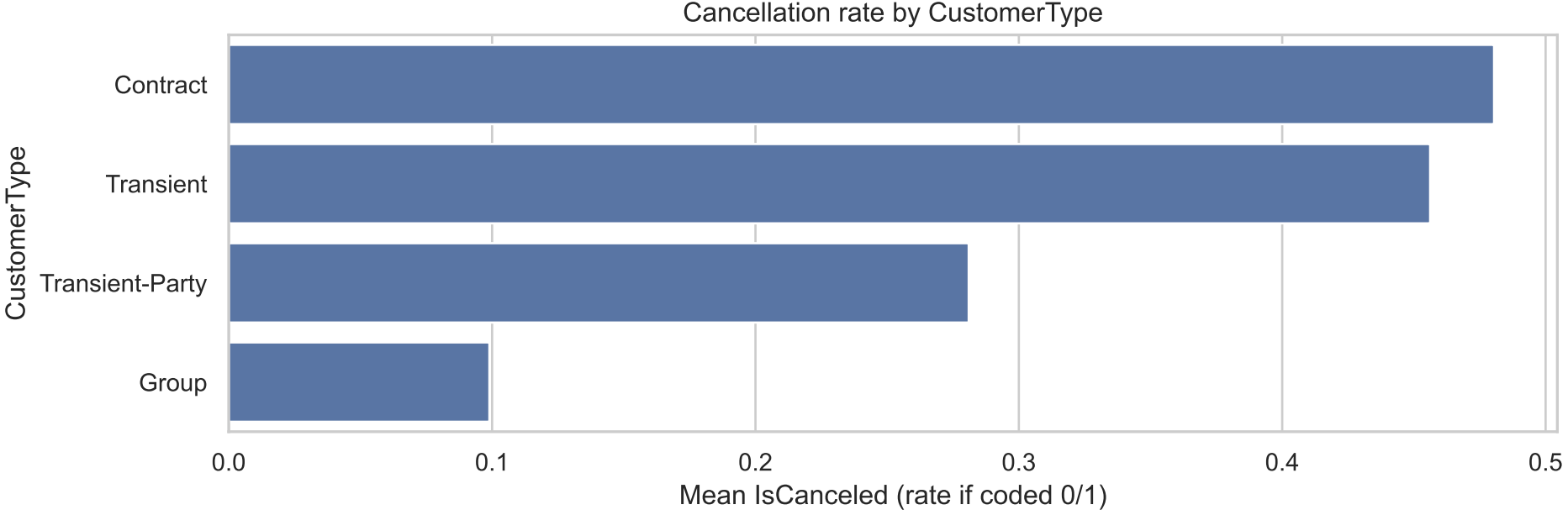

Cancellation rate by MarketSegment
MarketSegment
Undefined
Groups
Offline TA/TO
Online TA
Aviation
Corporate
Direct
Complementary
0.0
0.2
0.4
0.6
0.8
1.0
Mean IsCanceled (rate if coded 0/1)

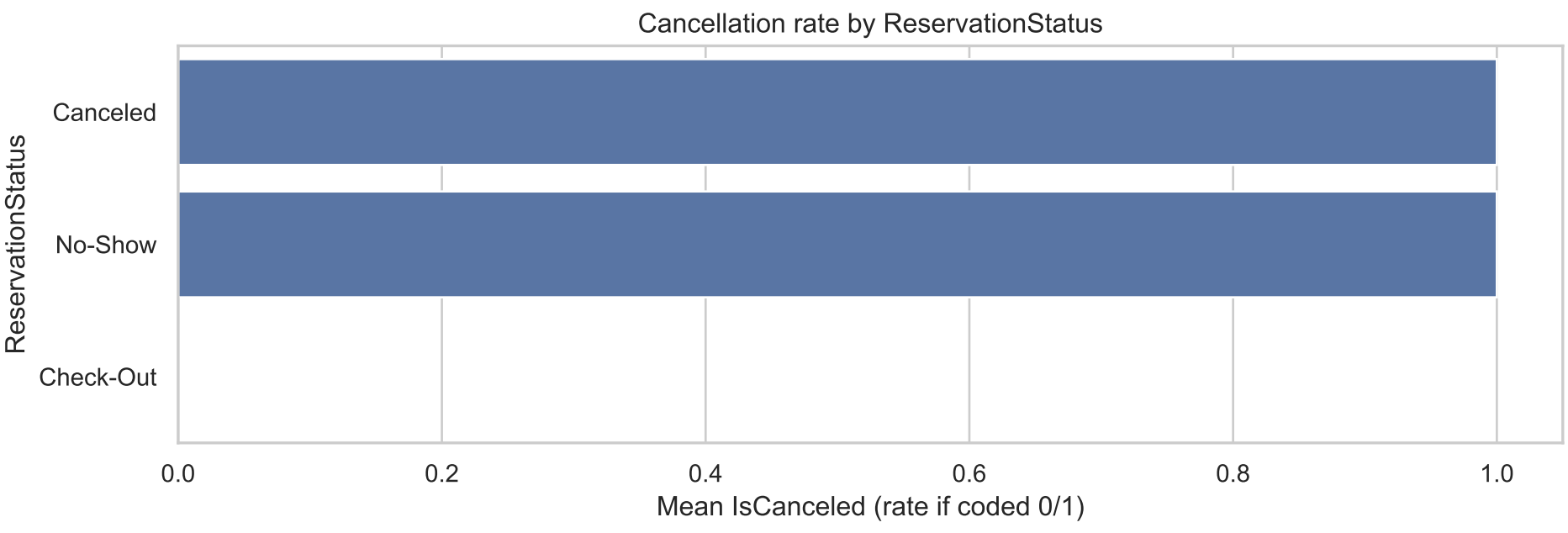

Cancellation rate by DistributionChannel
DistributionChannel
Undefined
TA/TO
Corporate
GDS
Direct
0.0
0.2
0.4
0.6
0.8
1.0
Mean IsCanceled (rate if coded 0/1)

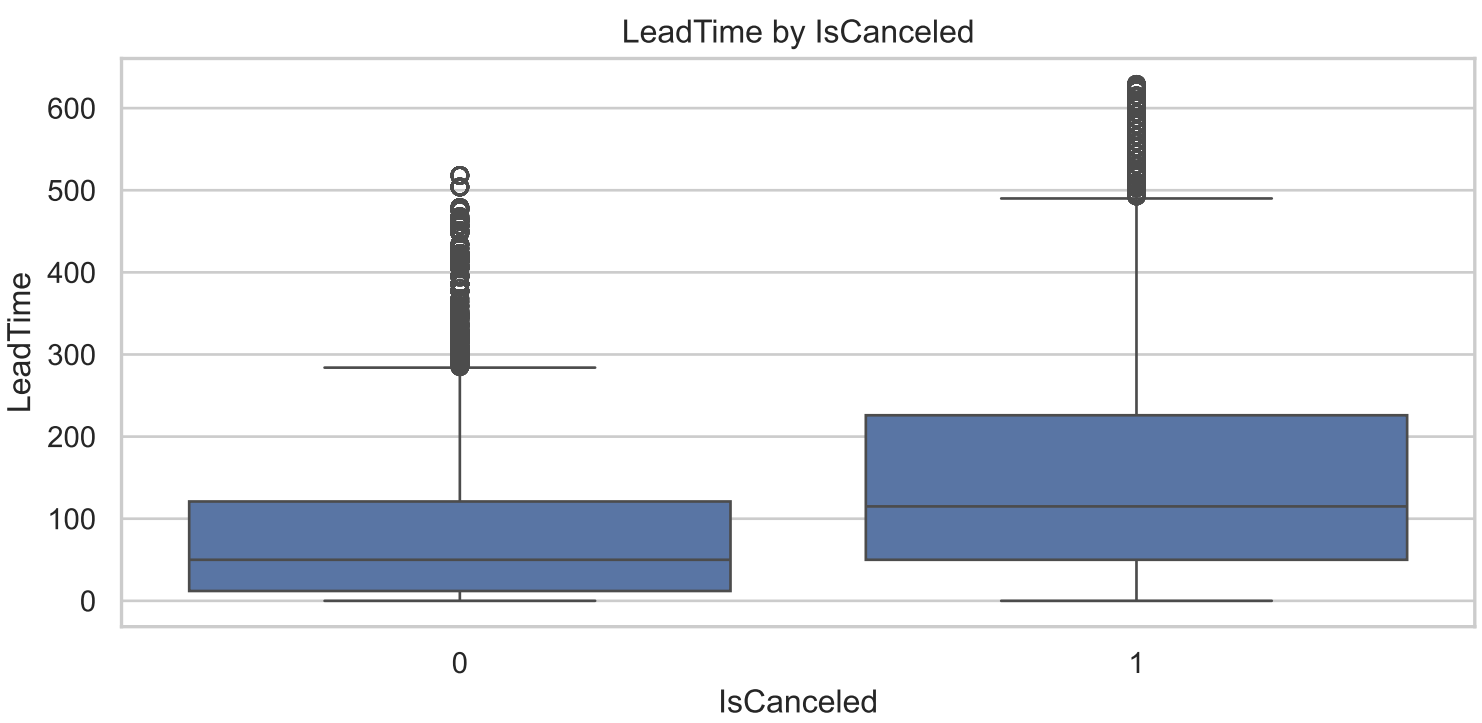

Cancellation rate by ReservedRoomType
ReservedRoomType
P
A
F
C
D
B
E
G
0.0
0.2
0.4
0.6
0.8
1.0
Mean IsCanceled (rate if coded 0/1)

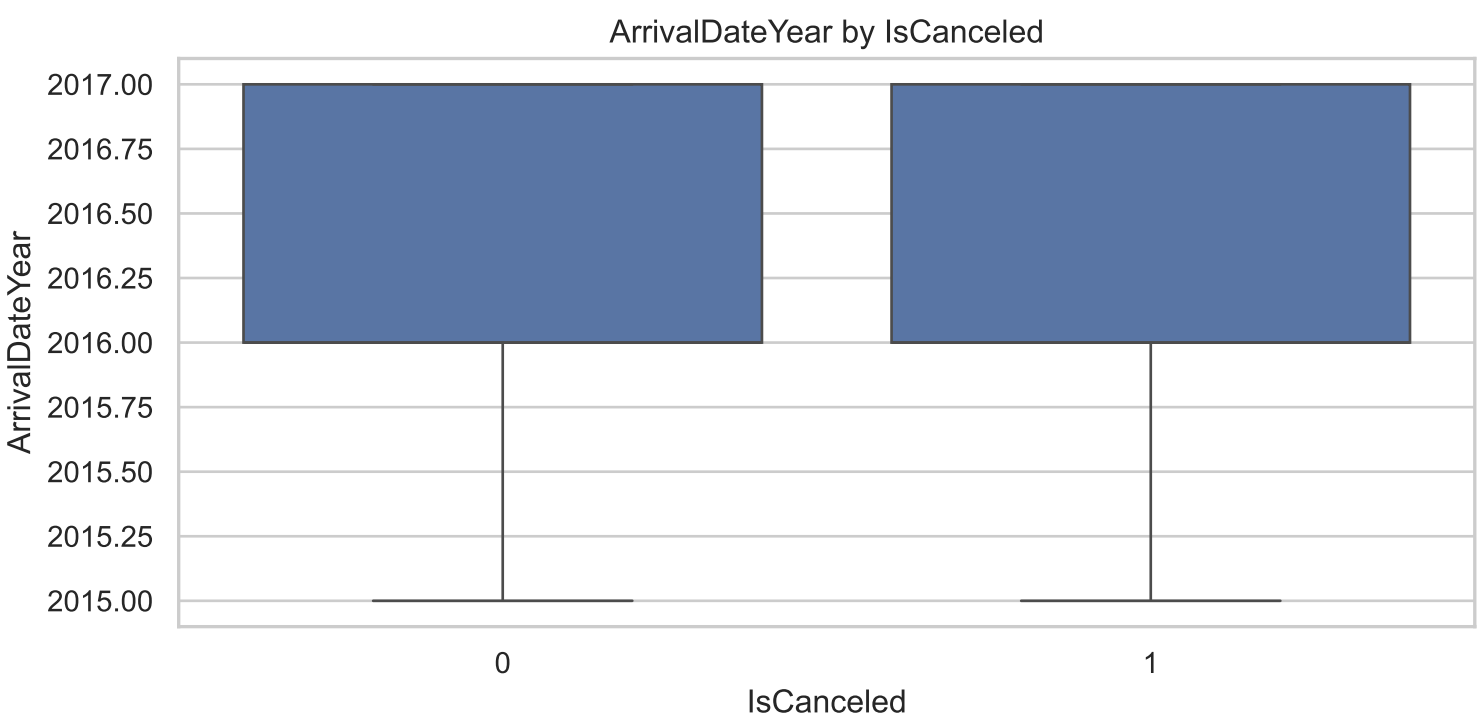

Cancellation rate by AssignedRoomType
AssignedRoomType
P
A
F
D
B
E
G
C
K
0.0
0.2
0.4
0.6
0.8
1.0
Mean IsCanceled (rate if coded 0/1)

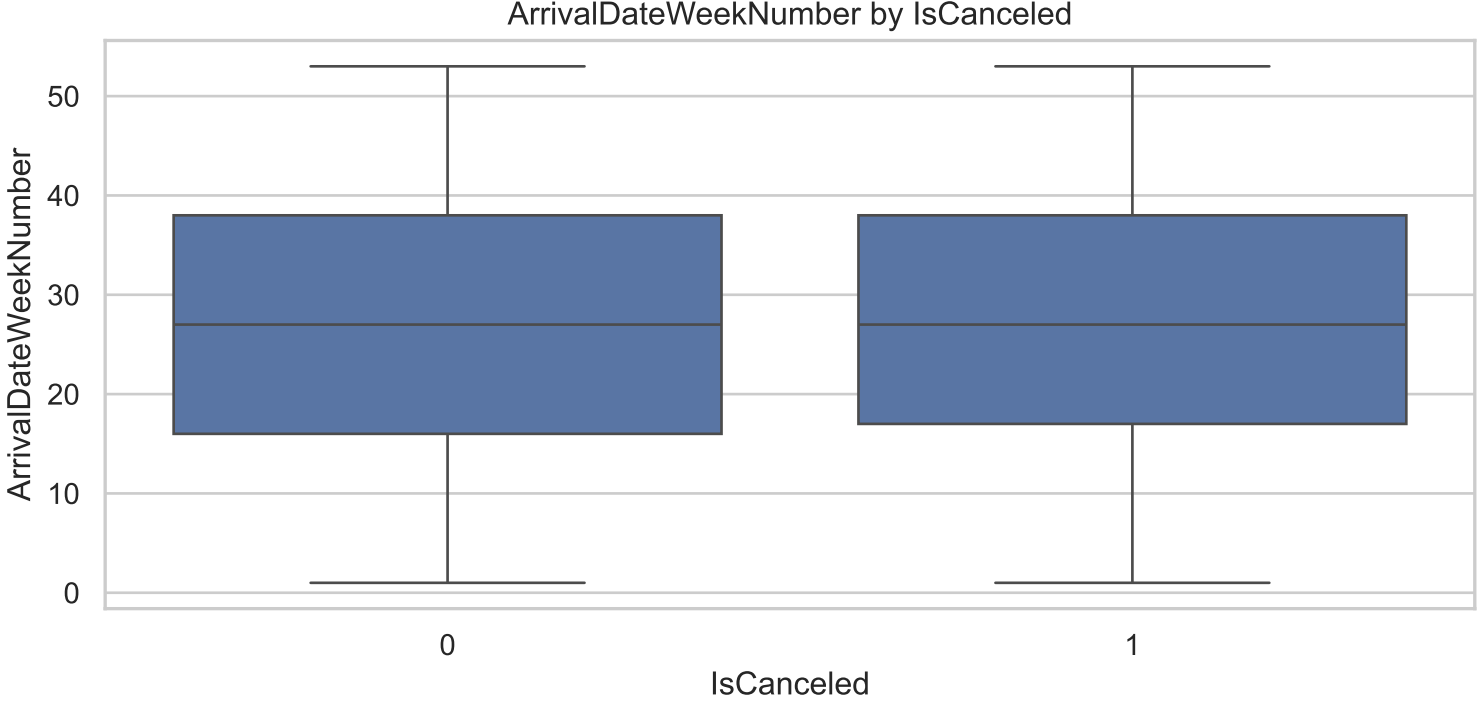

Cancellation rate by DepositType
DepositType
Non Refund
Refundable
No Deposit
0.0
0.2
0.4
0.6
0.8
1.0
Mean IsCanceled (rate if coded 0/1)

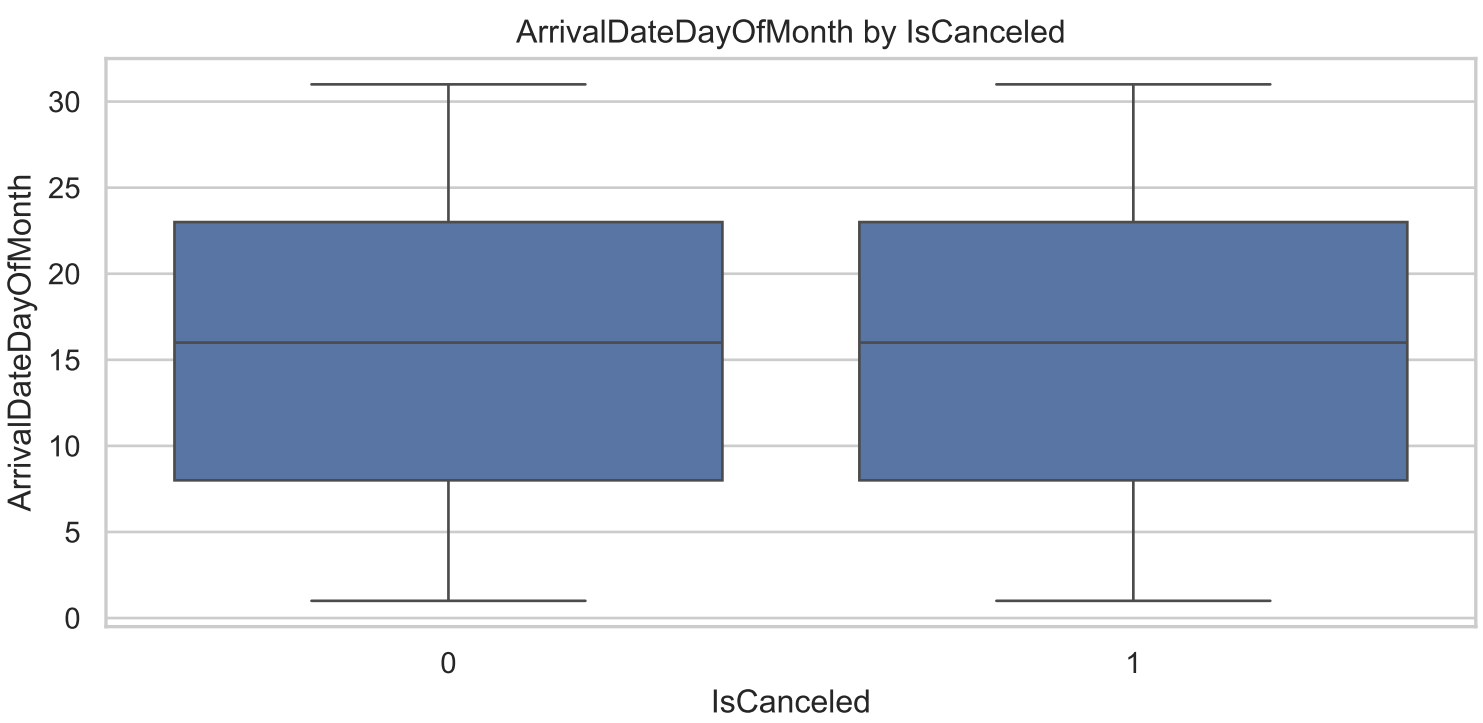

Cancellation rate by CustomerType
CustomerType
Contract
Transient
Transient-Party
Group
0.0
0.1
0.2
0.3
0.4
0.5
Mean IsCanceled (rate if coded 0/1)

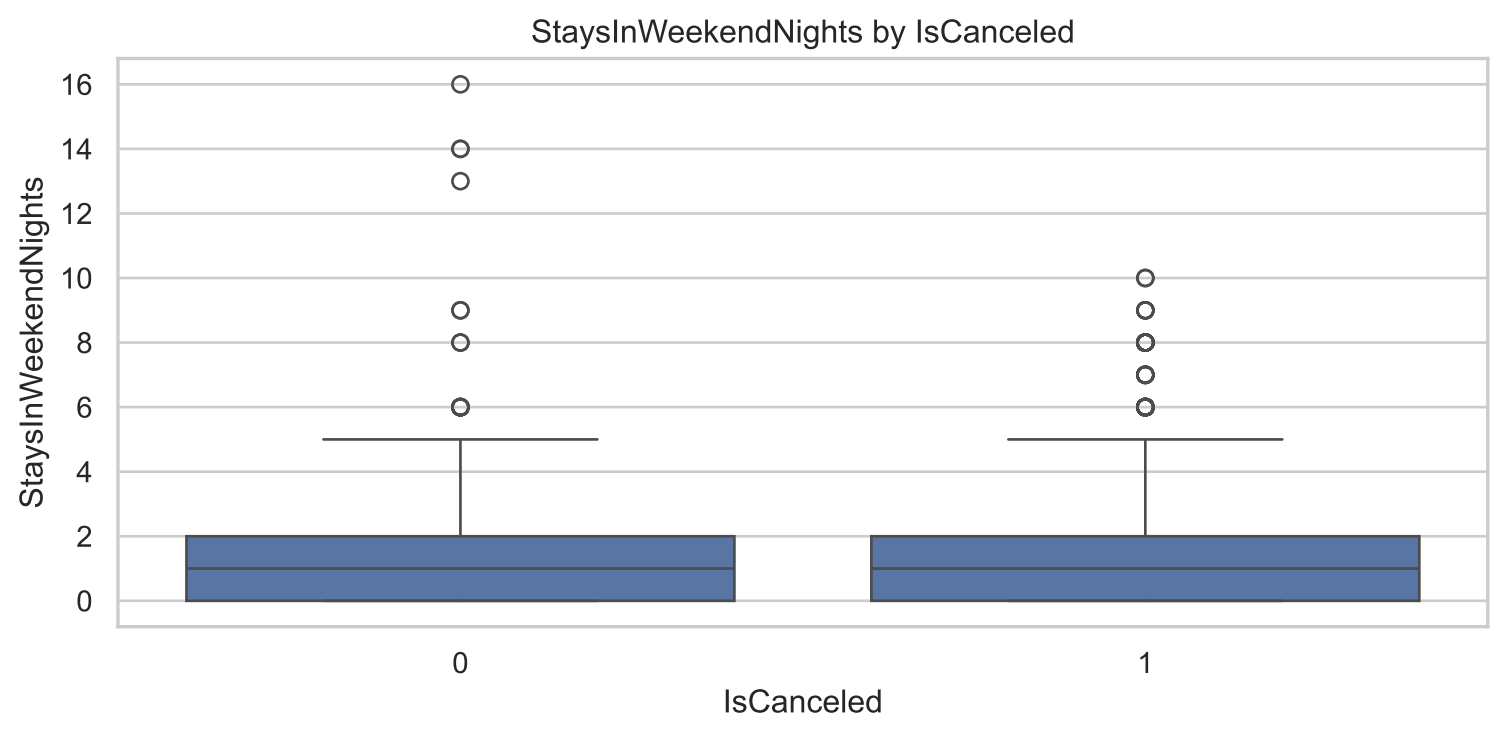

Cancellation rate by ReservationStatus
ReservationStatus
Canceled
No-Show
Check-Out
0.0
0.2
0.4
0.6
0.8
1.0
Mean IsCanceled (rate if coded 0/1)

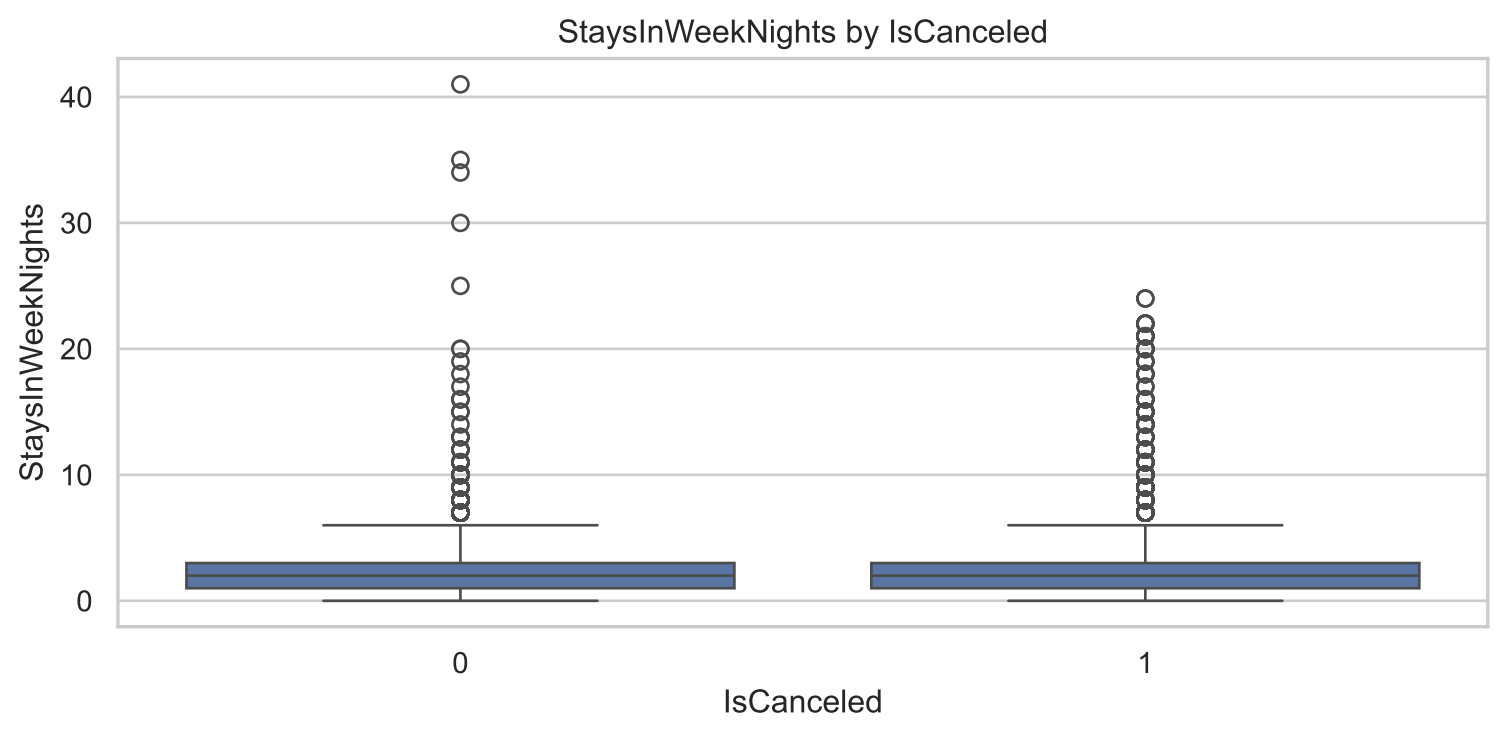

LeadTime by IsCanceled
LeadTime
600
500
400
300
200
100
0
0
1
IsCanceled

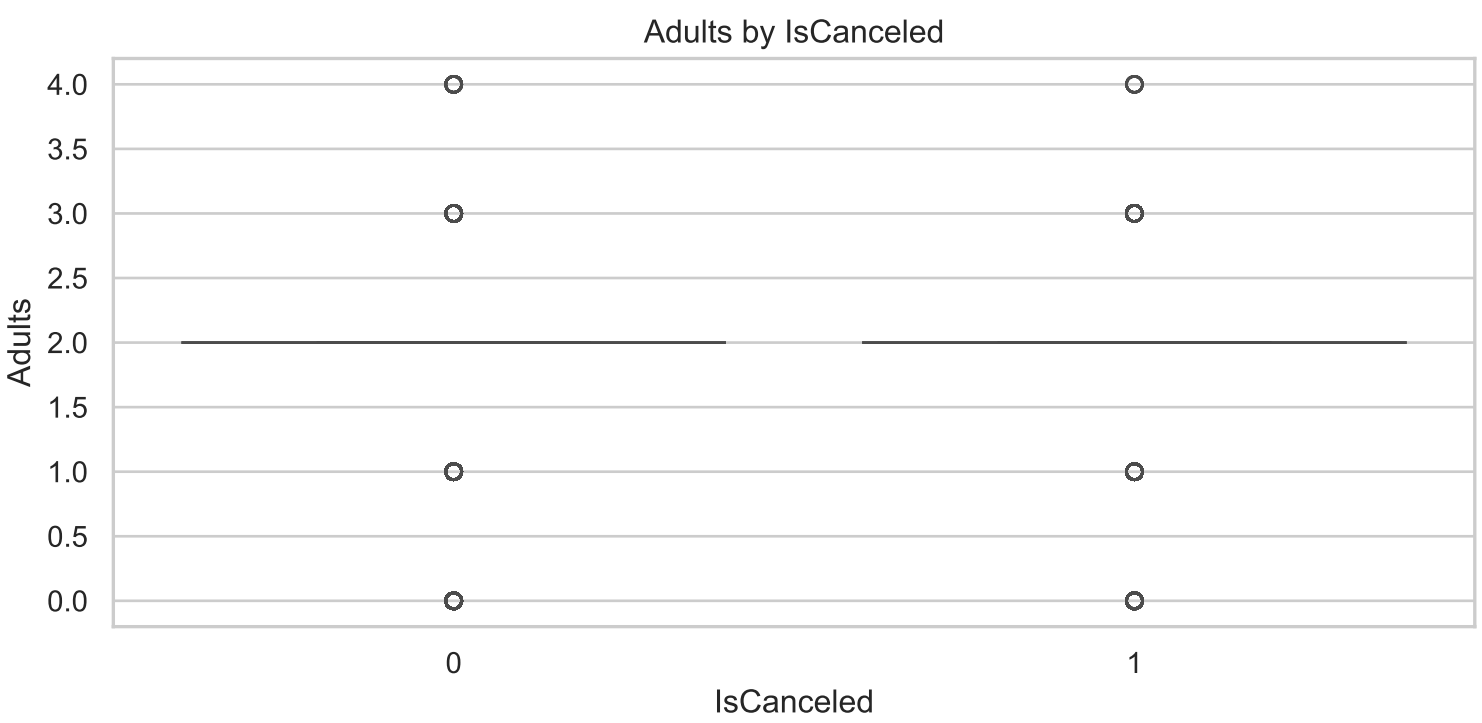

ArrivalDateYear by IsCanceled
2017.00
2016.75
2016.50
2016.25
2016.00
2015.75
2015.50
2015.25
2015.00
ArrivalDateYear
0
1
IsCanceled

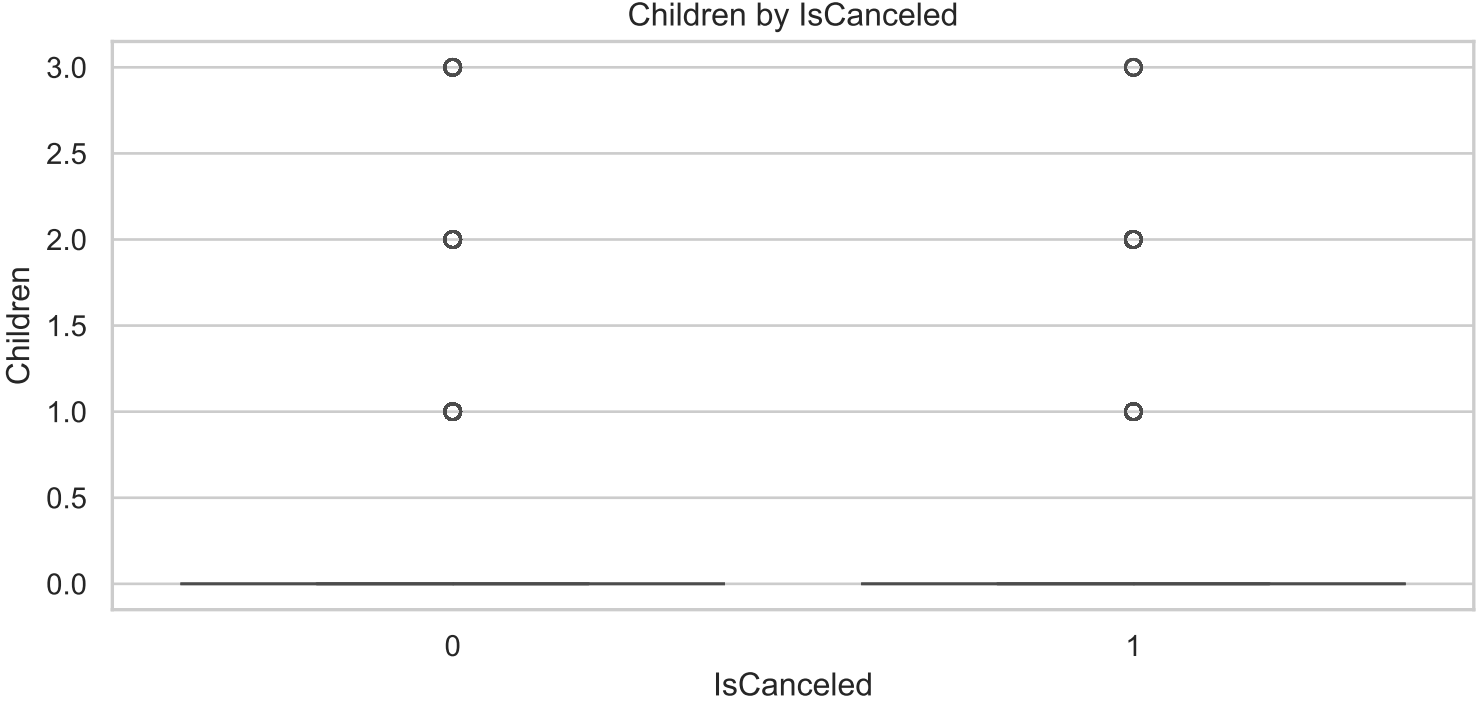

ArrivalDateWeekNumber by IsCanceled
ArrivalDateWeekNumber
0
10
20
30
40
50
0
1
IsCanceled

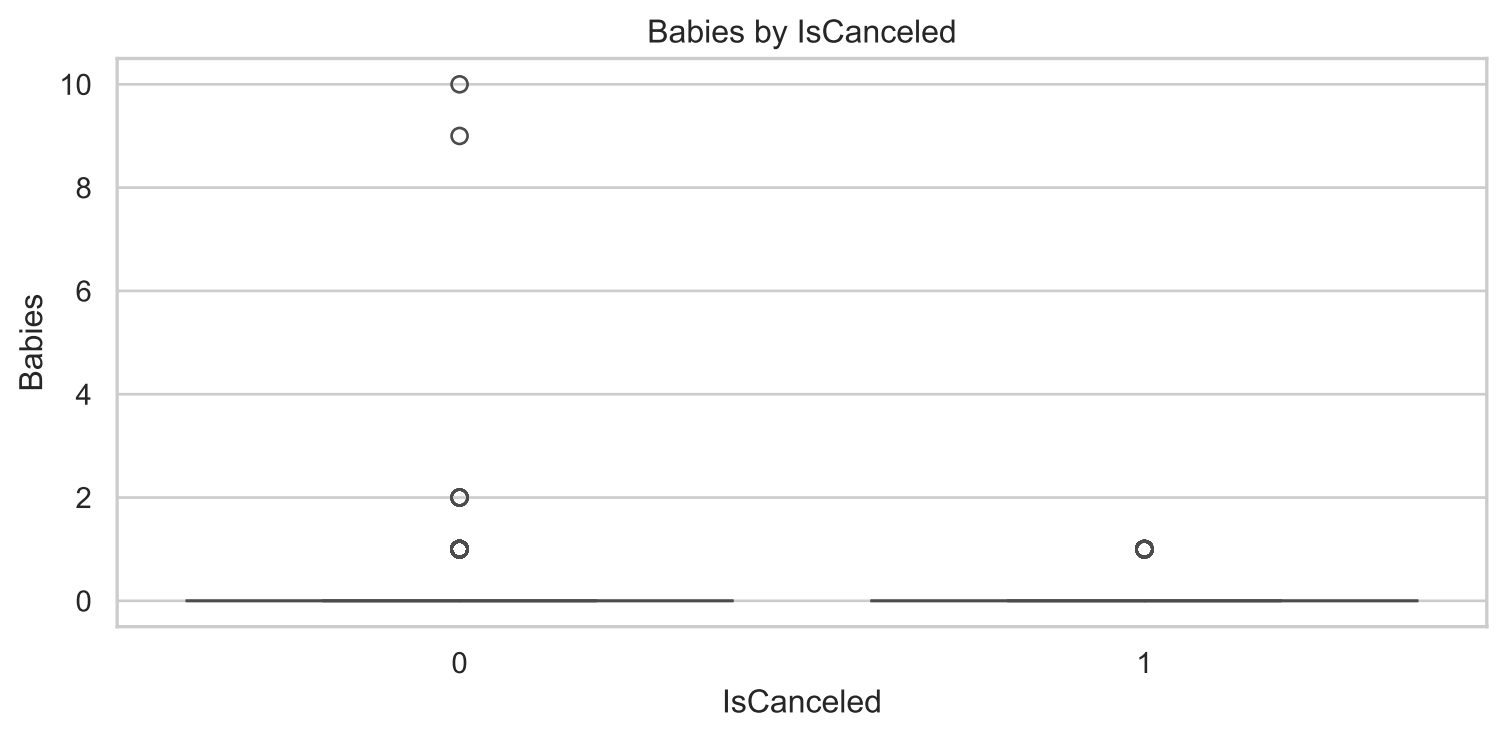

ArrivalDateDayOfMonth by IsCanceled
ArrivalDateDayOfMonth
0
5
10
15
20
25
30
0
1
IsCanceled

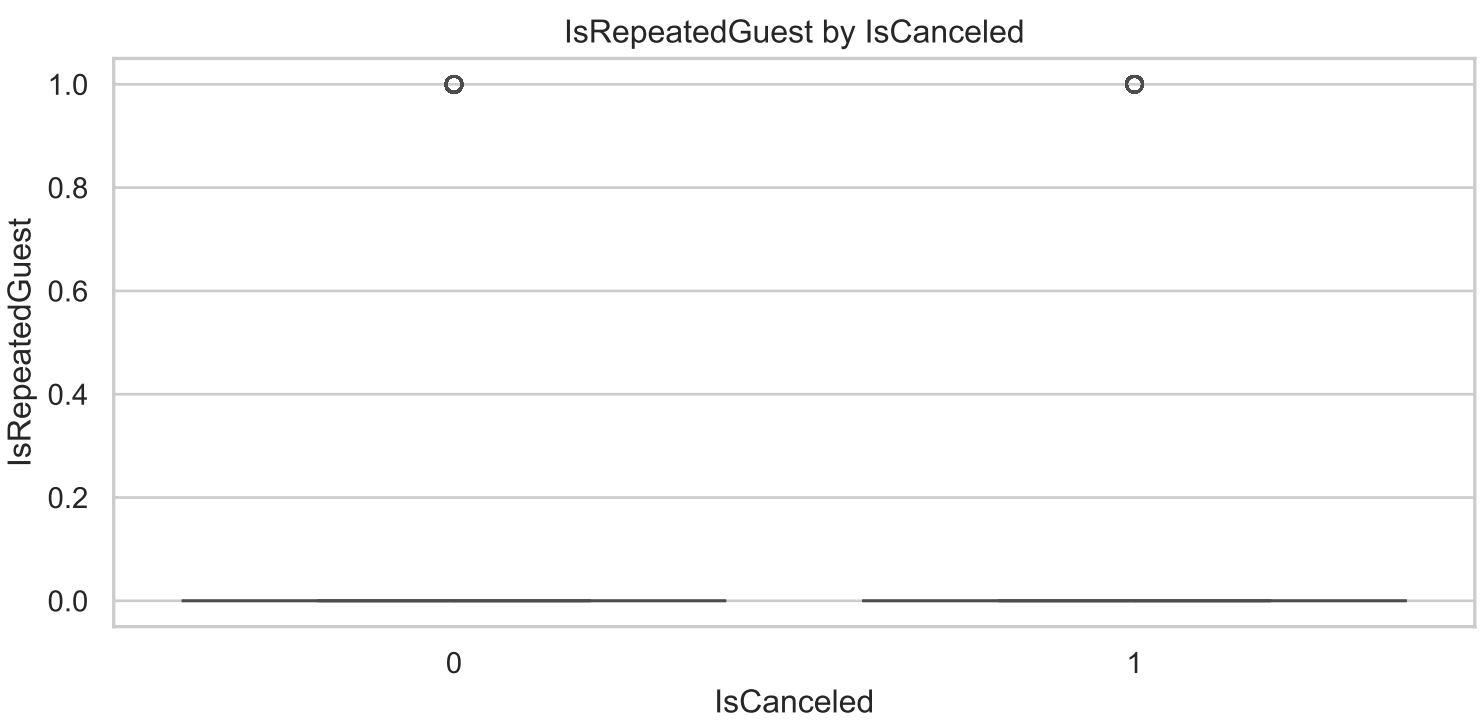

StaysInWeekendNights by IsCanceled
StaysInWeekendNights
16
14
12
10
8
6
4
2
0
0
1
IsCanceled

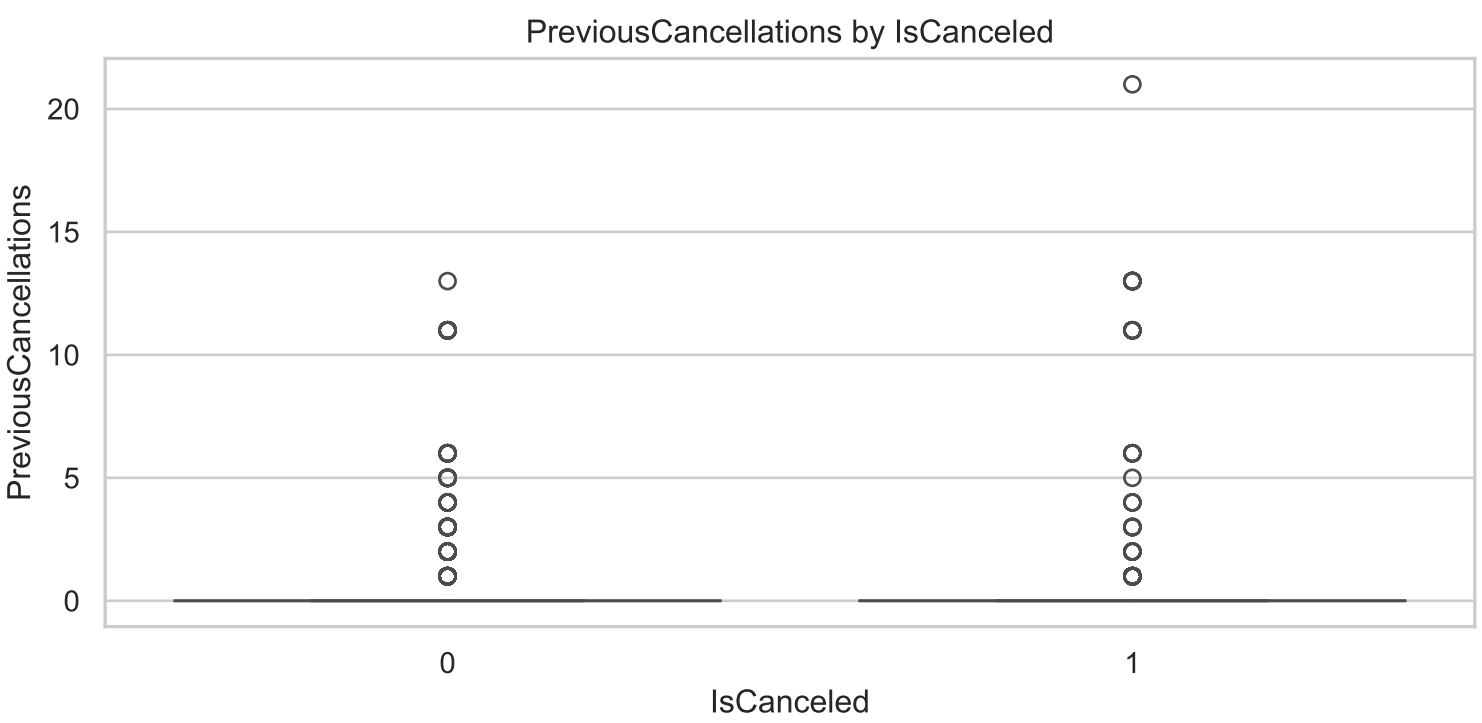

StaysInWeekNights by IsCanceled
StaysInWeekNights
40
30
20
10
0
0
1
IsCanceled

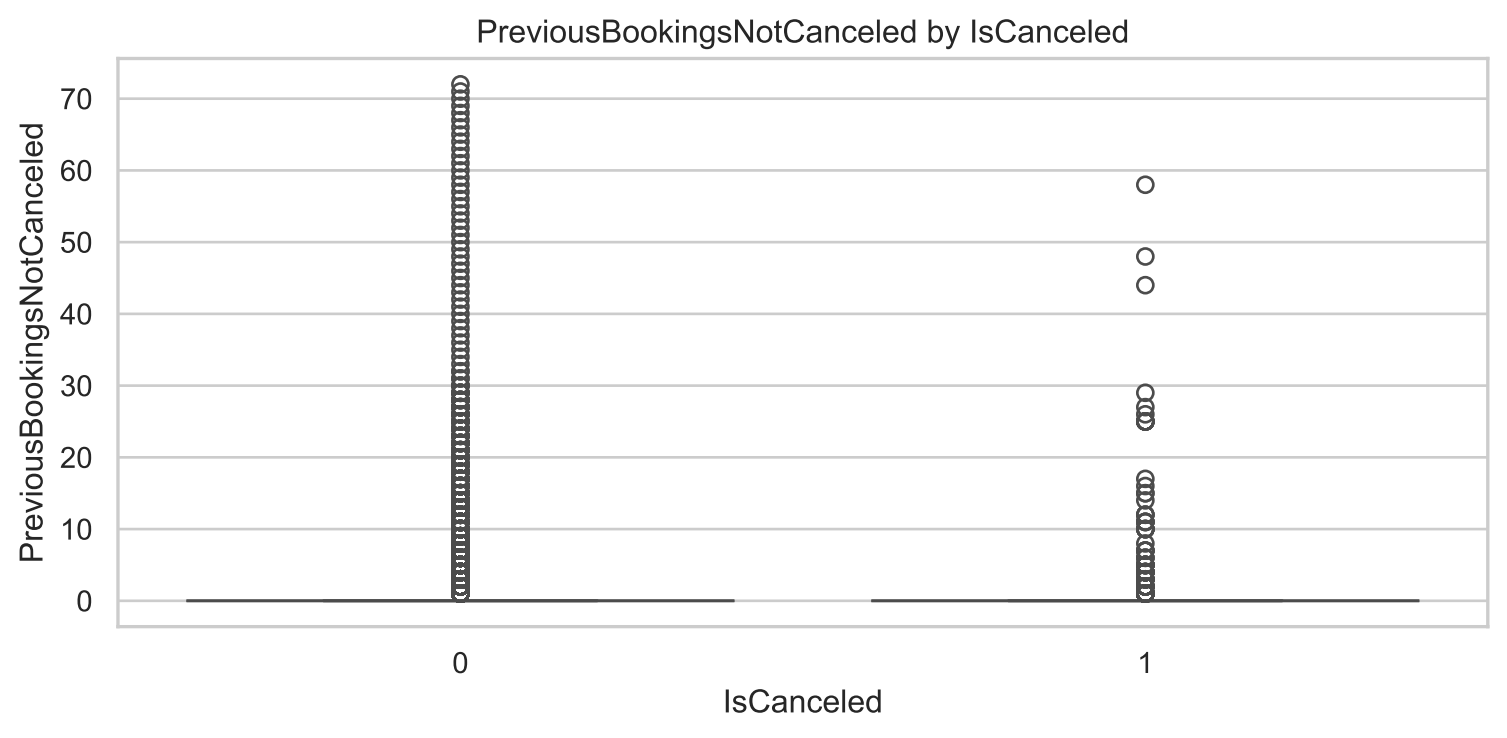

Adults by IsCanceled
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Adults
0
1
IsCanceled

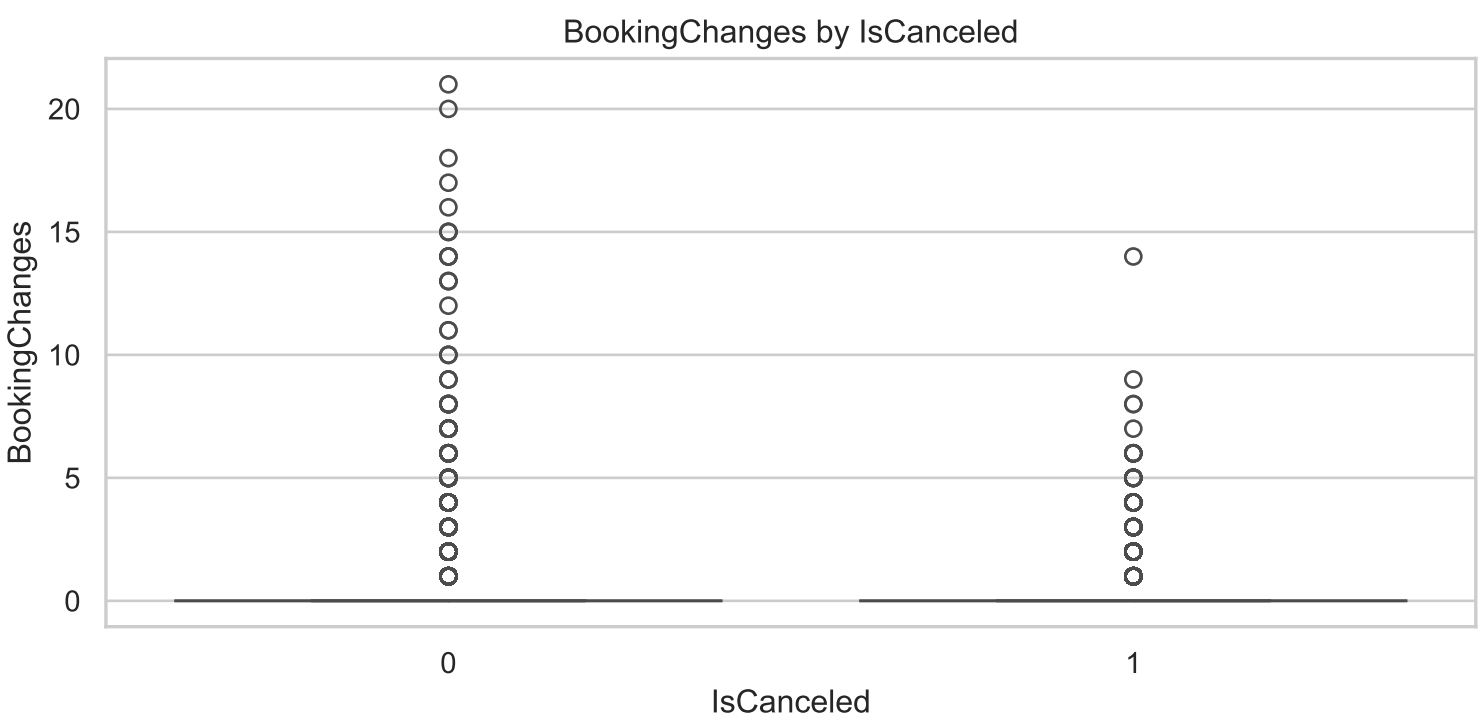

Children by IsCanceled
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Children
0
1
IsCanceled

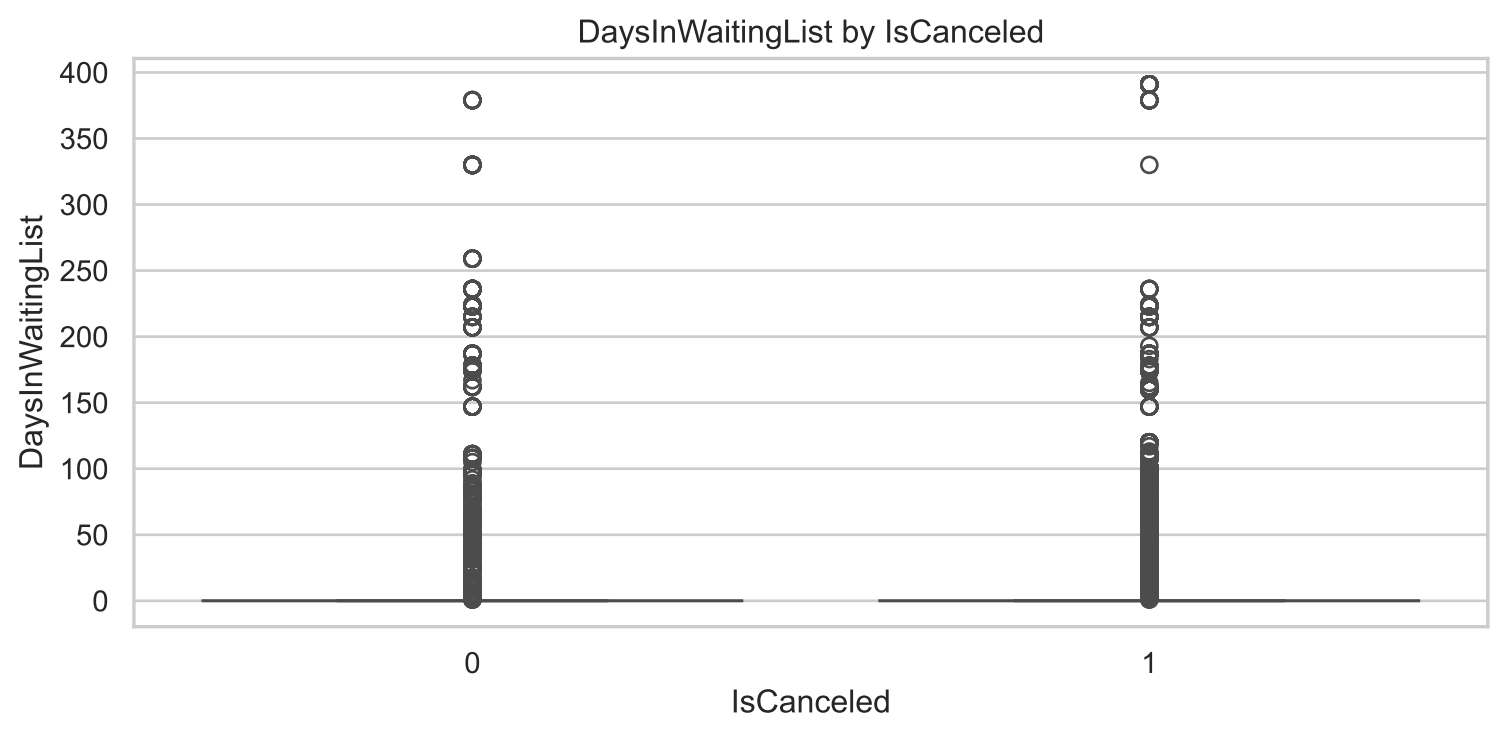

Babies by IsCanceled
Babies
10
8
6
4
2
0
0
1
IsCanceled

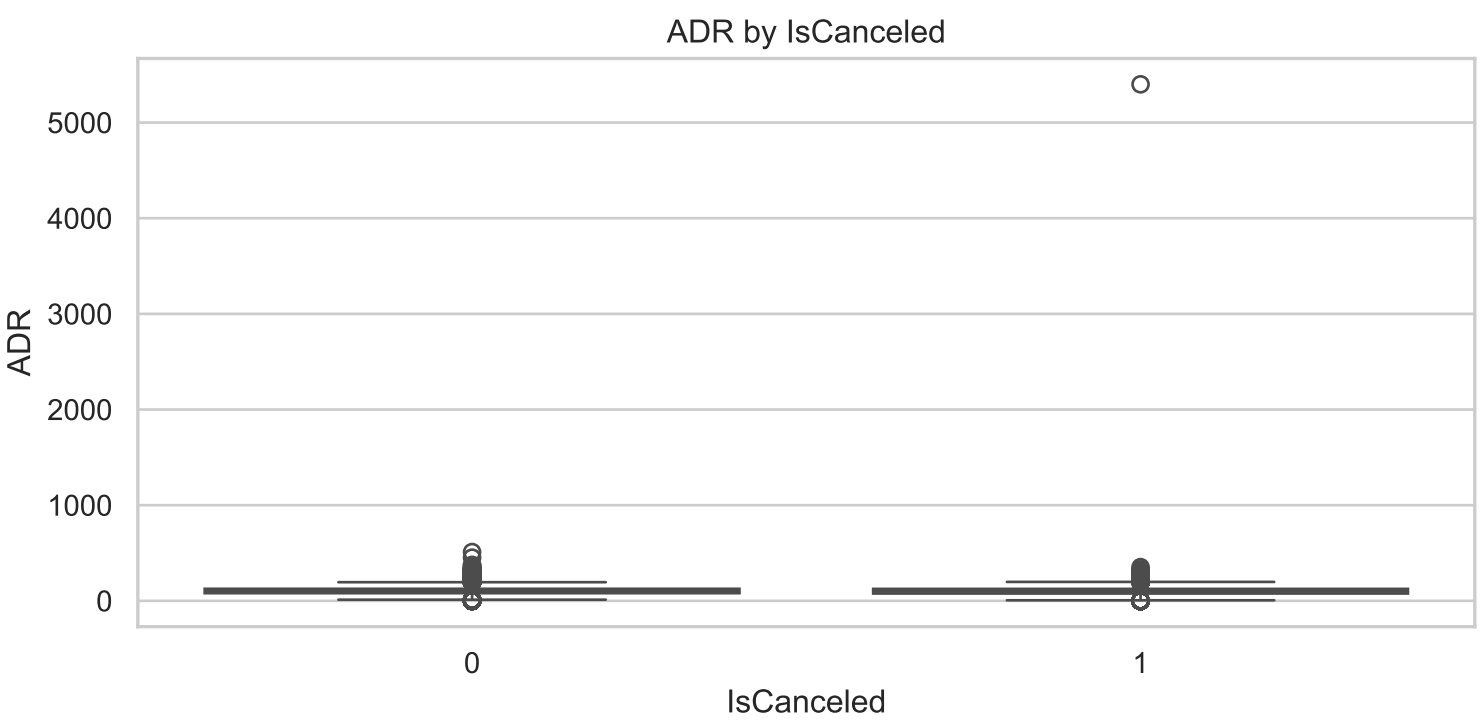

IsRepeatedGuest by IsCanceled
1.0
0.8
0.6
0.4
0.2
0.0
IsRepeatedGuest
0
1
IsCanceled

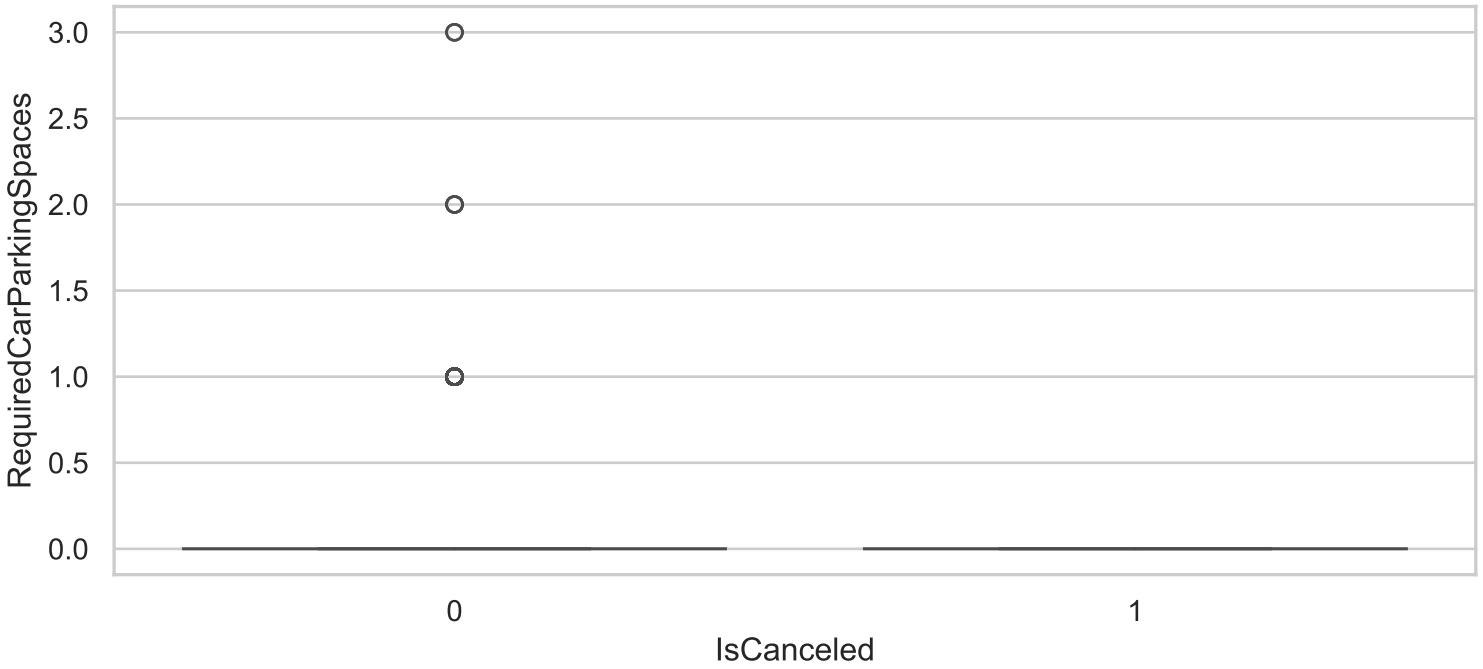

PreviousCancellations by IsCanceled
PreviousCancellations
20
15
10
5
0
0
1
IsCanceled

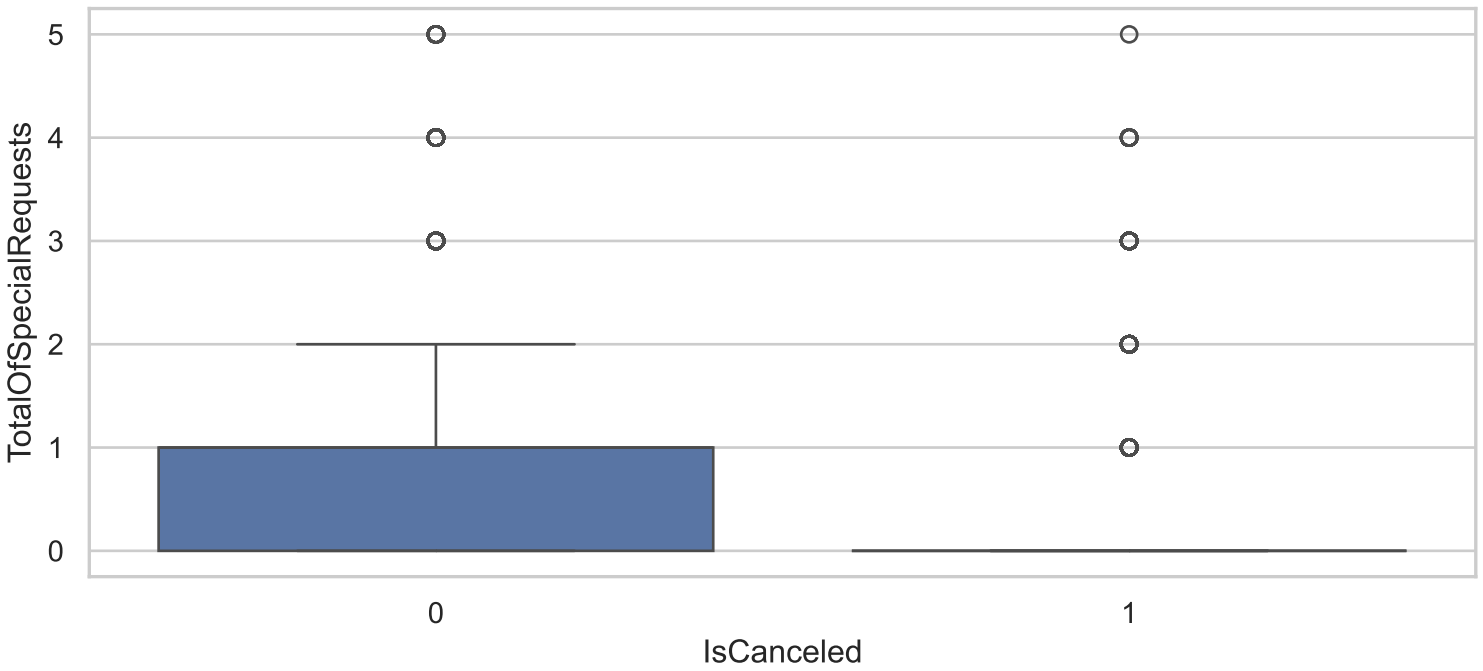

PreviousBookingsNotCanceled by IsCanceled
PreviousBookingsNotCanceled
70
60
50
40
30
20
10
0
0
1
IsCanceled

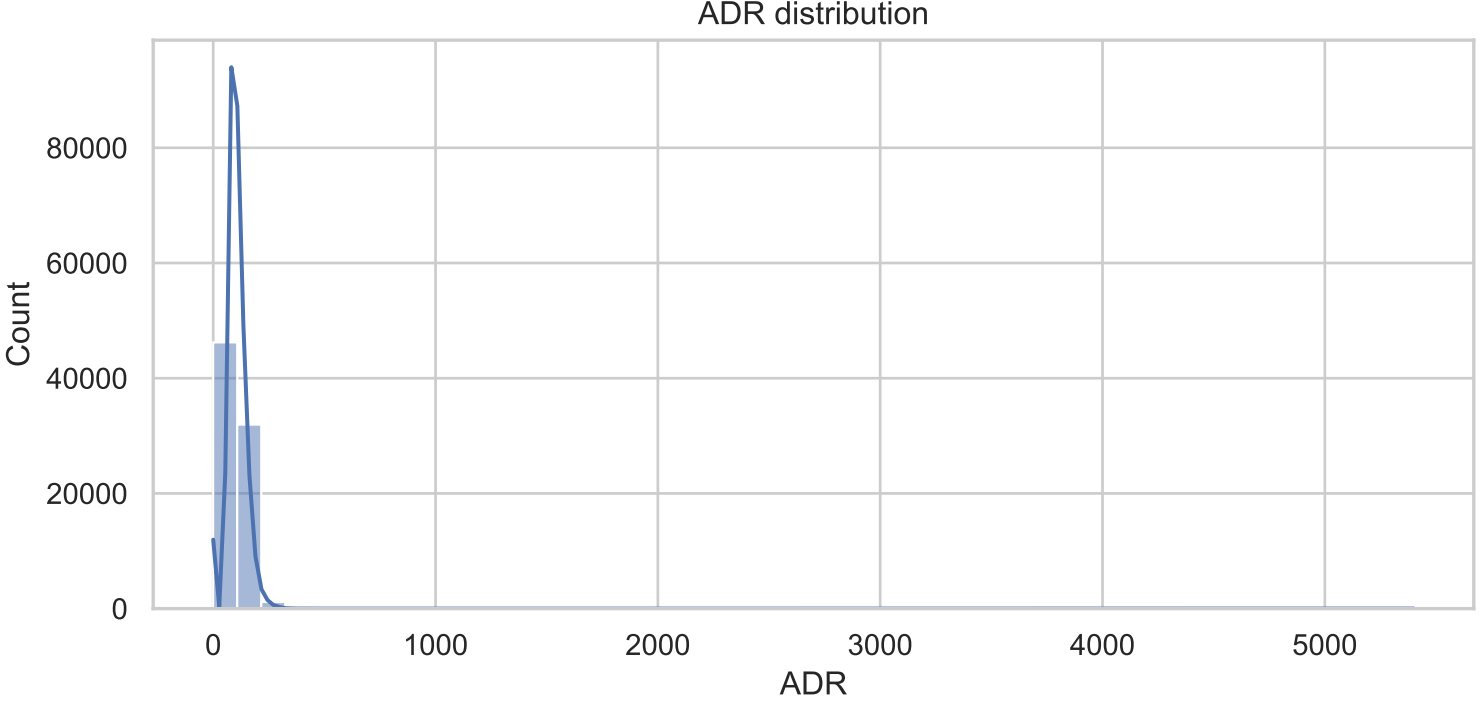

BookingChanges by IsCanceled
BookingChanges
20
15
10
5
0
0
1
IsCanceled

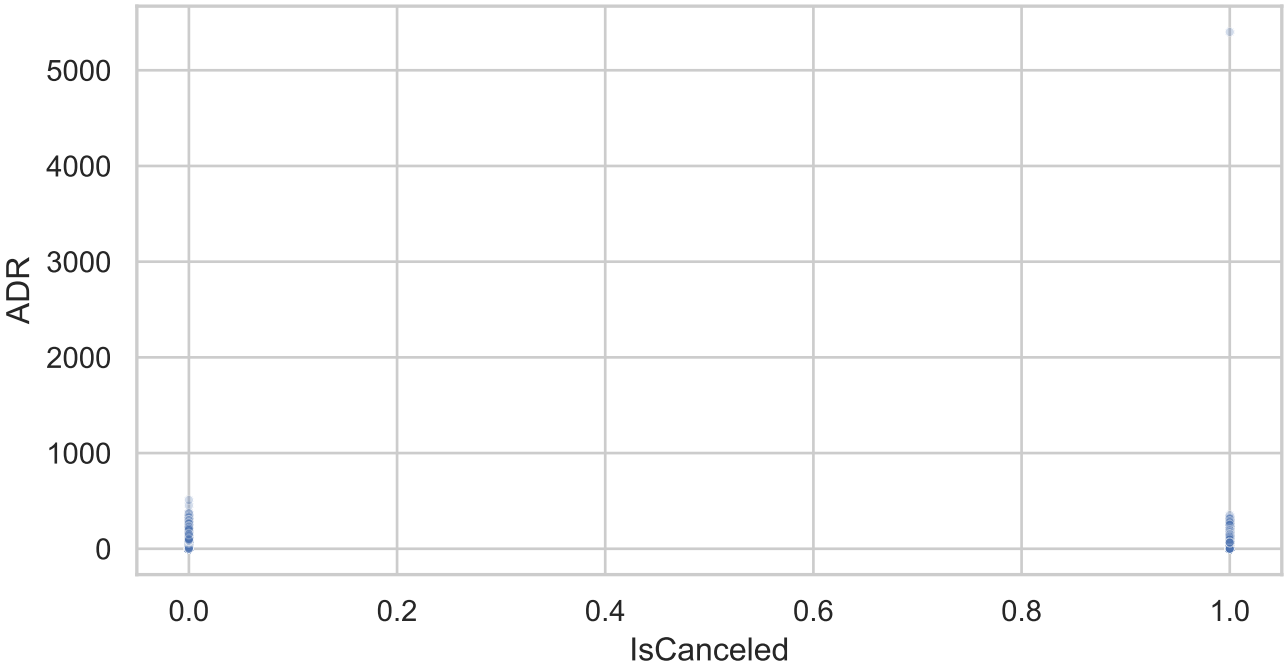

DaysInWaitingList by IsCanceled
DaysInWaitingList
400
350
300
250
200
150
100
50
0
0
1
IsCanceled

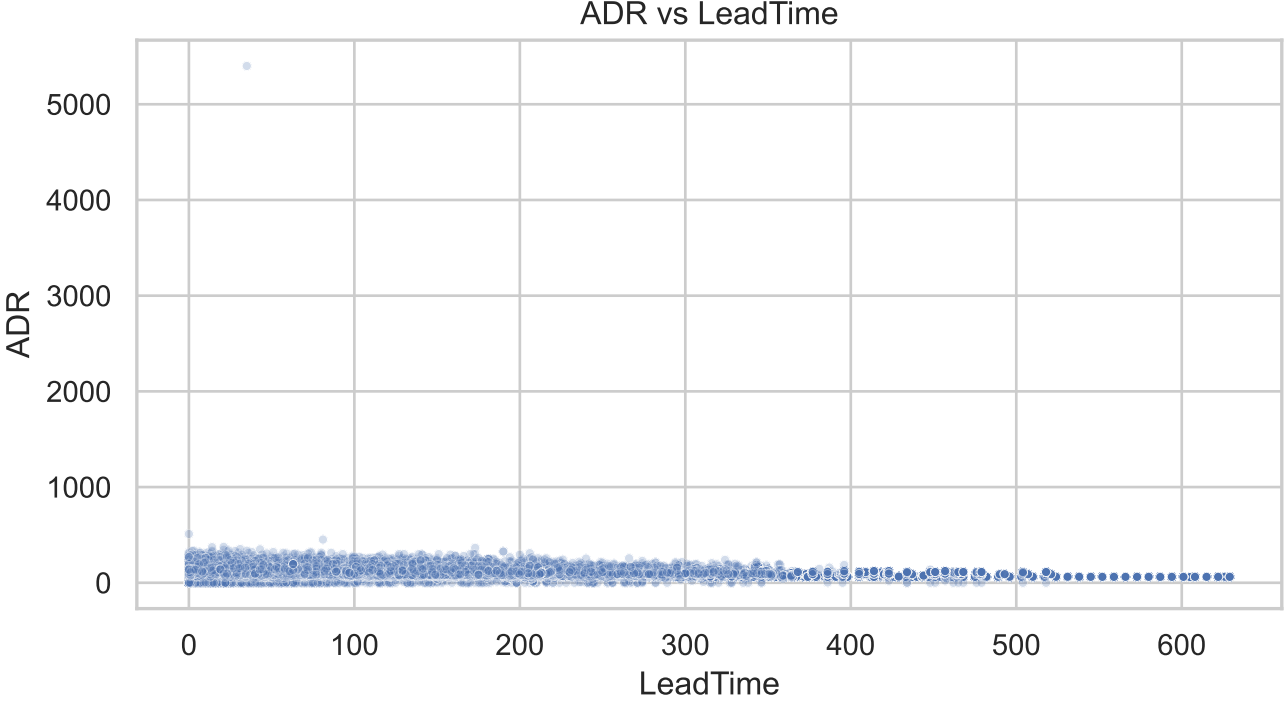

ADR by IsCanceled
ADR
5000
4000
3000
2000
1000
0
0
1
IsCanceled

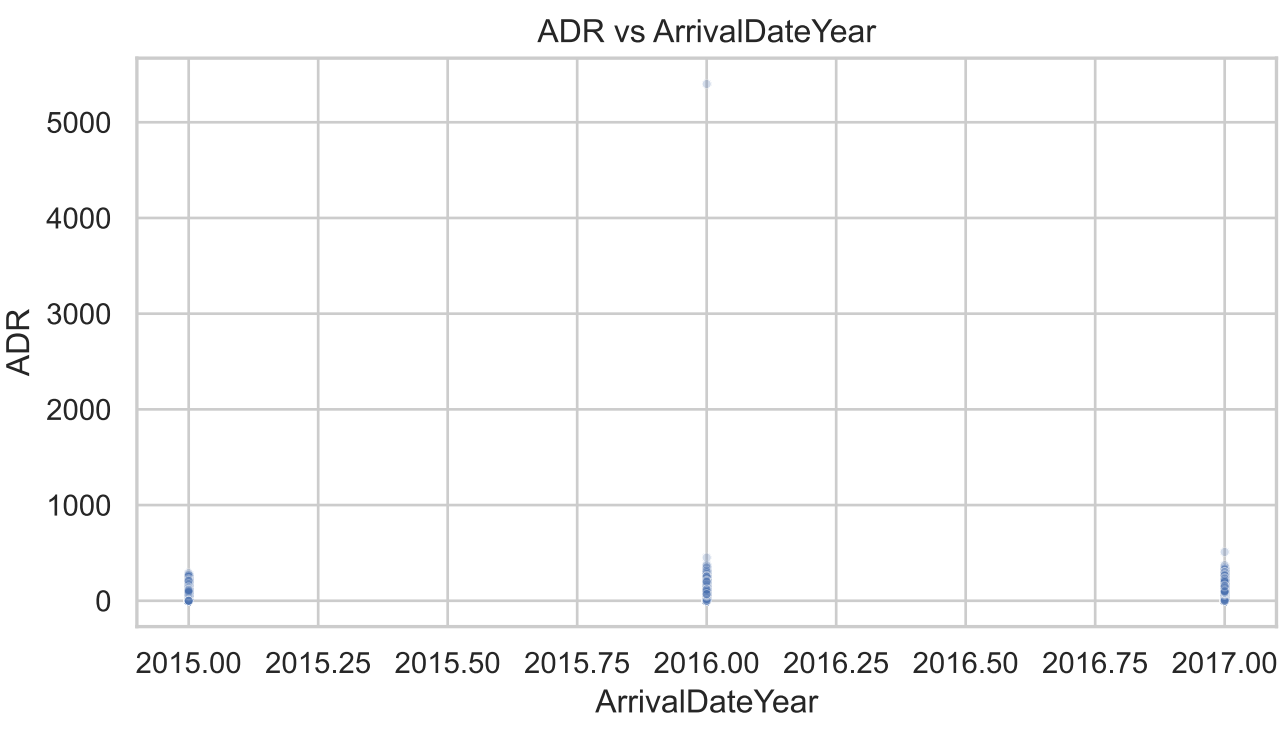

RequiredCarParkingSpaces by IsCanceled
RequiredCarParkingSpaces
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
1
IsCanceled

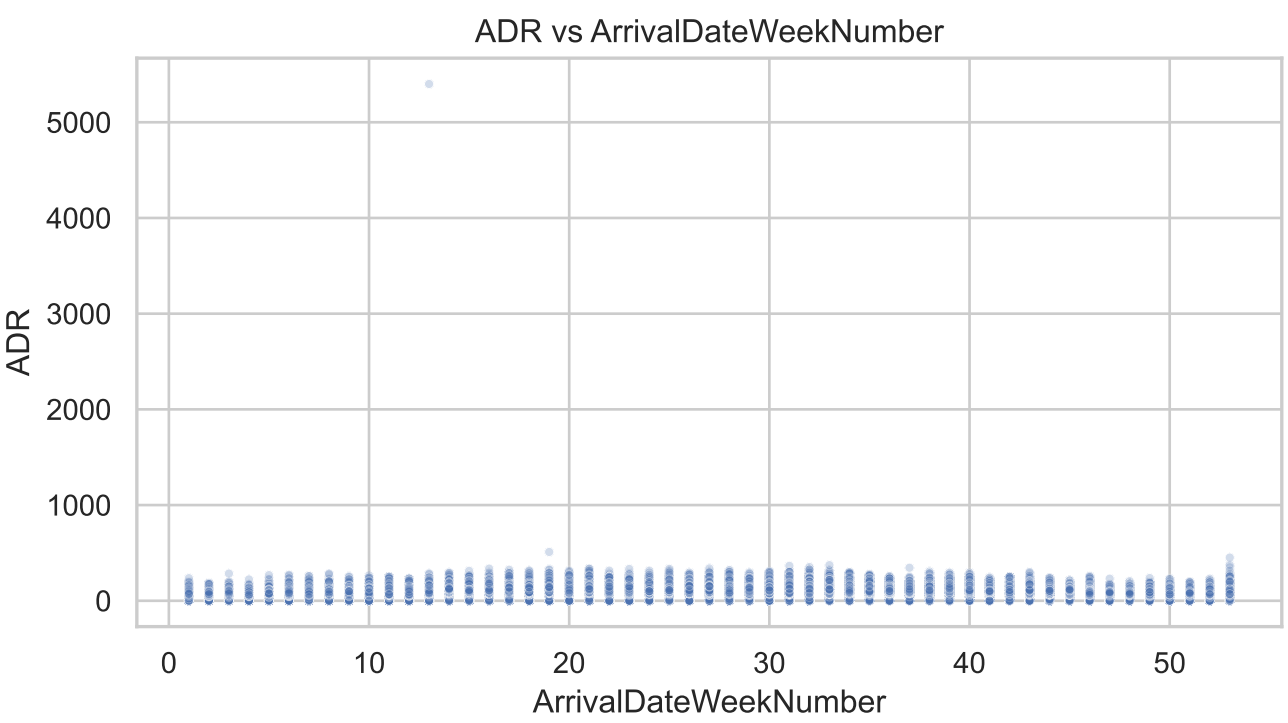

TotalOfSpecialRequests by IsCanceled
TotalOfSpecialRequests
0
1
2
3
4
5
0
1
IsCanceled

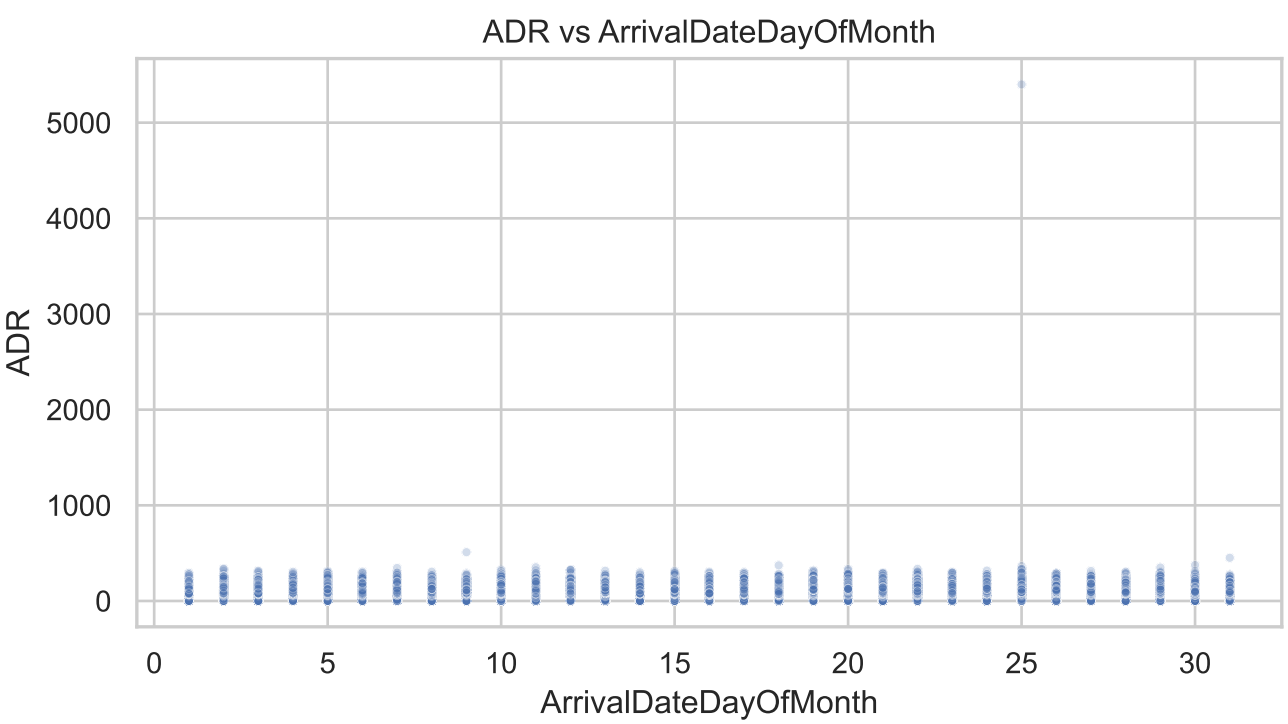

ADR distribution
Count
80000
60000
40000
20000
0
0
1000
2000
3000
4000
5000
ADR

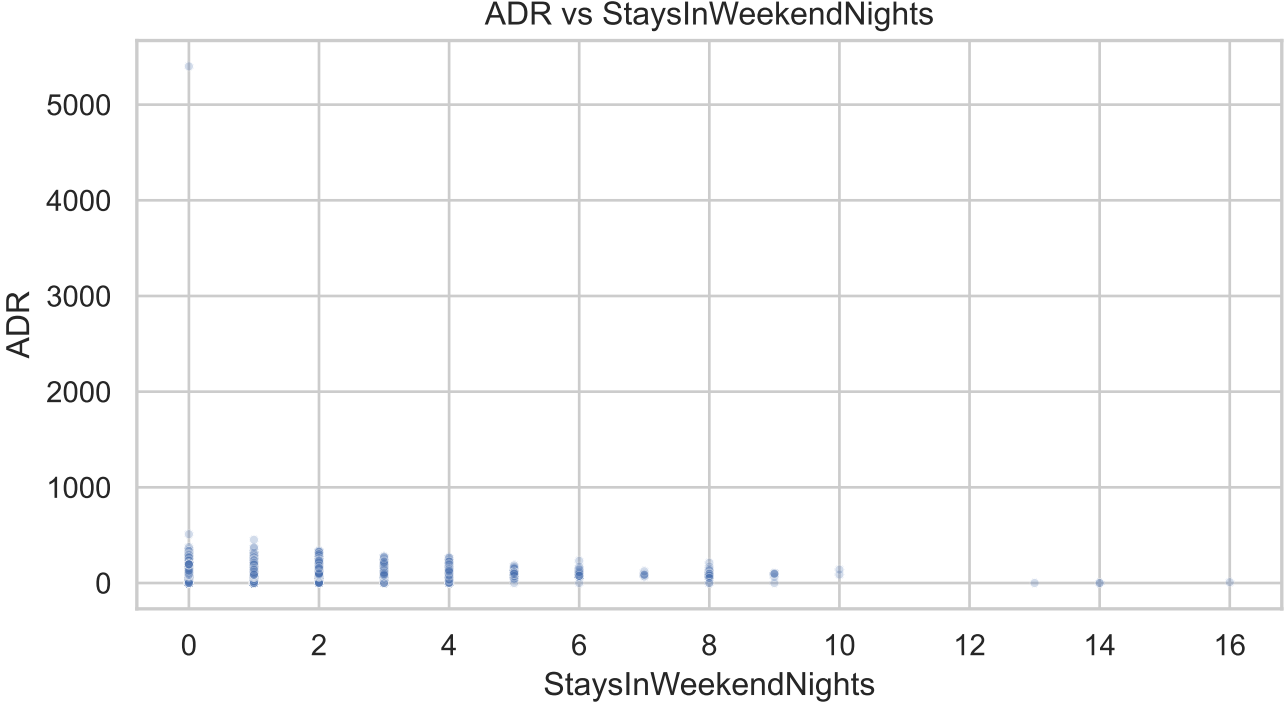

ADR vs IsCanceled
ADR
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
IsCanceled

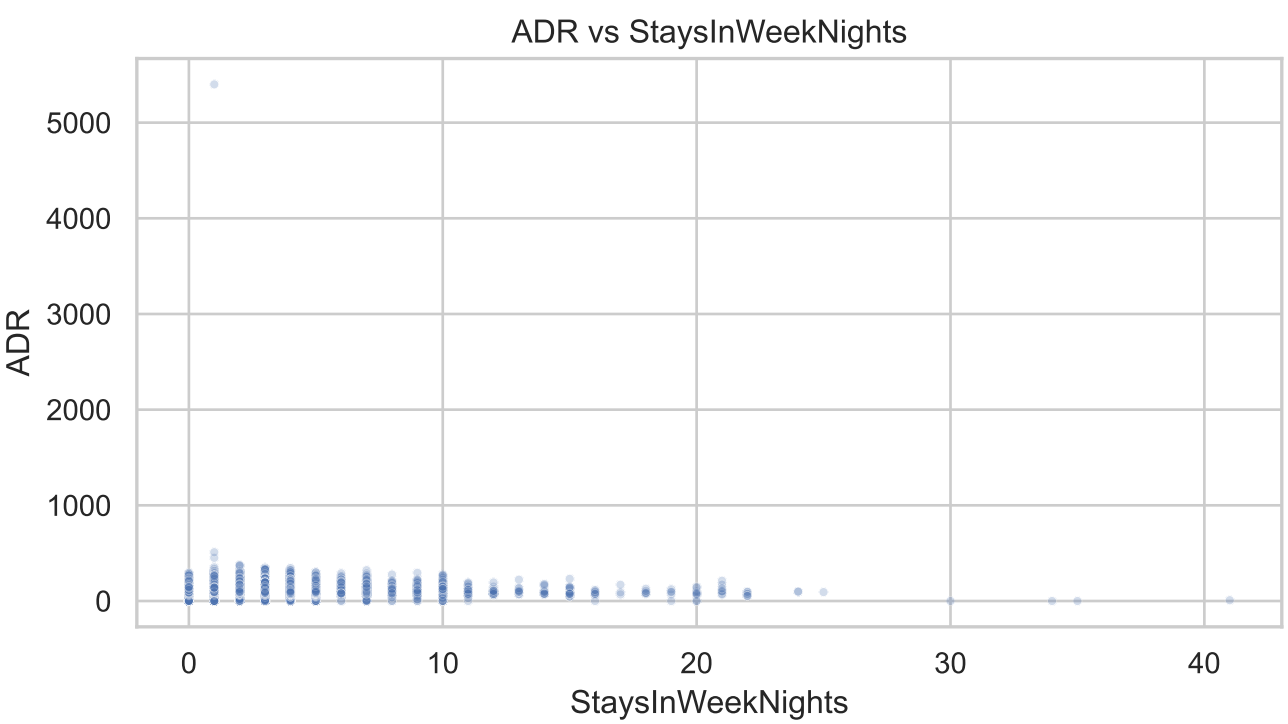

ADR vs LeadTime
ADR
LeadTime
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
600

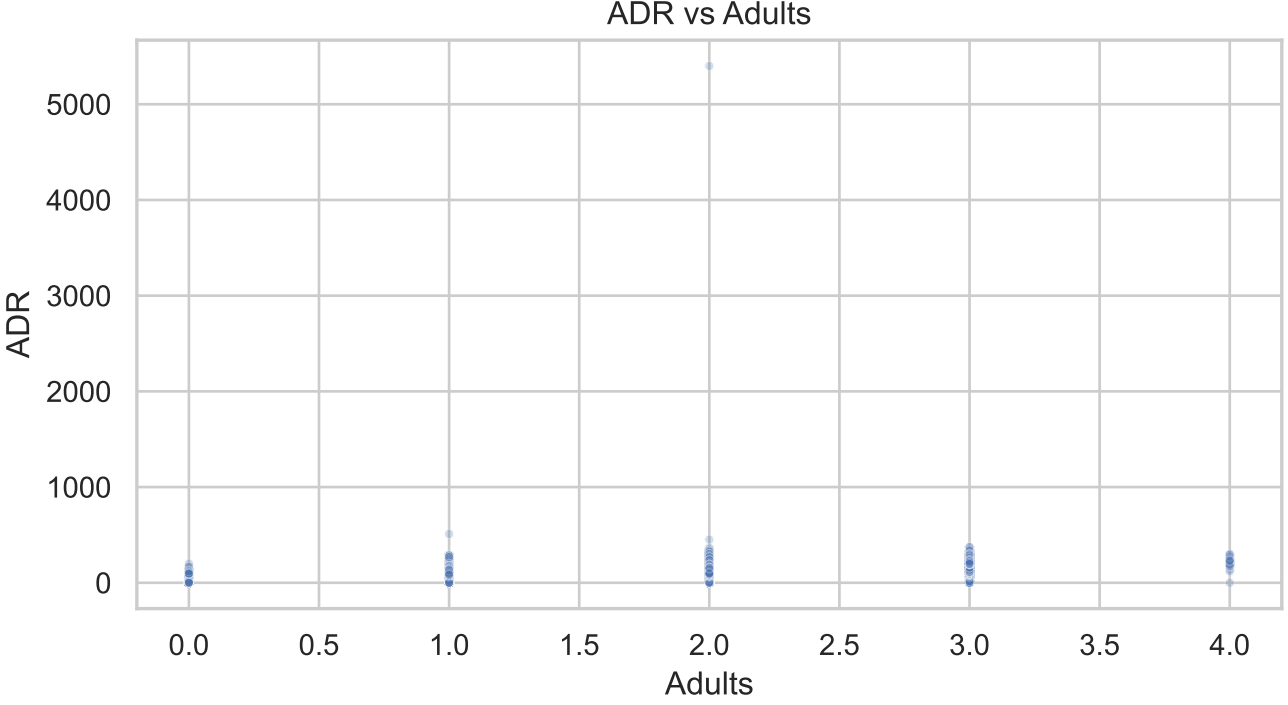

ADR vs ArrivalDateYear
ADR
5000
4000
3000
2000
1000
0
2015.00
2015.25
2015.50
2015.75
2016.00
2016.25
2016.50
2016.75
2017.00
ArrivalDateYear

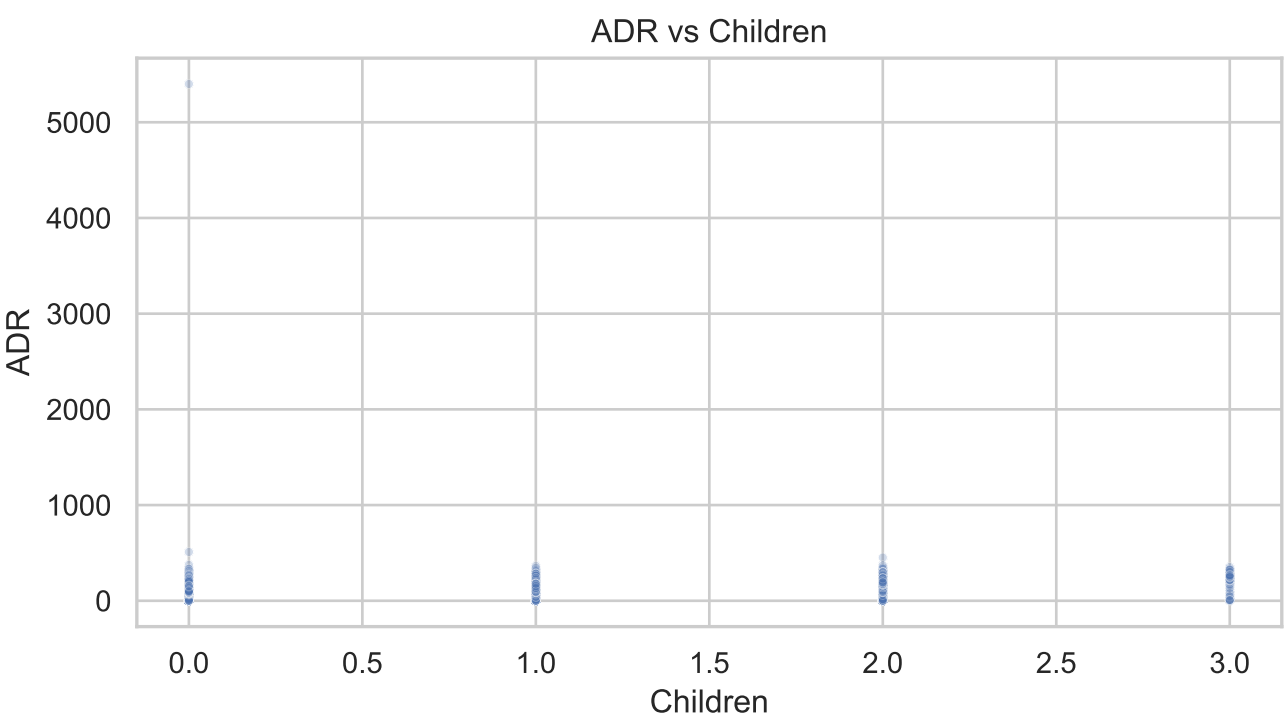

ADR vs ArrivalDateWeekNumber
ADR
5000
4000
3000
2000
1000
0
0
10
20
30
40
50
ArrivalDateWeekNumber

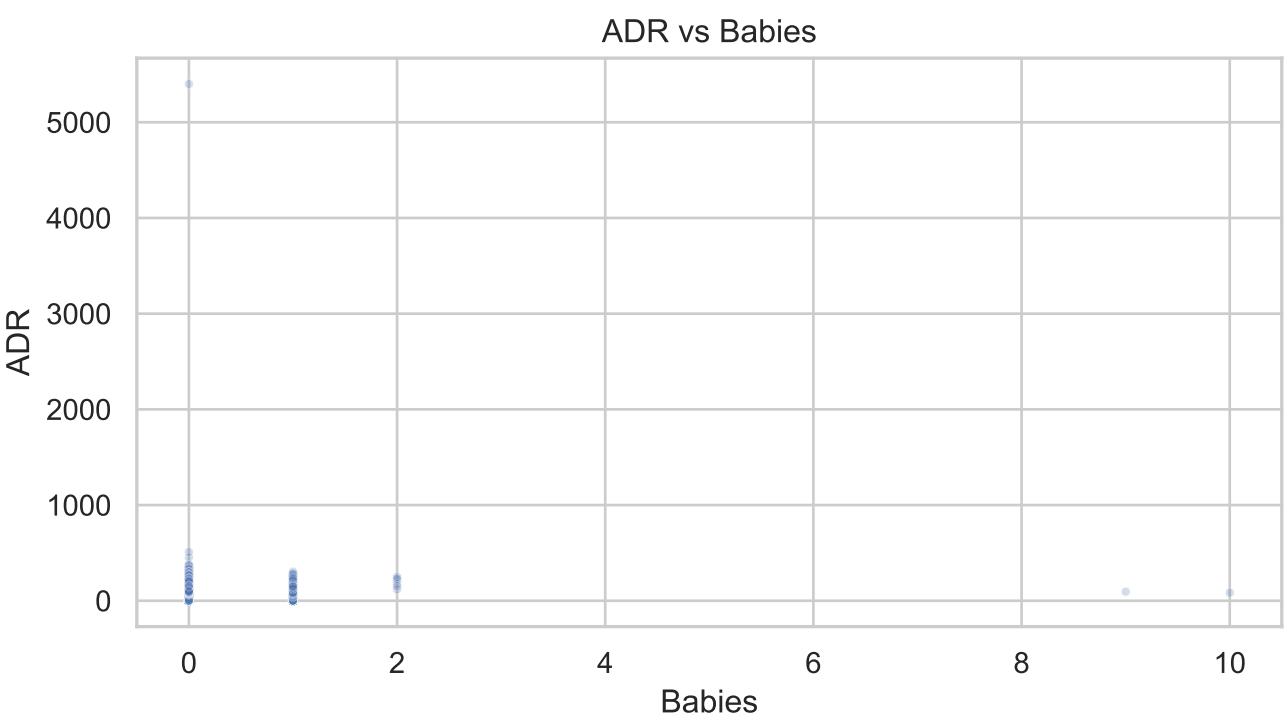

ADR vs ArrivalDateDayOfMonth
ADR
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
ArrivalDateDayOfMonth

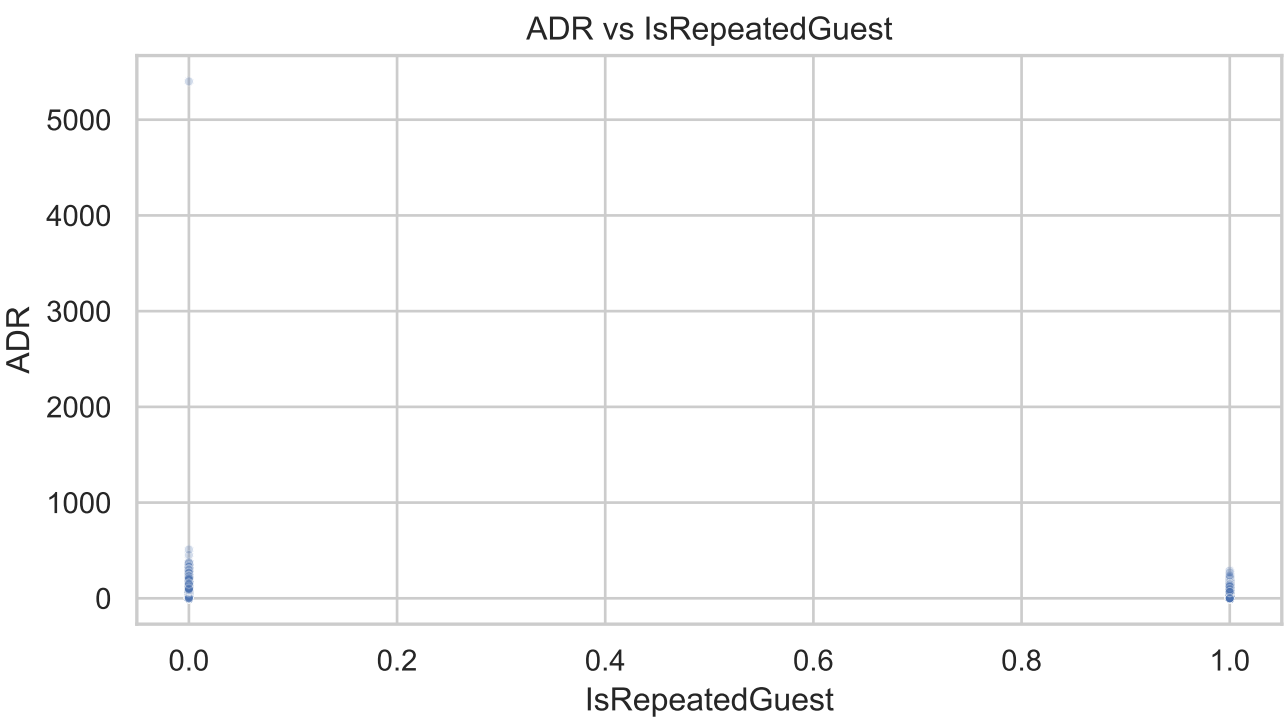

ADR vs StaysInWeekendNights
ADR
5000
4000
3000
2000
1000
0
0
2
4
6
8
10
12
14
16
StaysInWeekendNights

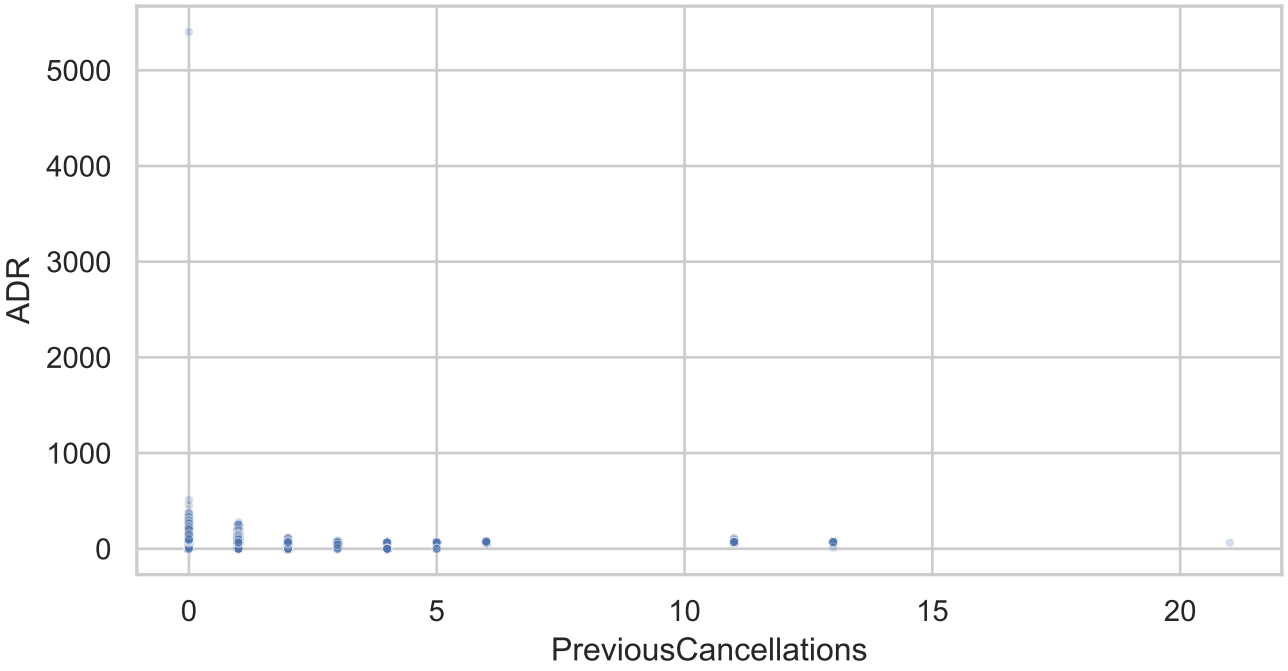

ADR vs StaysInWeekNights
ADR
5000
4000
3000
2000
1000
0
0
10
20
30
40
StaysInWeekNights

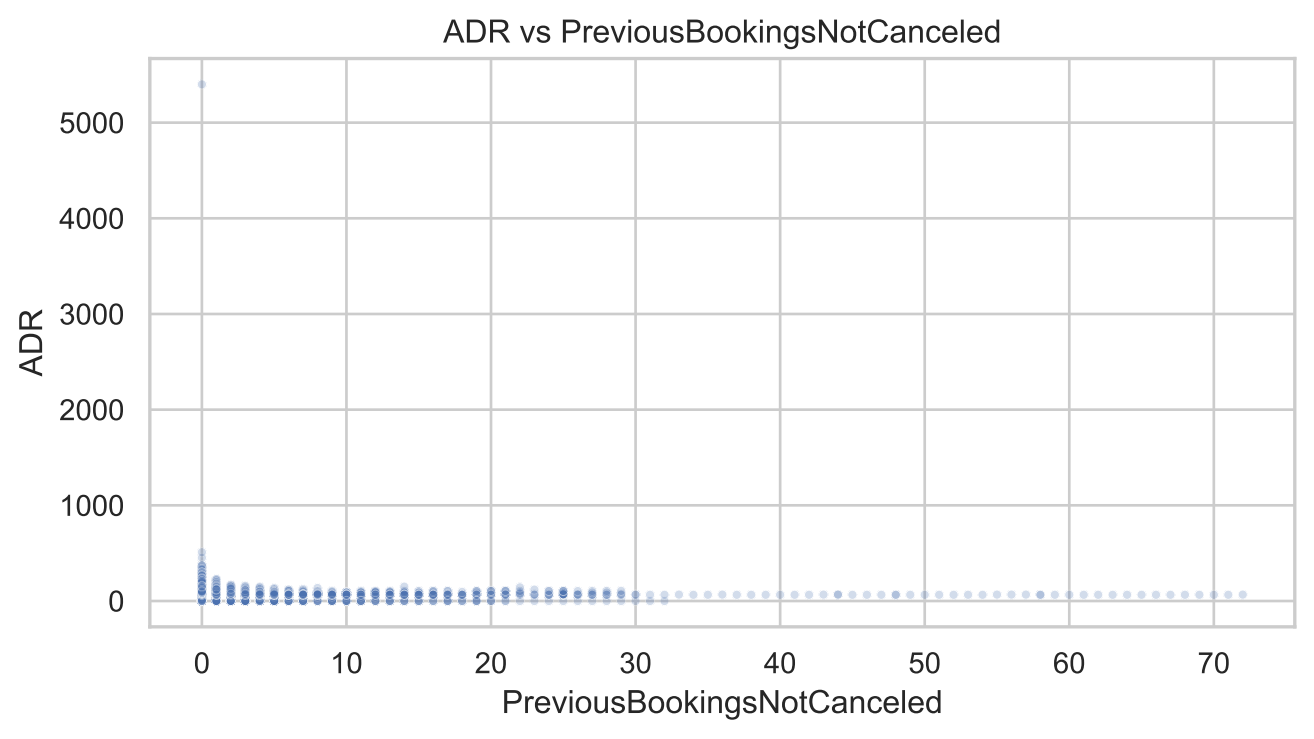

ADR vs Adults
ADR
5000
4000
3000
2000
1000
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Adults

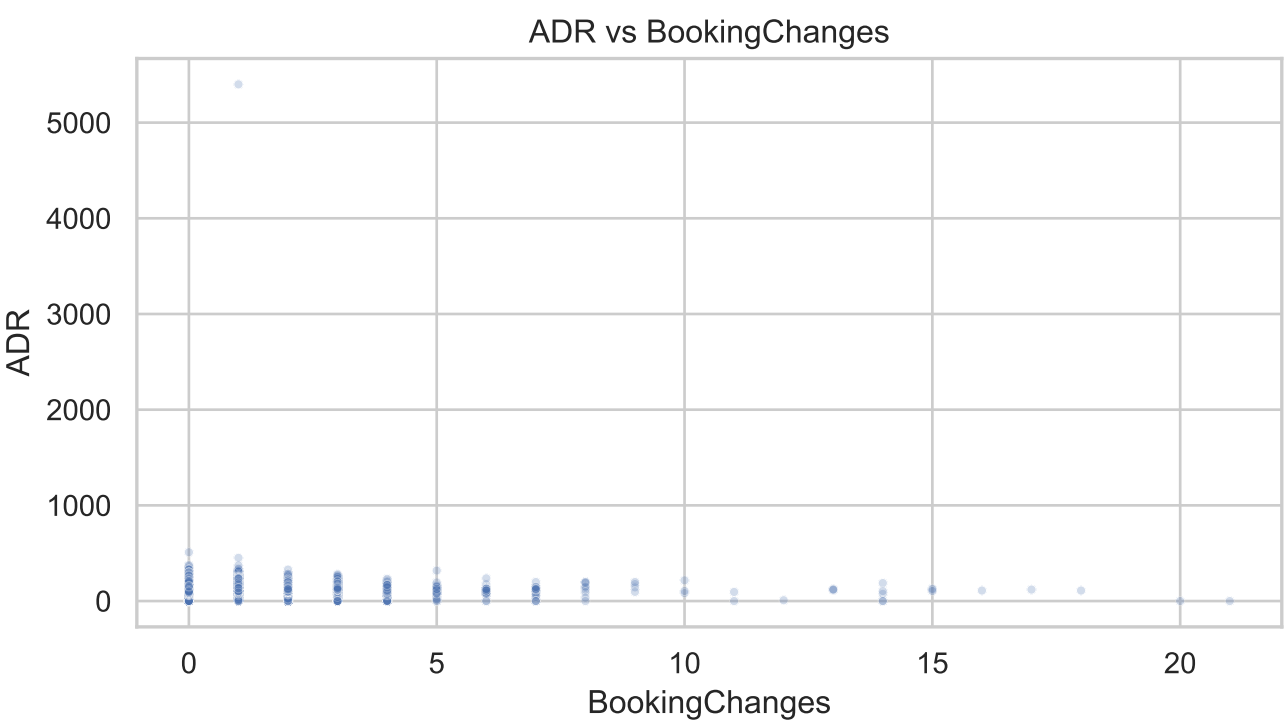

ADR vs Children
ADR
5000
4000
3000
2000
1000
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Children

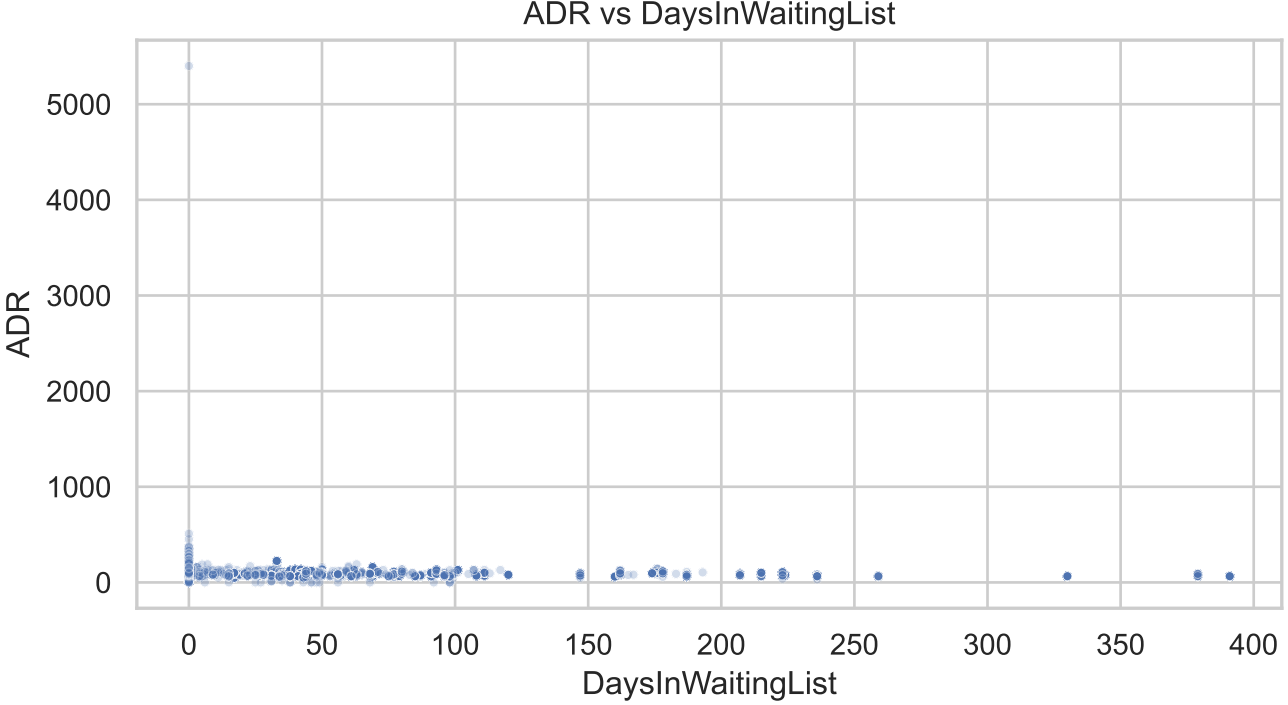

ADR vs Babies
ADR
5000
4000
3000
2000
1000
0
0
2
4
6
8
10
Babies

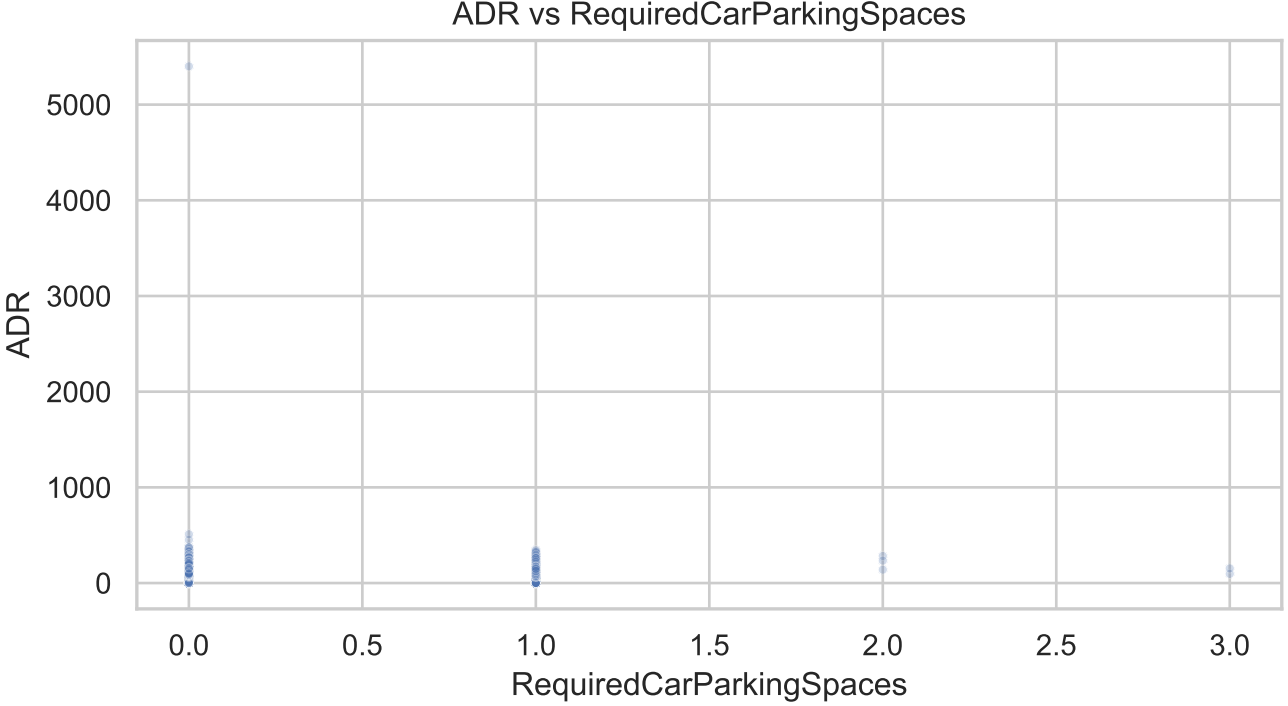

ADR vs IsRepeatedGuest
ADR
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
IsRepeatedGuest

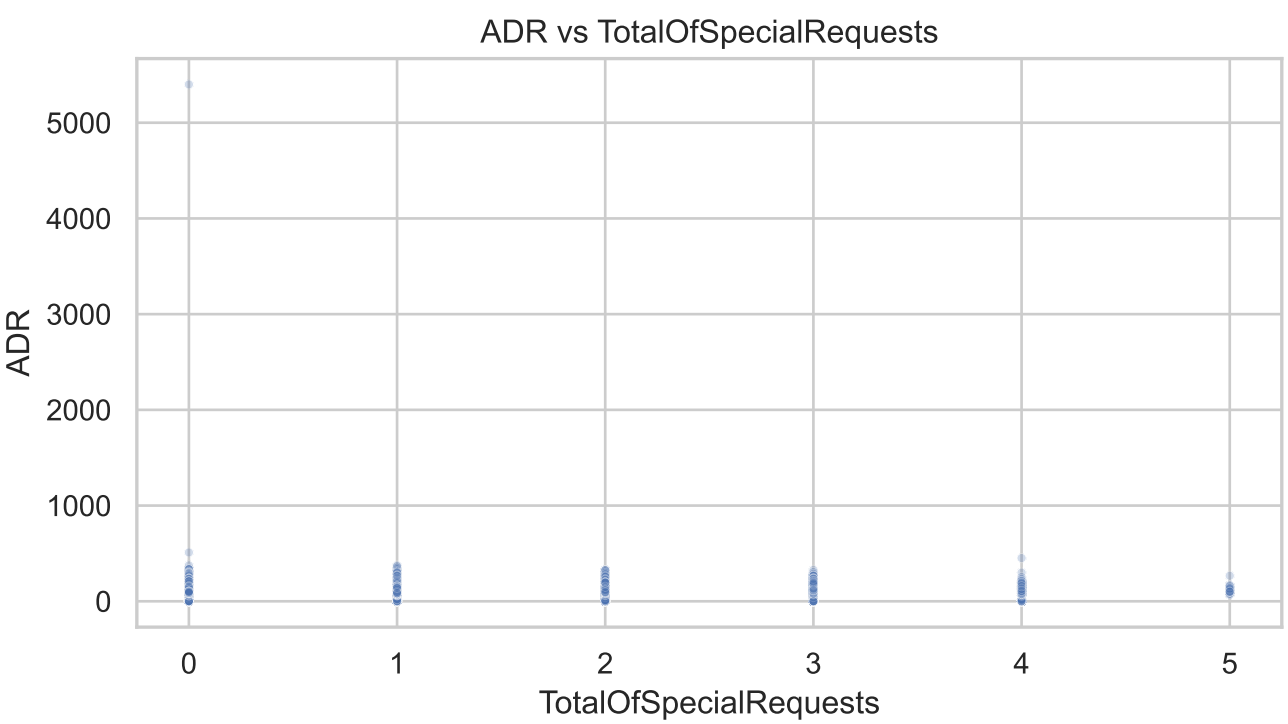

ADR vs PreviousCancellations
ADR
5000
4000
3000
2000
1000
0
0
5
10
15
20
PreviousCancellations

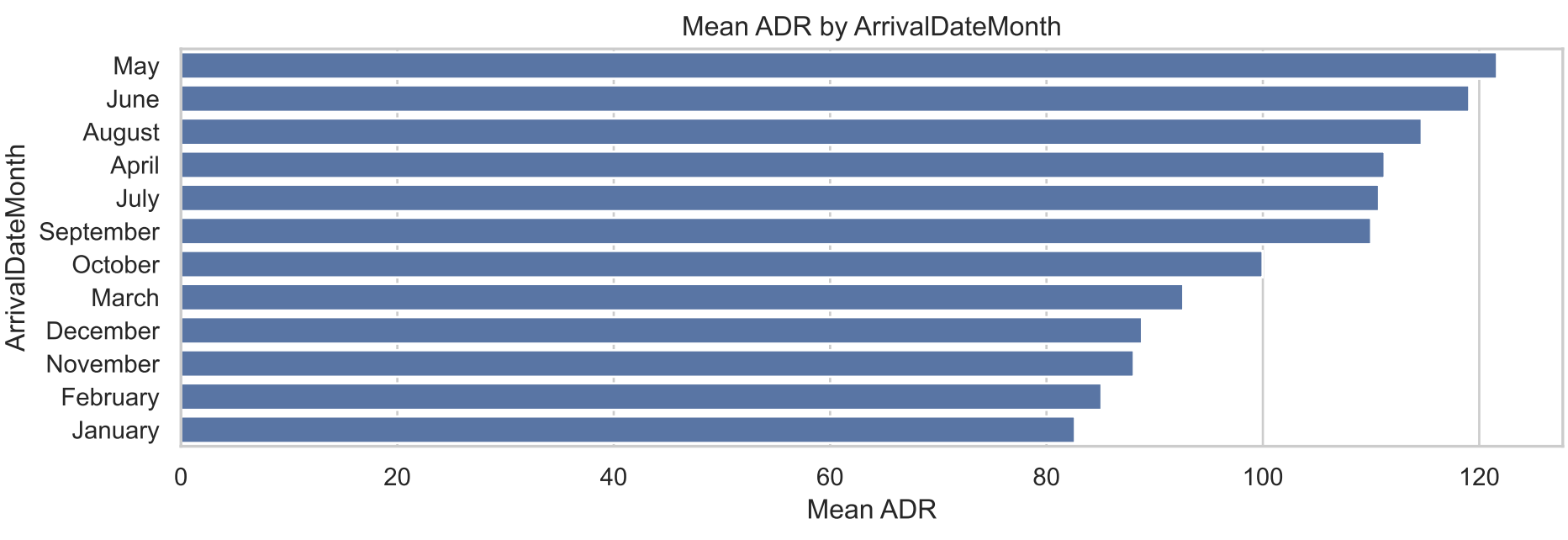

ADR vs PreviousBookingsNotCanceled
ADR
5000
4000
3000
2000
1000
0
0
10
20
30
40
50
60
70
PreviousBookingsNotCanceled

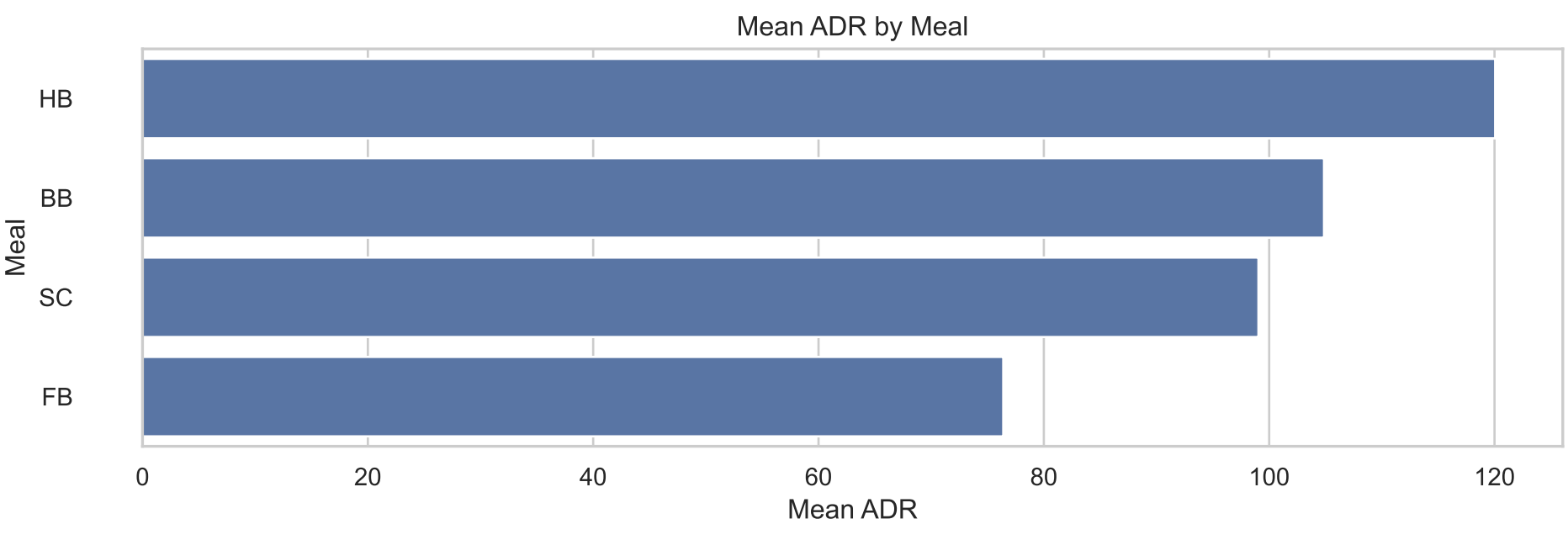

ADR vs BookingChanges
ADR
5000
4000
3000
2000
1000
0
0
5
10
15
20
BookingChanges

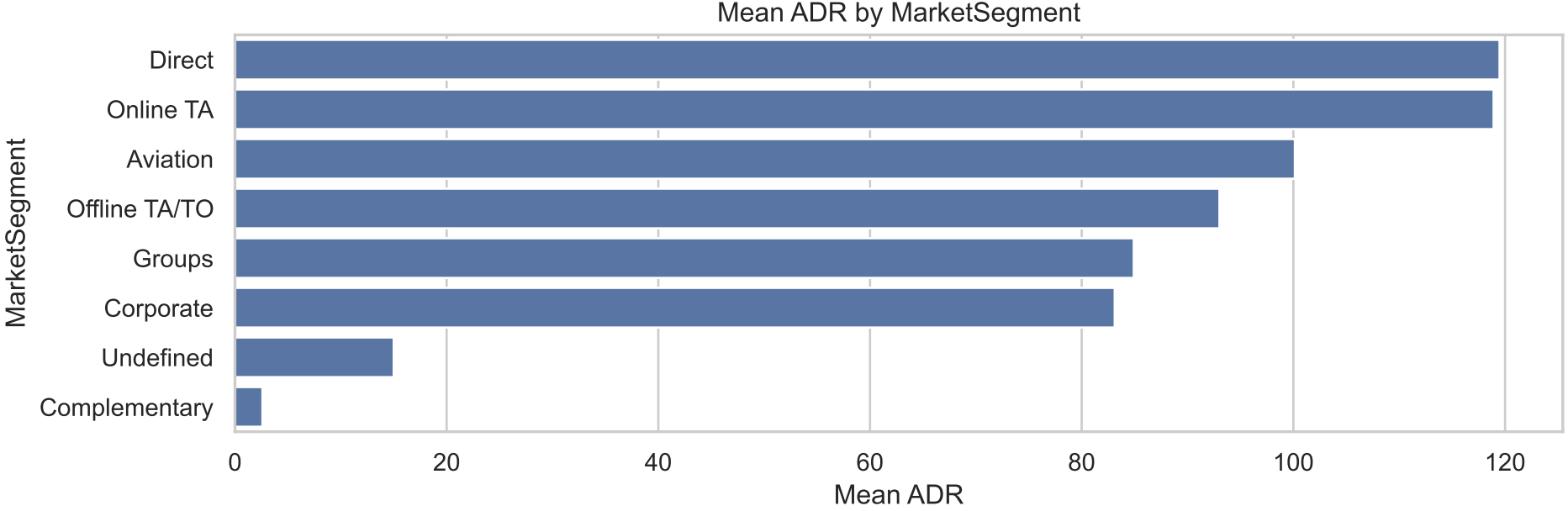

ADR vs DaysInWaitingList
ADR
5000
4000
3000
2000
1000
0
0
50
100
150
200
250
300
350
400
DaysInWaitingList

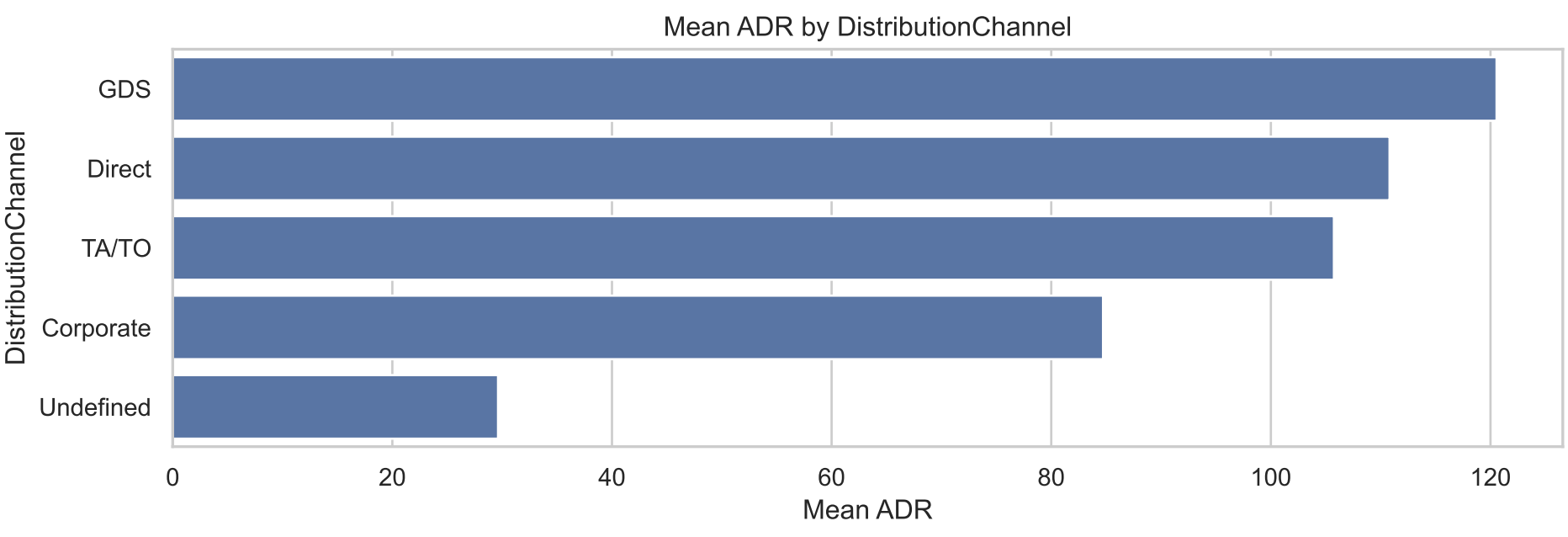

ADR vs RequiredCarParkingSpaces
ADR
5000
4000
3000
2000
1000
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
RequiredCarParkingSpaces

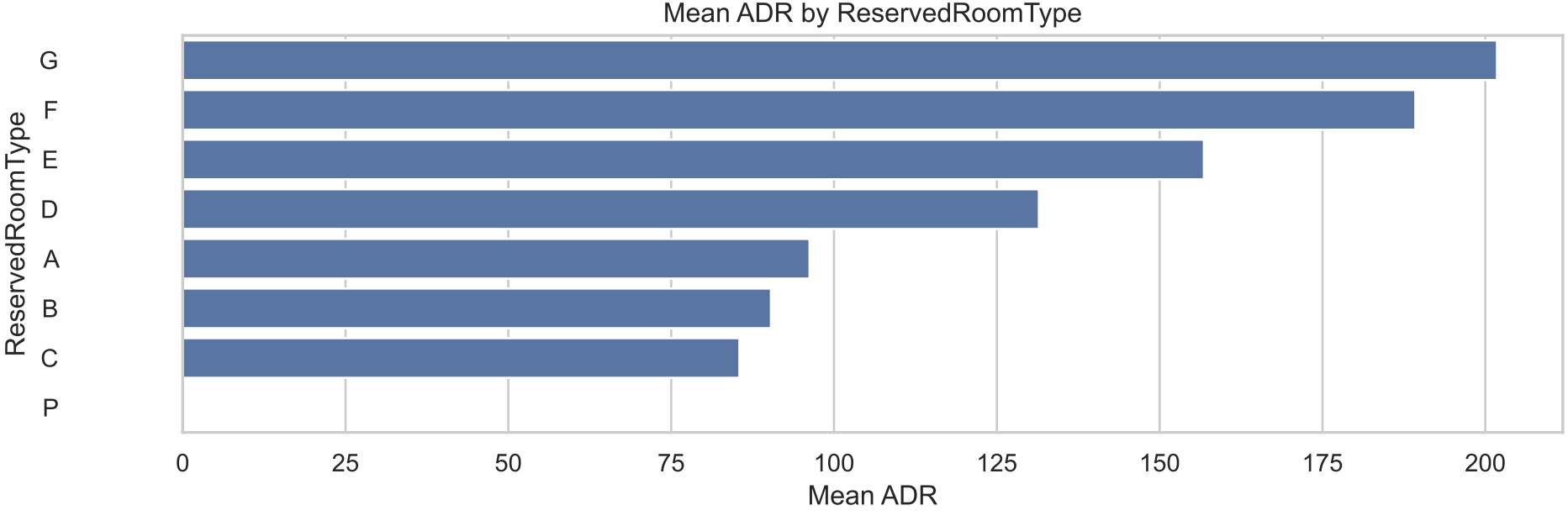

ADR vs TotalOfSpecialRequests
ADR
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
TotalOfSpecialRequests

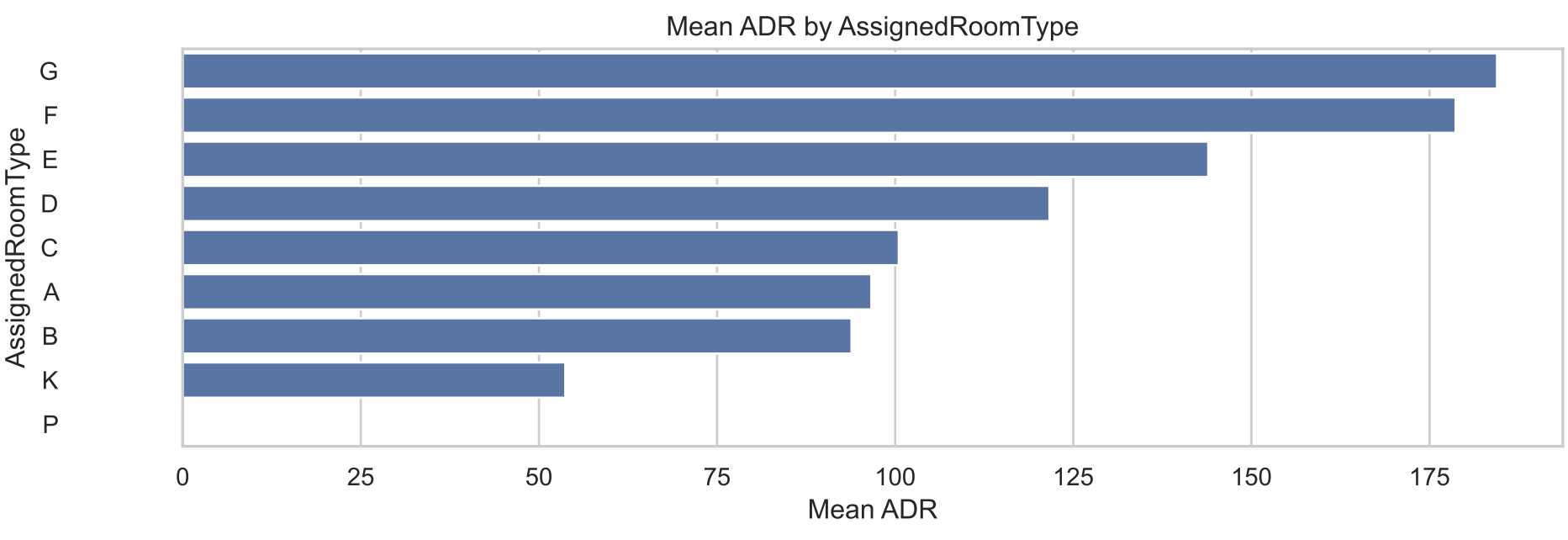

Mean ADR by ArrivalDateMonth
ArrivalDateMonth
May
June
August
April
July
September
October
March
December
November
February
January
0
20
40
60
80
100
120
Mean ADR

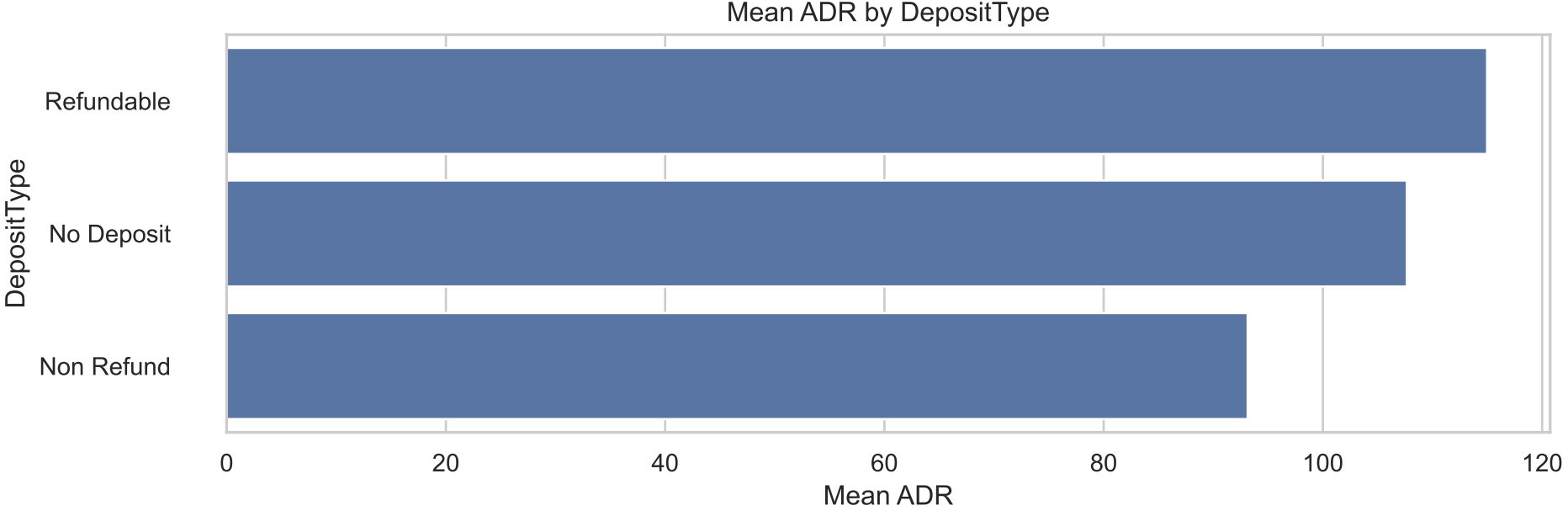

Mean ADR by Meal
Meal
HB
BB
SC
FB
0
20
40
60
80
100
120
Mean ADR

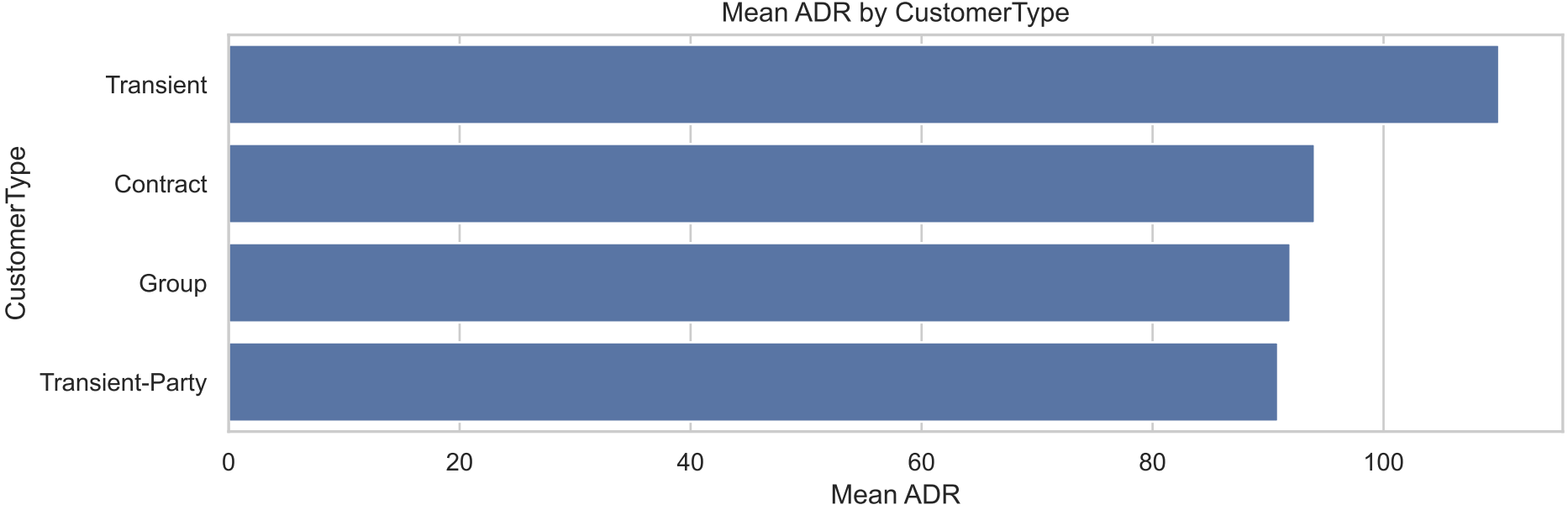

Mean ADR by MarketSegment
MarketSegment
Direct
Online TA
Aviation
Offline TA/TO
Groups
Corporate
Undefined
Complementary
0
20
40
60
80
100
120
Mean ADR

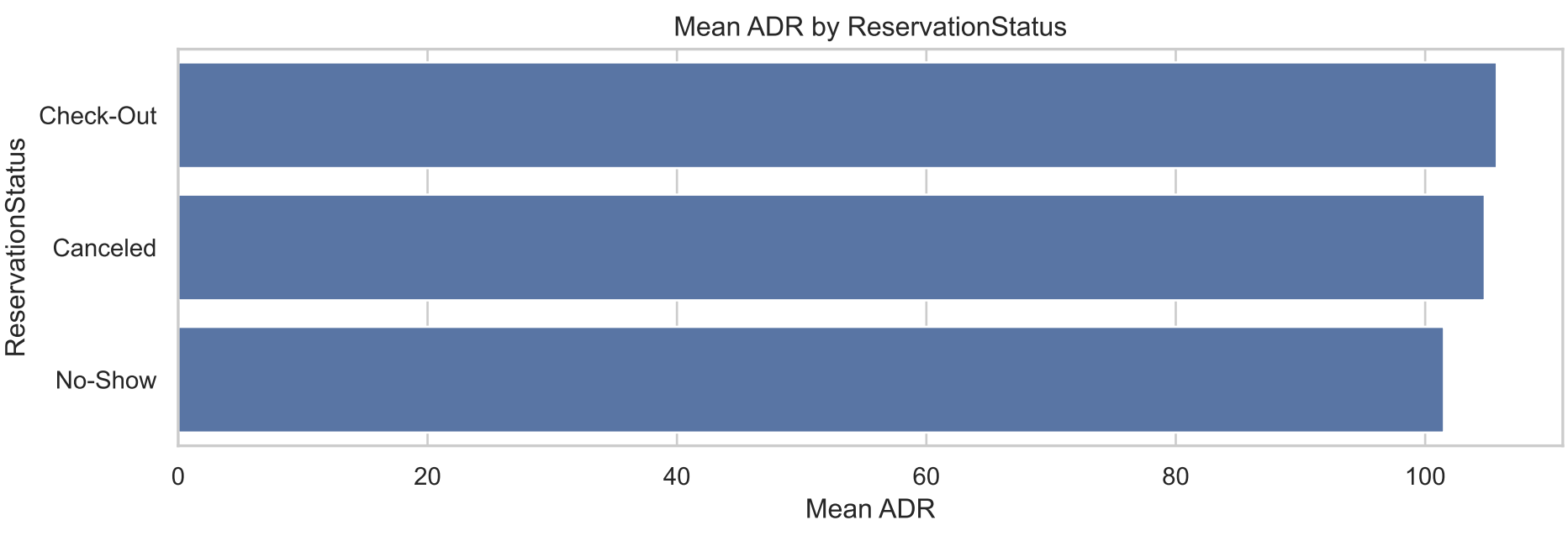

Mean ADR by DistributionChannel
DistributionChannel
GDS
Direct
TA/TO
Corporate
Undefined
0
20
40
60
80
100
120
Mean ADR


Mean ADR by ReservedRoomType
ReservedRoomType
G
F
E
D
A
B
C
P
0
25
50
75
100
125
150
175
200
Mean ADR


Mean ADR by AssignedRoomType
AssignedRoomType
G
F
E
D
C
A
B
K
P
0
25
50
75
100
125
150
175
Mean ADR

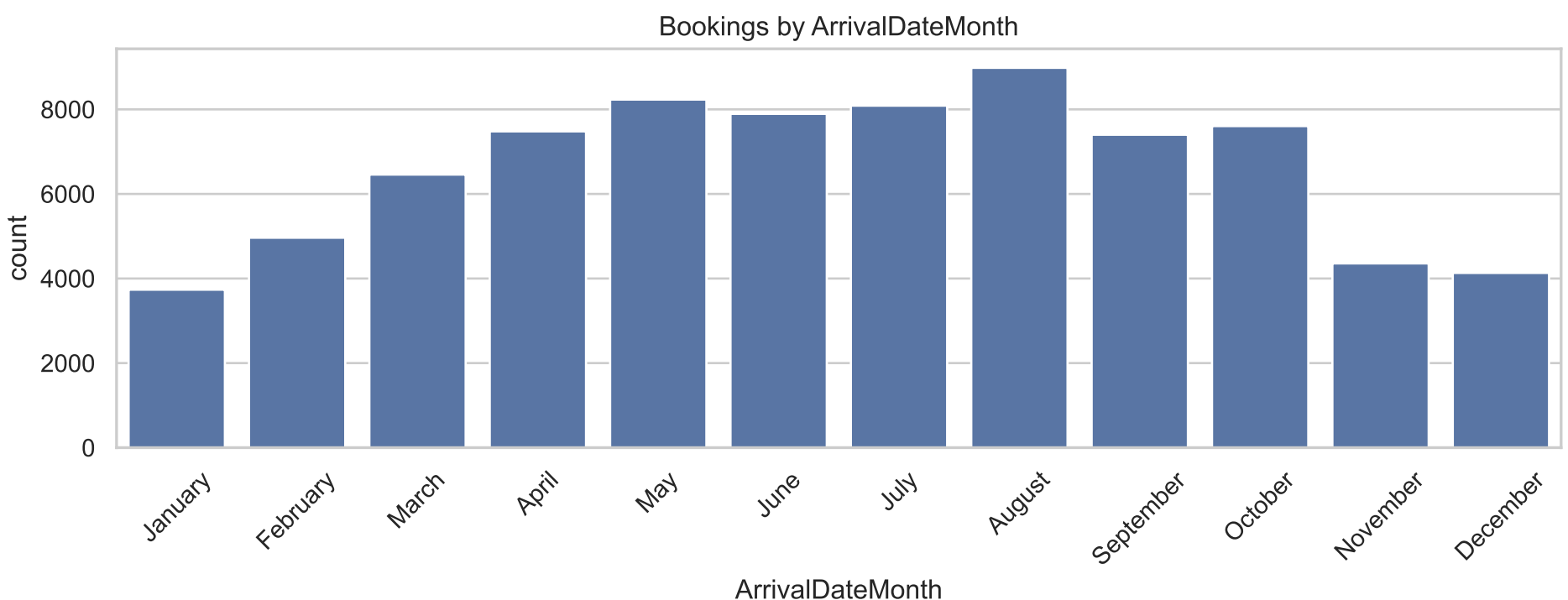

Mean ADR by DepositType
DepositType
Refundable
No Deposit
Non Refund
0
20
40
60
80
100
120
Mean ADR

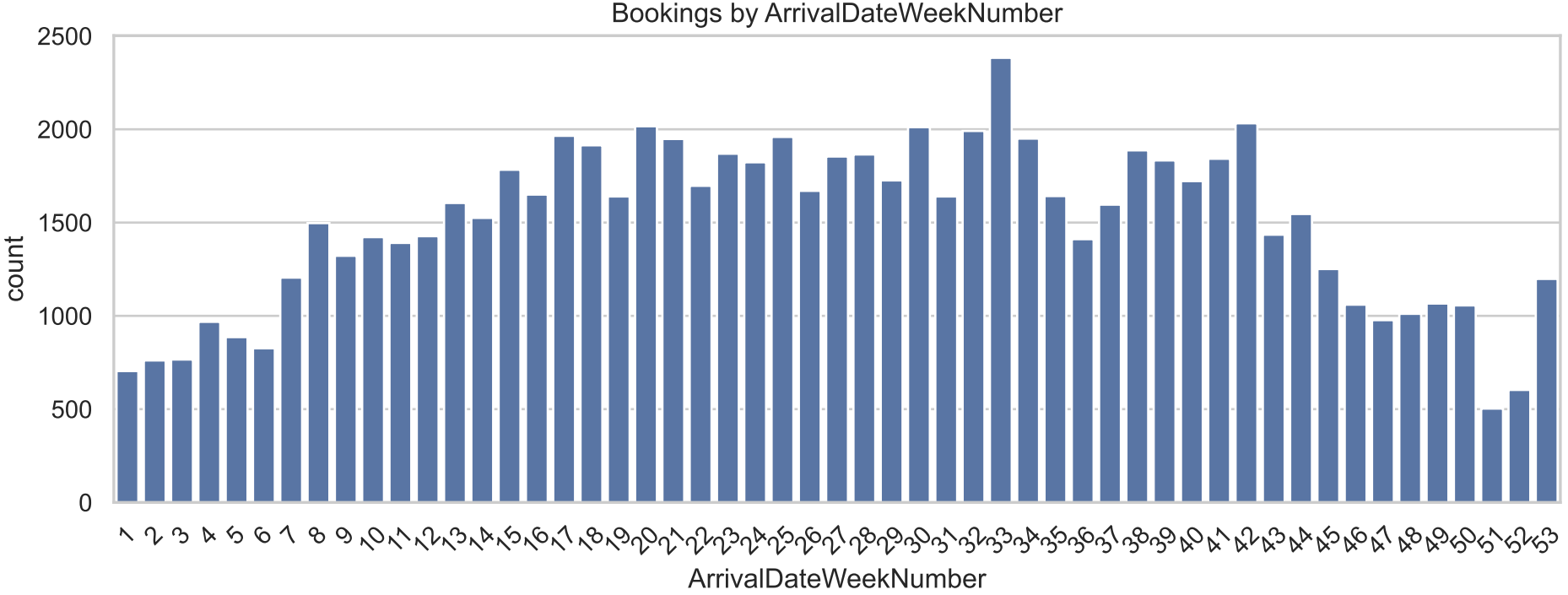

Mean ADR by CustomerType
CustomerType
Transient
Contract
Group
Transient-Party
0
20
40
60
80
100
Mean ADR

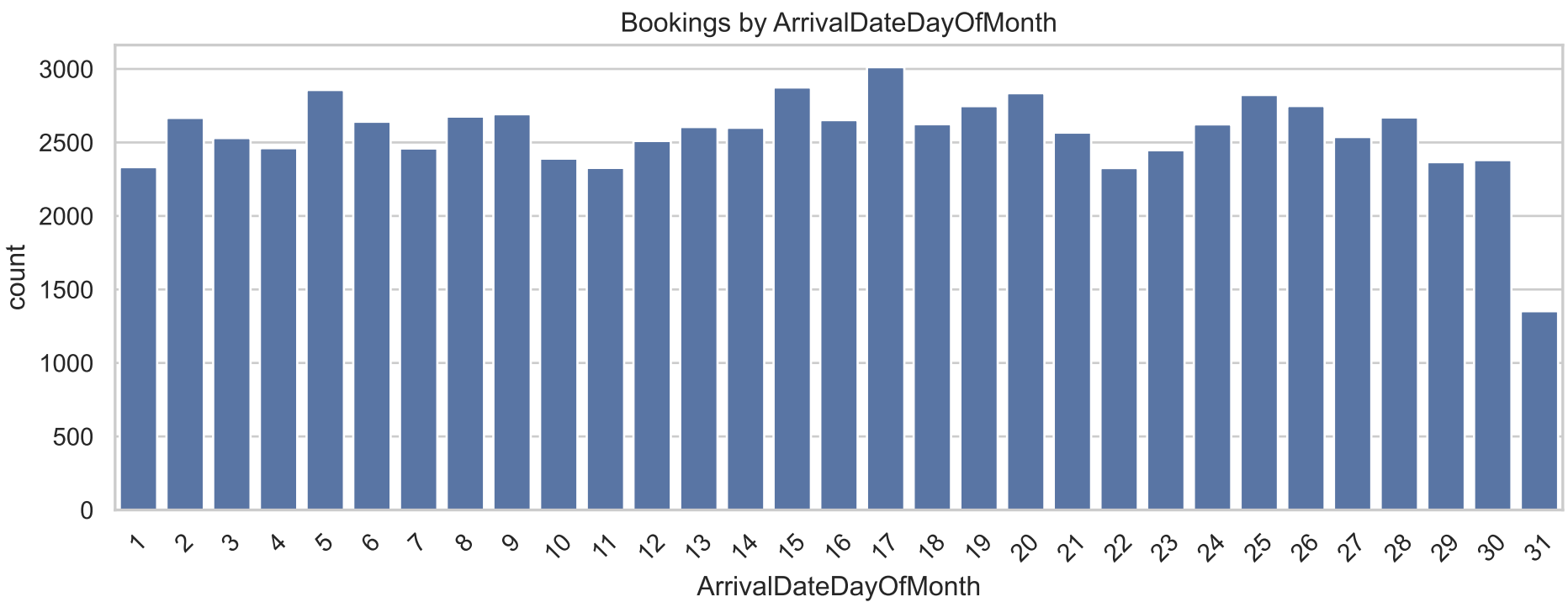

Mean ADR by ReservationStatus
ReservationStatus
Check-Out
Canceled
No-Show
0
20
40
60
80
100
Mean ADR

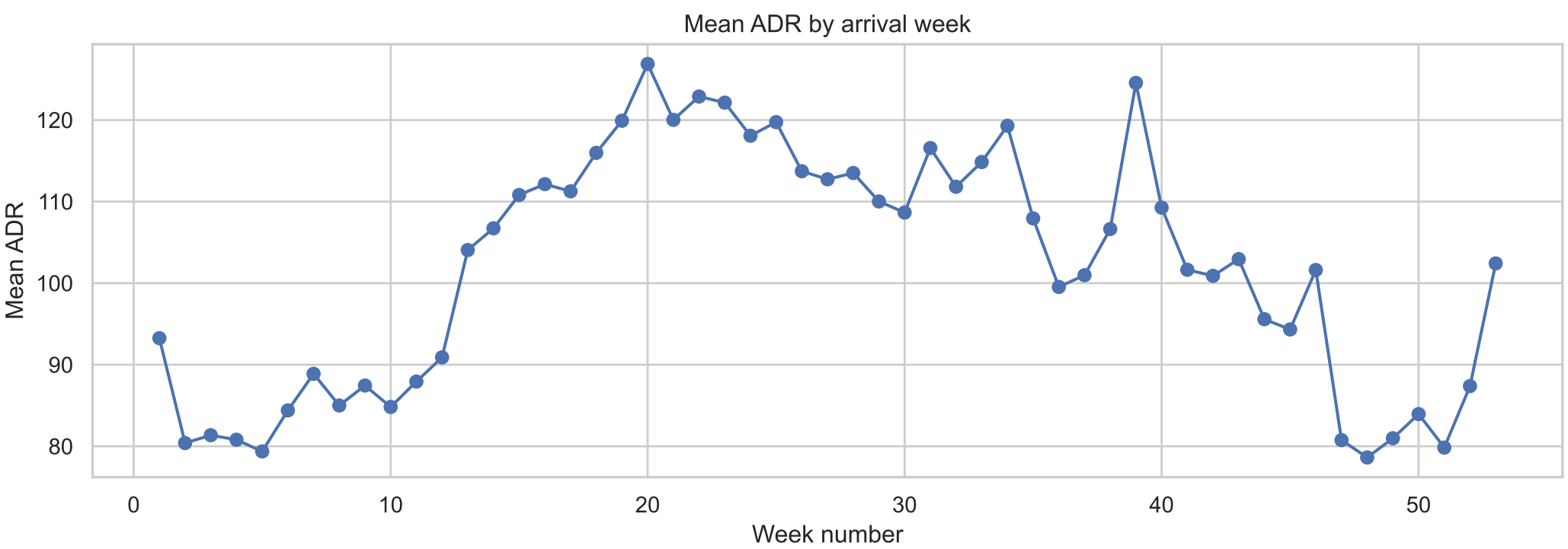
=== 9. ARRIVAL DATE FEATURES ===

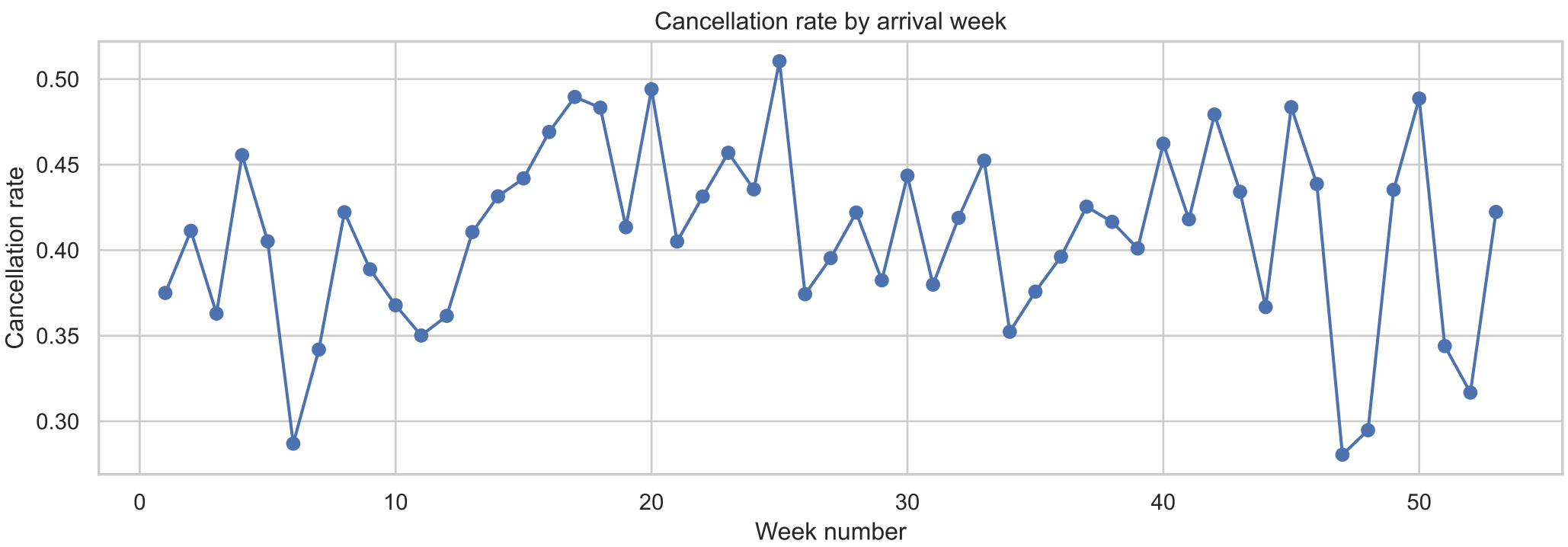

Bookings by ArrivalDateYear
40000
35000
30000
25000
20000
15000
10000
5000
0
count
2015
2016
2017
ArrivalDateYear


Bookings by ArrivalDateMonth
count
0
2000
4000
6000
8000
January
February
March
April
May
June
July
August
September
October
November
December
ArrivalDateMonth


Bookings by ArrivalDateWeekNumber
count
2500
2000
1500
1000
500
0
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53
ArrivalDateWeekNumber

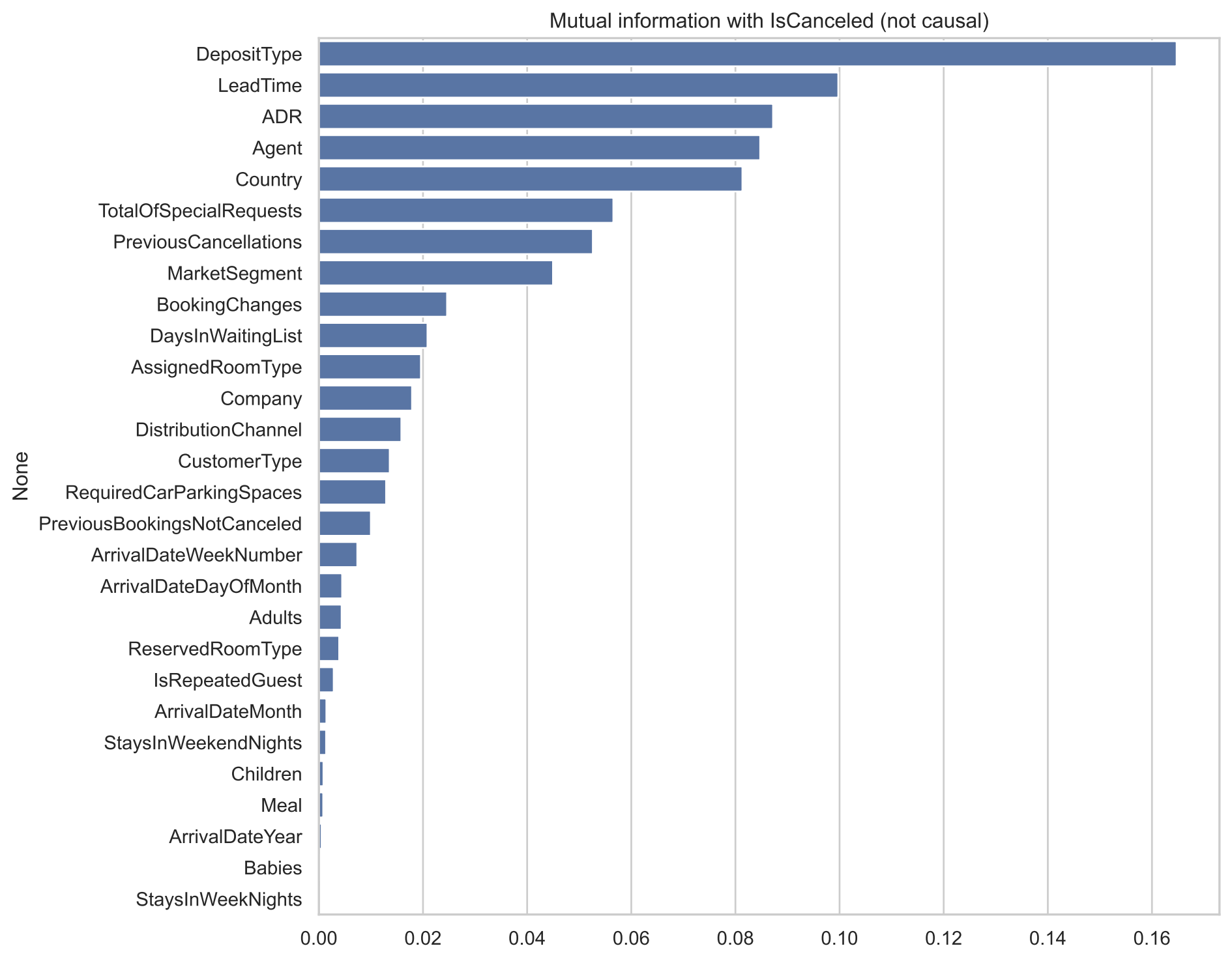

Bookings by ArrivalDateDayOfMonth
count
3000
2500
2000
1500
1000
500
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
ArrivalDateDayOfMonth


Mean ADR by arrival week
Mean ADR
120
110
100
90
80
0
10
20
30
40
50
Week number

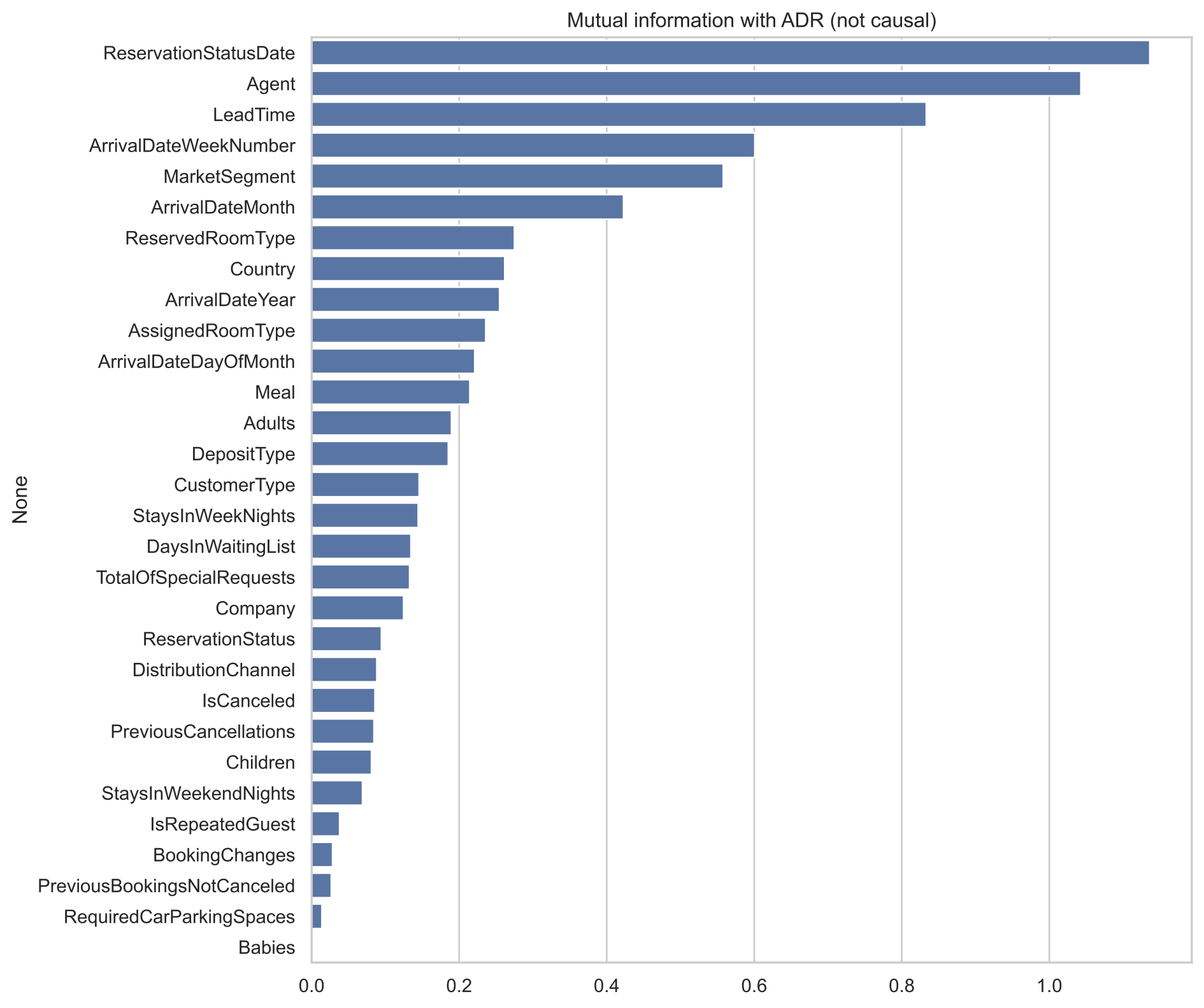

Cancellation rate by arrival week
Cancellation rate
0.50
0.45
0.40
0.35
0.30
0
10
20
30
40
50
Week number

```
=== 10. MUTUAL INFORMATION (OPTIONAL) ===
Excluded likely leakage fields for IsCanceled MI: ['ReservationStatus', 'ReservationStatusDate']

Mutual information ranking for IsCanceled:
 DepositType                   0.164729
LeadTime                       0.099753
ADR                            0.087227
Agent                          0.084759
Country                        0.081341
TotalOfSpecialRequests         0.056554
PreviousCancellations          0.052615
MarketSegment                  0.044963
BookingChanges                 0.024620
DaysInWaitingList              0.020856
AssignedRoomType               0.019598
Company                        0.017894
DistributionChannel            0.015840
CustomerType                   0.013606
RequiredCarParkingSpaces       0.012906
PreviousBookingsNotCanceled    0.009985
ArrivalDateWeekNumber          0.007380
ArrivalDateDayOfMonth          0.004468
Adults                         0.004367
ReservedRoomType               0.003899
IsRepeatedGuest                0.002849
ArrivalDateMonth               0.001439
StaysInWeekendNights           0.001385
Children                       0.000880
Meal                           0.000836
ArrivalDateYear                0.000533
Babies                         0.000322
StaysInWeekNights              0.000000
dtype: float64
```


Mutual information with IsCanceled (not causal)
DepositType
LeadTime
ADR
Agent
Country
TotalOfSpecialRequests
PreviousCancellations
MarketSegment
BookingChanges
DaysInWaitingList
AssignedRoomType
Company
DistributionChannel
CustomerType
RequiredCarParkingSpaces
PreviousBookingsNotCanceled
ArrivalDateWeekNumber
ArrivalDateDayOfMonth
Adults
ReservedRoomType
IsRepeatedGuest
ArrivalDateMonth
StaysInWeekendNights
Children
Meal
ArrivalDateYear
Babies
StaysInWeekNights
None
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16

```
Mutual information ranking for ADR:
 ReservationStatusDate          1.136258
Agent                           1.042873
LeadTime                        0.833282
ArrivalDateWeekNumber           0.600833
MarketSegment                   0.558021
ArrivalDateMonth                0.422869
ReservedRoomType                0.274923
Country                         0.261810
ArrivalDateYear                 0.254821
AssignedRoomType                0.236105
ArrivalDateDayOfMonth           0.221185
Meal                            0.214564
Adults                          0.189600
DepositType                     0.185374
CustomerType                    0.145828
StaysInWeekNights               0.144515
DaysInWaitingList               0.134702
TotalOfSpecialRequests          0.132901
Company                         0.124768
ReservationStatus               0.094662
DistributionChannel             0.088485
IsCanceled                      0.085929
PreviousCancellations           0.084581
Children                        0.081195
StaysInWeekendNights            0.069017
IsRepeatedGuest                 0.037983
BookingChanges                  0.028463
PreviousBookingsNotCanceled     0.026727
RequiredCarParkingSpaces        0.014047
Babies                          0.000448
dtype: float64
```


Mutual information with ADR (not causal)
ReservationStatusDate
Agent
LeadTime
ArrivalDateWeekNumber
MarketSegment
ArrivalDateMonth
ReservedRoomType
Country
ArrivalDateYear
AssignedRoomType
ArrivalDateDayOfMonth
Meal
Adults
DepositType
CustomerType
StaysInWeekNights
DaysInWaitingList
TotalOfSpecialRequests
Company
ReservationStatus
DistributionChannel
IsCanceled
PreviousCancellations
Children
StaysInWeekendNights
IsRepeatedGuest
BookingChanges
PreviousBookingsNotCanceled
RequiredCarParkingSpaces
Babies
None
0.0
0.2
0.4
0.6
0.8
1.0

```
=== IMPORTANT INTERPRETATION NOTES ===
* Correlation is association, not causation.
* Raw variance depends on feature units; do not compare it blindly across columns.
* Outliers are not automatically errors.
* For cancellation prediction, ReservationStatus and ReservationStatusDate can leak the outcome.
* Fit imputers/encoders/scalers using training data only when building predictive models.

EDA complete.
```